

MatrixMultiplicationBias

January 22, 2026

1 Triton Kernel Profiling on AMD GPUs

1.1 Get the System Hardware Info

```
[1]: !amd-smi
```

```
+-----+
| AMD-SMI 26.0.2+39589fda      amdgpu version: 6.16.6   ROCm version: 7.0.3   |
| Platform: Linux Guest                                             |
+-----+
| BDF                        GPU-Name | Mem-Uti   Temp   UEC       Power-Usage |
| GPU  HIP-ID  OAM-ID  Partition-Mode | GFX-Uti   Fan    |           Mem-Usage |
+=====+
| 0000:ff:00.0 AMD Instinct MI300X VF | 0 %       36 °C   0         150/750 W |
| 0      0      7      SPX/NPS1      | 0 %       N/A    |           285/196288 MB |
+-----+
+-----+
| Processes:                                                         |
| GPU      PID  Process Name          GTT_MEM  VRAM_MEM  MEM_USAGE  CU % |
+=====+
| No running processes found                                         |
+-----+
```

1.2 Check profiling environment

```
[2]: !rocprow-compute --version
```

```
-----
rocprow-compute version: 3.2.3 (release)
Git revision:      1d722c14
-----
```

1.3 Triton MatMulBias Kernel Program

```
[3]: %%writefile triton/matmulbias.py
import argparse
import torch

import triton
```

```

import triton.language as tl
import triton.profiler as proton

def _matmul_launch_metadata(grid, kernel, args):
    ret = {}
    M, N, K, WS = args["M"], args["N"], args["K"], args.get("WARP_SPECIALIZE",
↪False)
    BM, BN, BK = args["BLOCK_SIZE_M"], args["BLOCK_SIZE_N"],
↪args["BLOCK_SIZE_K"]
    ws_str = "_ws" if WS else ""
    ret["name"] = f"{kernel.name}{ws_str} [M={M}, N={N}, K={K}] [BM={BM},
↪BN={BN} BK={BK}]"
    if "output_ptr" in args:
        bytes_per_elem = args["output_ptr"].element_size()
    else:
        bytes_per_elem = 2
    ret[f"flops{bytes_per_elem * 8}"] = 2. * M * N * K
    ret["bytes"] = bytes_per_elem * (M * K + N * K + M * N)
    return ret

def _bias_launch_metadata(grid, kernel, args):
    ret = {}
    M, N, WS = args["M"], args["N"], args.get("WARP_SPECIALIZE", False)
    BM, BN = args["BLOCK_SIZE_M"], args["BLOCK_SIZE_N"]
    ws_str = "_ws" if WS else ""
    ret["name"] = f"{kernel.name}{ws_str} [M={M}, N={N}] [BM={BM}, BN={BN}]"
    if "output_ptr" in args:
        bytes_per_elem = args["output_ptr"].element_size()
    else:
        bytes_per_elem = 2
    ret[f"flops{bytes_per_elem * 8}"] = 2. * M * N
    ret["bytes"] = bytes_per_elem * (M + N + M * N)
    return ret

def matmul_autotune_config(pre_hook=None):
    return [
        triton.Config({'BLOCK_SIZE_M': BM, 'BLOCK_SIZE_N': BN, "BLOCK_SIZE_K":
↪BK, "GROUP_SIZE_M": 8}, num_stages=s,
            num_warps=w, pre_hook=pre_hook)
        for BM in [128]
        for BN in [128, 256]
        for BK in [64, 128]
        for s in ([3, 4, 5])
        for w in [4, 8]
    ]

```

```

def bias_autotune_config(pre_hook=None):
    return [
        triton.Config({'BLOCK_SIZE_M': BM, 'BLOCK_SIZE_N': BN, "GROUP_SIZE_M": 8}, num_stages=s,
        ↪}, num_warps=w, pre_hook=pre_hook)
        for BM in [128]
        for BN in [128, 256]
        for s in ([3, 4, 5])
        for w in [4, 8]
    ]

# MatMul kernel
@triton.autotune(
    configs=matmul_autotune_config(),
    key=['M', 'N', 'K'],
)
@triton.jit(launch_metadata=_matmul_launch_metadata)
def matmul_kernel(
    a_ptr, b_ptr, c_ptr,
    M, N, K,
    stride_am, stride_ak,
    stride_bk, stride_bn,
    stride_cm, stride_cn,
    BLOCK_SIZE_M: tl.constexpr, BLOCK_SIZE_N: tl.constexpr, BLOCK_SIZE_K: 8,
    ↪tl.constexpr,
    GROUP_SIZE_M: tl.constexpr
):
    pid = tl.program_id(axis=0)
    num_pid_m = tl.cdiv(M, BLOCK_SIZE_M)
    num_pid_n = tl.cdiv(N, BLOCK_SIZE_N)
    num_pid_in_group = GROUP_SIZE_M * num_pid_n
    group_id = pid // num_pid_in_group
    first_pid_m = group_id * GROUP_SIZE_M
    group_size_m = min(num_pid_m - first_pid_m, GROUP_SIZE_M)
    pid_m = first_pid_m + ((pid % num_pid_in_group) % group_size_m)
    pid_n = (pid % num_pid_in_group) // group_size_m

    tl.assume(pid_m >= 0)
    tl.assume(pid_n >= 0)
    tl.assume(stride_am > 0)
    tl.assume(stride_ak > 0)
    tl.assume(stride_bn > 0)
    tl.assume(stride_bk > 0)
    tl.assume(stride_cm > 0)
    tl.assume(stride_cn > 0)

    offs_am = (pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)) % M

```

```

    offs_bn = (pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)) % N
    offs_k = tl.arange(0, BLOCK_SIZE_K)
    a_ptrs = a_ptr + (offs_am[:, None] * stride_am + offs_k[None, :] *
↪stride_ak)
    b_ptrs = b_ptr + (offs_k[:, None] * stride_bk + offs_bn[None, :] *
↪stride_bn)

    accumulator = tl.zeros((BLOCK_SIZE_M, BLOCK_SIZE_N), dtype=tl.float32)
    for k in range(0, tl.cdiv(K, BLOCK_SIZE_K)):
        a = tl.load(a_ptrs, mask=offs_k[None, :] < K - k * BLOCK_SIZE_K,
↪other=0.0)
        b = tl.load(b_ptrs, mask=offs_k[:, None] < K - k * BLOCK_SIZE_K,
↪other=0.0)
        accumulator = tl.dot(a, b, accumulator)
        a_ptrs += BLOCK_SIZE_K * stride_ak
        b_ptrs += BLOCK_SIZE_K * stride_bk
    c = accumulator.to(tl.float16)

    offs_cm = pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)
    offs_cn = pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)
    c_ptrs = c_ptr + stride_cm * offs_cm[:, None] + stride_cn * offs_cn[None, :]
    c_mask = (offs_cm[:, None] < M) & (offs_cn[None, :] < N)
    tl.store(c_ptrs, c, mask=c_mask)

# Bias kernel
@triton.autotune(
    configs=bias_autotune_config(),
    key=['M', 'K'],
)
@triton.jit(launch_metadata=_bias_launch_metadata)
def bias_kernel(
    a_ptr,
    bias_ptr,
    output_ptr,
    M, N,
    stride_m, stride_n,
    BLOCK_SIZE_M: tl.constexpr, BLOCK_SIZE_N: tl.constexpr, GROUP_SIZE_M:
↪tl.constexpr,
):
    pid = tl.program_id(axis=0)
    num_pid_m = tl.cdiv(M, BLOCK_SIZE_M)
    num_pid_n = tl.cdiv(N, BLOCK_SIZE_N)
    num_pid_in_group = GROUP_SIZE_M * num_pid_n
    group_id = pid // num_pid_in_group
    first_pid_m = group_id * GROUP_SIZE_M
    group_size_m = min(num_pid_m - first_pid_m, GROUP_SIZE_M)
    pid_m = first_pid_m + ((pid % num_pid_in_group) % group_size_m)

```

```

pid_n = (pid % num_pid_in_group) // group_size_m

offs_m = (pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)) % M
offs_n = (pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)) % N
a_ptrs = a_ptr + (offs_m[:, None] * stride_m + offs_n[None, :] * stride_n)

a = tl.load(a_ptrs)

bias = tl.load(bias_ptr + offs_n, mask=offs_n < N, other=0.0)

output = a + bias[None, :]

offs_m = pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)
offs_n = pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)
output_ptrs = output_ptr + stride_m * offs_m[:, None] + stride_n *
↪offs_n[None, :]
output_mask = (offs_m[:, None] < M) & (offs_n[None, :] < N)
tl.store(output_ptrs, output, mask=output_mask)

# MatMulBias Fusion kernel
@triton.autotune(
    configs=matmul_autotune_config(),
    key=['M', 'N', 'K'],
)
@triton.jit(launch_metadata=_matmul_launch_metadata)
def matmulbias_kernel(
    a_ptr, b_ptr, c_ptr,
    bias_ptr,
    M, N, K,
    stride_am, stride_ak,
    stride_bk, stride_bn,
    stride_cm, stride_cn,
    BLOCK_SIZE_M: tl.constexpr, BLOCK_SIZE_N: tl.constexpr, BLOCK_SIZE_K:
↪tl.constexpr,
    GROUP_SIZE_M: tl.constexpr
):
    pid = tl.program_id(axis=0)
    num_pid_m = tl.cdiv(M, BLOCK_SIZE_M)
    num_pid_n = tl.cdiv(N, BLOCK_SIZE_N)
    num_pid_in_group = GROUP_SIZE_M * num_pid_n
    group_id = pid // num_pid_in_group
    first_pid_m = group_id * GROUP_SIZE_M
    group_size_m = min(num_pid_m - first_pid_m, GROUP_SIZE_M)
    pid_m = first_pid_m + ((pid % num_pid_in_group) % group_size_m)
    pid_n = (pid % num_pid_in_group) // group_size_m

    tl.assert(pid_m >= 0)

```

```

tl.assume(pid_n >= 0)
tl.assume(stride_am > 0)
tl.assume(stride_ak > 0)
tl.assume(stride_bn > 0)
tl.assume(stride_bk > 0)
tl.assume(stride_cm > 0)
tl.assume(stride_cn > 0)

offs_am = (pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)) % M
offs_bn = (pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)) % N
offs_k = tl.arange(0, BLOCK_SIZE_K)
a_ptrs = a_ptr + (offs_am[:, None] * stride_am + offs_k[None, :] *
↪stride_ak)
b_ptrs = b_ptr + (offs_k[:, None] * stride_bk + offs_bn[None, :] *
↪stride_bn)

accumulator = tl.zeros((BLOCK_SIZE_M, BLOCK_SIZE_N), dtype=tl.float32)
for k in range(0, tl.cdiv(K, BLOCK_SIZE_K)):
    a = tl.load(a_ptrs, mask=offs_k[None, :] < K - k * BLOCK_SIZE_K,
↪other=0.0)
    b = tl.load(b_ptrs, mask=offs_k[:, None] < K - k * BLOCK_SIZE_K,
↪other=0.0)
    accumulator = tl.dot(a, b, accumulator)
    a_ptrs += BLOCK_SIZE_K * stride_ak
    b_ptrs += BLOCK_SIZE_K * stride_bk
    c = accumulator.to(tl.float16)

bias = tl.load(bias_ptr + offs_bn, mask=offs_bn < N, other=0.0)
output = c + bias[None, :]

offs_cm = pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)
offs_cn = pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)
c_ptrs = c_ptr + stride_cm * offs_cm[:, None] + stride_cn * offs_cn[None, :]
c_mask = (offs_cm[:, None] < M) & (offs_cn[None, :] < N)
tl.store(c_ptrs, output, mask=c_mask)

# MatMulBias kernel wrapper function
def nonfusion_matmulbias(a: torch.Tensor, b: torch.Tensor, bias: torch.Tensor):
    assert a.shape[1] == b.shape[0], "Incompatible dimensions"
    assert a.is_contiguous(), "Matrix A must be contiguous"
    M, K = a.shape
    K, N = b.shape
    assert bias.shape[0] == N, "BIAS has incompatible dimensions"
    c = torch.empty((M, N), device=a.device, dtype=torch.float16)
    matmul_grid = lambda META: (triton.cdiv(M, META['BLOCK_SIZE_M']) * triton.
↪cdiv(N, META['BLOCK_SIZE_N']), )

```

```

matmul_kernel[matmul_grid](
    a, b, c,
    M, N, K,
    a.stride(0), a.stride(1),
    b.stride(0), b.stride(1),
    c.stride(0), c.stride(1),
)

o = torch.empty((M, N), device=c.device, dtype=torch.float16)
bias_grid = lambda META: (triton.cdiv(M, META['BLOCK_SIZE_M']) * triton.
↪cdiv(N, META['BLOCK_SIZE_N']), )
bias_kernel[bias_grid](
    c, bias, o,
    M, N,
    c.stride(0), c.stride(1),
)

return o

# MatMulBias kernel wrapper function
def fusion_matmulbias(a: torch.Tensor, b: torch.Tensor, bias: torch.Tensor):
    assert a.shape[1] == b.shape[0], "Incompatible dimensions"
    assert a.is_contiguous(), "Matrix A must be contiguous"
    M, K = a.shape
    K, N = b.shape
    assert bias.shape[0] == N, "BIAS has incompatible dimensions"
    o = torch.empty((M, N), device=a.device, dtype=torch.float16)
    grid = lambda META: (triton.cdiv(M, META['BLOCK_SIZE_M']) * triton.cdiv(N, ↪
↪META['BLOCK_SIZE_N']), )
    matmulbias_kernel[grid](
        a, b, o,
        bias,
        M, N, K,
        a.stride(0), a.stride(1),
        b.stride(0), b.stride(1),
        o.stride(0), o.stride(1),
    )

    return o

def get_rtol():
    target = triton.runtime.driver.active.get_current_target()
    if target.backend == "hip":
        if target.arch == "gfx90a":
            return 1e-2
        elif target.arch == "gfx942":

```

```

        return 1e-2
    else:
        return 0

def show_profile(profile_name):
    import triton.profiler.viewer as proton_viewer
    metric_names = ["time/ms", "tflop16/s"]
    file_name = f"profiles/{profile_name}.hatchet"
    tree, metrics = proton_viewer.parse(metric_names, file_name)

    print(f"Proton profile results for {profile_name}")
    proton_viewer.print_tree(tree, metrics)

if __name__ == "__main__":
    parser = argparse.ArgumentParser()
    parser.add_argument("--profile", action="store_true")
    parser.add_argument("--verify", action="store_true")
    parser.add_argument("kernel", choices=["fusion", "nonfusion"])
    args = parser.parse_args()

    # Test matrices
    torch.manual_seed(0)
    M = 1024
    N = 1024
    a = torch.randn((M, N), device='cuda', dtype=torch.float16)
    b = torch.randn((N, M), device='cuda', dtype=torch.float16)
    bias = torch.randn(N, device='cuda', dtype=torch.float16)

    if args.profile:
        print(f"Profiling the {args.kernel} matmulbias kernel ...")
        proton.start(f"profiles/{args.kernel}_matmulbias", hook="triton")
    else:
        print(f"Running the {args.kernel} matmulbias kernel ...")

    if args.kernel == "fusion":
        triton_output = fusion_matmulbias(a, b, bias)
    elif args.kernel == "nonfusion":
        triton_output = nonfusion_matmulbias(a, b, bias)

    if args.profile:
        proton.finalize()
    print(f"triton_output: {triton_output}")

    if args.profile:
        show_profile(f"{args.kernel}_matmulbias")

```



```

if args.verify:
    print(f"Running the torch matmulbias kernel ...")
    torch_output = torch.addmm(bias, a, b)
    print(f"torch_output: {torch_output}")
    print("Verifying triton results with torch ...")
    if torch.allclose(triton_output, torch_output, atol=1e-2,
↪rtol=get_rtol()):
        print("Verified")
    else:
        print("Differ")

```

Writing triton/matmulbias.py

1.4 Non-Fusion MatMulBias Kernel

1.4.1 Verify the Non-Fusion MatMulBias Kernel

This also ensures the kernel has been built and cached.

[4]: `!python triton/matmulbias.py --verify nonfusion`

Running the nonfusion matmulbias kernel ...

```

triton_output: tensor([[ 2.7480, 43.6250, 30.9531, ..., 26.2500,  8.8203,
 6.4609],
        [ 7.1562, -21.3906, 17.7969, ..., -16.8125, 29.6562, -48.3125],
        [ 13.8125, -6.4609,  0.5146, ..., 48.0000, 25.1562, -27.5156],
        ...,
        [-54.4375, -1.4375, -0.4214, ...,  5.4297, 17.2812, -39.0625],
        [ 9.6094, -52.2500, 43.0000, ..., -45.8438, -49.3125, 32.6875],
        [-12.8516,  9.5391, 53.3750, ..., -9.3438,  7.6797, 54.1250]],
        device='cuda:0', dtype=torch.float16)

```

Running the torch matmulbias kernel ...

```

torch_output: tensor([[ 2.7480, 43.6562, 30.9531, ..., 26.2500,  8.8203,
 6.4609],
        [ 7.1602, -21.3906, 17.7969, ..., -16.8125, 29.6406, -48.3125],
        [ 13.8125, -6.4609,  0.5142, ..., 48.0312, 25.1406, -27.5156],
        ...,
        [-54.4375, -1.4375, -0.4214, ...,  5.4297, 17.2812, -39.0625],
        [ 9.6094, -52.2500, 43.0000, ..., -45.8125, -49.3125, 32.7188],
        [-12.8516,  9.5391, 53.3750, ..., -9.3438,  7.6797, 54.1562]],
        device='cuda:0', dtype=torch.float16)

```

Verifying triton results with torch ...

Verified

1.4.2 Profile the Non-Fusion MatMulBias Kernel

Using the Triton Proton CLI Generates a hatchet file, `nonfusion_matmulbias.hatchet`, under the profiles directory.

```
[5]: !proton -n profiles/nonfusion_matmulbias -k triton triton/matmulbias.py ↵
      ↪nonfusion
```

Running the nonfusion matmulbias kernel ...

```
triton_output: tensor([[ 2.7480, 43.6250, 30.9531, ..., 26.2500,  8.8203,
 6.4609],
      [ 7.1562, -21.3906, 17.7969, ..., -16.8125, 29.6562, -48.3125],
      [13.8125, -6.4609,  0.5146, ..., 48.0000, 25.1562, -27.5156],
      ...,
      [-54.4375, -1.4375, -0.4214, ...,  5.4297, 17.2812, -39.0625],
      [ 9.6094, -52.2500, 43.0000, ..., -45.8438, -49.3125, 32.6875],
      [-12.8516,  9.5391, 53.3750, ..., -9.3438,  7.6797, 54.1250]],
      device='cuda:0', dtype=torch.float16)
```

Using the Triton Proton API Generates a hatchet file, nonfusion_matmulbias.hatchet, under the profiles directory and outputs the profile.

```
[6]: !python triton/matmulbias.py --profile nonfusion
```

Profiling the nonfusion matmulbias kernel ...

```
triton_output: tensor([[ 2.7480, 43.6250, 30.9531, ..., 26.2500,  8.8203,
 6.4609],
      [ 7.1562, -21.3906, 17.7969, ..., -16.8125, 29.6562, -48.3125],
      [13.8125, -6.4609,  0.5146, ..., 48.0000, 25.1562, -27.5156],
      ...,
      [-54.4375, -1.4375, -0.4214, ...,  5.4297, 17.2812, -39.0625],
      [ 9.6094, -52.2500, 43.0000, ..., -45.8438, -49.3125, 32.6875],
      [-12.8516,  9.5391, 53.3750, ..., -9.3438,  7.6797, 54.1250]],
      device='cuda:0', dtype=torch.float16)
```

Proton profile results for nonfusion_matmulbias

```
1004.262 3.629 ROOT
  875.640 nan _ZN2at6native29vectorized_elementwise_kern
elILi4ENS0_11FillFunctorIiEESt5arrayIPcLm1EEEEviT0_T1_
  43.242 0.584 bias_kernel [M=1024, N=1024] [BM=128,
BN=128]
  47.288 0.552 bias_kernel [M=1024, N=1024] [BM=128,
BN=256]
  38.093 94.316 matmul_kernel [M=1024, N=1024, K=1024]
[BM=128, BN=128 BK=64]
```

Legend (Metric: time/ms (inc) Min: 38.09 Max: 1004.26)

```
907.64 - 1004.26
714.41 - 907.64
521.18 - 714.41
327.94 - 521.18
134.71 - 327.94
38.09 - 134.71
```

```
name User code      Only in left graph
Only in right graph
```

Using the ROCm Systems Compute profiler Generates files found under the workloads/nonfusion_matmulbias/ directory.

```
[7]: !ROCPROF=rocprofiler-sdk rocprof-compute profile -n nonfusion_matmulbias --  
python triton/matmulbias.py nonfusion
```

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 | | | () | (_ | |) | | | () | _ | _ _ | (_ | () | | | | | | | | | | | | | | | |
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 \ _ , _ | \ _ \ _ _ |
 | _ |
 | _ |

WARNING DEPRECATION WARNING: rocm-smi is deprecated in ROCm 7.0 and will be removed from rocprof-compute in ROCm 7.1. Please migrate to amd-smi for compute partition parsing. For migration help, see <https://github.com/ROCm/amdsmi>

```
INFO Rocprofiler-Compute version: 3.2.3
INFO Profiler choice: rocprofiler-sdk
INFO Path: /workspace/user/workloads/nonfusion_matmulbias/MI300X_A1
INFO Target: MI300X_A1
INFO Command: python triton/matmulbias.py nonfusion
INFO Kernel Selection: None
INFO Dispatch Selection: None
INFO Hardware Blocks: []
INFO Report Sections: []
INFO
INFO ~~~~~
INFO Collecting Performance Counters
INFO ~~~~~
INFO
INFO [Run 1/12][Approximate profiling time left: pending first
measurement...]
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias/MI300X_A1/perfmon/SQ_IFETCH_LEVEL.txt
```

```

INFO      |-> [rocprofiler-sdk] W20260122 20:27:30.184826
139937287896640 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.178832 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:27:30.184929
139937287896640 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:27:31.448975
139937287896640 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260122 20:27:35.250550
139937287896640 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      5.065621 sec

INFO      |-> [rocprofiler-sdk] E20260122 20:27:35.263923
139937287896640 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/144_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:27:35.284560
139937287896640 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/144_counter_collectio
n.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:27:37.583360
139937287896640 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/144_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260122 20:27:37.586515
139937287896640 simple_timer.cpp:55] [rocprofv3] output generation ::
2.330345 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:27:37.587609
139937287896640 simple_timer.cpp:55] [rocprofv3] tool finalization ::
2.331503 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 2/12] [Approximate profiling time left: 2 minutes and 36
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias/MI300X_A1/perfmon/SQ_INST_LEVEL_LDS.txt
INFO      |-> [rocprofiler-sdk] W20260122 20:27:45.878277
140234000236096 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.177411 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:27:45.878380
140234000236096 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:27:47.146935
140234000236096 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...

```

```

INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO      |-> [rocprofiler-sdk] W20260122 20:27:50.117837

140234000236096 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.239457 sec

INFO      |-> [rocprofiler-sdk] E20260122 20:27:50.131321

140234000236096 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/166_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:27:50.152148

140234000236096 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/166_counter_collectio
n.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:27:52.475139

140234000236096 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/166_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260122 20:27:52.478252

140234000236096 simple_timer.cpp:55] [rocprofv3] output generation ::
2.354716 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:27:52.479340

140234000236096 simple_timer.cpp:55] [rocprofv3] tool finalization ::
2.355876 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 3/12] [Approximate profiling time left: 2 minutes and 17
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias/MI300X_A1/perfmon/SQ_INST_LEVEL_SMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260122 20:28:00.695546
140440655150656 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.179741 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:00.695655
140440655150656 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:01.959743
140440655150656 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260122 20:28:04.914543
140440655150656 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.218888 sec

INFO      |-> [rocprofiler-sdk] E20260122 20:28:04.928047
140440655150656 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/187_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:28:04.948923
140440655150656 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/187_counter_collectio
n.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:28:07.304178
140440655150656 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/187_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260122 20:28:07.307317
140440655150656 simple_timer.cpp:55] [rocprofv3] output generation ::
2.387110 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:07.308462
140440655150656 simple_timer.cpp:55] [rocprofv3] tool finalization ::
2.388341 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 4/12] [Approximate profiling time left: 2 minutes and 1
second]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias/MI300X_A1/perfmon/SQ_INST_LEVEL_VMEM.txt
INFO      |-> [rocprofiler-sdk] W20260122 20:28:15.465879
140052409167424 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.153315 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:15.465993
140052409167424 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:16.734109
140052409167424 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...

```



```

INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO      |-> [rocprofiler-sdk] W20260122 20:28:19.491114

140052409167424 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.025121 sec

INFO      |-> [rocprofiler-sdk] E20260122 20:28:19.504463

140052409167424 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/209_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:28:19.526631

140052409167424 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/209_counter_collectio
n.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:28:20.901374

140052409167424 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/209_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260122 20:28:20.904656

140052409167424 simple_timer.cpp:55] [rocprofv3] output generation ::
1.407987 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:20.905731

140052409167424 simple_timer.cpp:55] [rocprofv3] tool finalization ::
1.409128 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 5/12] [Approximate profiling time left: 1 minute and 38
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias/MI300X_A1/perfmon/SQ_LEVEL_WAVES.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260122 20:28:25.912185
139975956831808 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128887 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:25.912290
139975956831808 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:27.179465
139975956831808 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260122 20:28:29.942982
139975956831808 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.030692 sec

INFO      |-> [rocprofiler-sdk] E20260122 20:28:29.956397
139975956831808 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/235_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:28:29.979023
139975956831808 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/235_counter_collectio
n.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:28:30.112555
139975956831808 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/235_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260122 20:28:30.115674
139975956831808 simple_timer.cpp:55] [rocprofv3] output generation ::
0.167078 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:30.116655
139975956831808 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.168130 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 6/12] [Approximate profiling time left: 1 minute and 14
seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/nonfusion_matmulbias/MI300X_A1/perfmon/pmc_perf_0.txt
INFO      |-> [rocprofiler-sdk] W20260122 20:28:31.812627
139929221279296 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128607 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:31.812725
139929221279296 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:33.071747
139929221279296 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...

```

```

INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO      |-> [rocprofiler-sdk] W20260122 20:28:35.828657

139929221279296 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.015931 sec

INFO      |-> [rocprofiler-sdk] E20260122 20:28:35.842105

139929221279296 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/255_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:28:35.864739

139929221279296 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/255_counter_collectio
n.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:28:35.986214

139929221279296 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/255_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260122 20:28:35.989236

139929221279296 simple_timer.cpp:55] [rocprofv3] output generation ::
0.154941 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:35.990238

139929221279296 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.156020 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 7/12][Approximate profiling time left: 56 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/nonfusion_matmulbias/MI300X_A1/perfmon/pmc_perf_1.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260122 20:28:37.647933
140564051975744 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128418 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:37.648033
140564051975744 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:38.908202
140564051975744 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260122 20:28:41.666682
140564051975744 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.018650 sec

INFO      |-> [rocprofiler-sdk] E20260122 20:28:41.680103
140564051975744 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/276_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:28:41.702892
140564051975744 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/276_counter_collectio
n.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:28:41.824484
140564051975744 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/276_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260122 20:28:41.827570
140564051975744 simple_timer.cpp:55] [rocprofv3] output generation ::
0.155286 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:41.828549
140564051975744 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.156335 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 8/12] [Approximate profiling time left: 42 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/nonfusion_matmulbias/MI300X_A1/perfmon/pmc_perf_2.txt
INFO      |-> [rocprofiler-sdk] W20260122 20:28:43.499791
140517184089664 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.127974 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:43.499893
140517184089664 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:44.758914
140517184089664 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,

```

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43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO    |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO    |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO    |-> [rocprofiler-sdk] ...,
INFO    |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO    |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO    |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO    |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO    |-> [rocprofiler-sdk] W20260122 20:28:47.512353

140517184089664 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.012461 sec

INFO    |-> [rocprofiler-sdk] E20260122 20:28:47.525691

140517184089664 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/296_kernel_trace.csv

INFO    |-> [rocprofiler-sdk] E20260122 20:28:47.548454

140517184089664 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/296_counter_collectio
n.csv

INFO    |-> [rocprofiler-sdk] E20260122 20:28:47.669886

140517184089664 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/296_agent_info.csv

INFO    |-> [rocprofiler-sdk] W20260122 20:28:47.672891

140517184089664 simple_timer.cpp:55] [rocprofv3] output generation ::
0.154981 sec

INFO    |-> [rocprofiler-sdk] W20260122 20:28:47.673858

140517184089664 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.156012 sec

INFO    |-> [rocprofiler-sdk]
INFO [Run 9/12][Approximate profiling time left: 29 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/nonfusion_matmulbias/MI300X_A1/perfmon/pmc_perf_3.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260122 20:28:49.347521
139764957724224 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128249 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:49.347624
139764957724224 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:50.606428
139764957724224 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```



```

INFO      |-> [rocprofiler-sdk] W20260122 20:28:53.401512
139764957724224 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.053888 sec

INFO      |-> [rocprofiler-sdk] E20260122 20:28:53.414962
139764957724224 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/317_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:28:53.441580
139764957724224 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/317_counter_collectio
n.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:28:53.584808
139764957724224 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/317_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260122 20:28:53.587894
139764957724224 simple_timer.cpp:55] [rocprofv3] output generation ::
0.180748 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:53.588888
139764957724224 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.181810 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 10/12] [Approximate profiling time left: 18 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/nonfusion_matmulbias/MI300X_A1/perfmon/pmc_perf_4.txt
INFO      |-> [rocprofiler-sdk] W20260122 20:28:55.362329
140053257760320 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128407 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:55.362433
140053257760320 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:28:56.623748
140053257760320 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,

```

```

43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO    |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO    |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO    |-> [rocprofiler-sdk] ...,
INFO    |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO    |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO    |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO    |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO    |-> [rocprofiler-sdk] W20260122 20:28:59.489691

140053257760320 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.127257 sec

INFO    |-> [rocprofiler-sdk] E20260122 20:28:59.503373

140053257760320 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/337_kernel_trace.csv

INFO    |-> [rocprofiler-sdk] E20260122 20:28:59.539683

140053257760320 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/337_counter_collectio
n.csv

INFO    |-> [rocprofiler-sdk] E20260122 20:28:59.745136

140053257760320 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/337_agent_info.csv

INFO    |-> [rocprofiler-sdk] W20260122 20:28:59.748436

140053257760320 simple_timer.cpp:55] [rocprofv3] output generation ::
0.252901 sec

INFO    |-> [rocprofiler-sdk] W20260122 20:28:59.749394

140053257760320 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.253931 sec

INFO    |-> [rocprofiler-sdk]
INFO [Run 11/12] [Approximate profiling time left: 9 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/nonfusion_matmulbias/MI300X_A1/perfmon/pmc_perf_5.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260122 20:29:01.765836
140500135538240 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128593 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:29:01.765949
140500135538240 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:29:03.025712
140500135538240 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260122 20:29:05.891798
140500135538240 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      4.125849 sec

INFO      |-> [rocprofiler-sdk] E20260122 20:29:05.905426
140500135538240 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/358_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:29:05.941749
140500135538240 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/358_counter_collectio
n.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:29:06.143989
140500135538240 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/358_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260122 20:29:06.147169
140500135538240 simple_timer.cpp:55] [rocprofv3] output generation ::
0.249502 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:29:06.148168
140500135538240 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.250570 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 12/12] [Approximate profiling time left: 0 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/nonfusion_matmulbias/MI300X_A1/perfmon/pmc_perf_6.txt
INFO      |-> [rocprofiler-sdk] W20260122 20:29:08.140533
139756268912192 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128293 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:29:08.140637
139756268912192 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:29:09.403071
139756268912192 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,

```

```

43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO    |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO    |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO    |-> [rocprofiler-sdk] ...,
INFO    |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO    |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO    |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO    |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO    |-> [rocprofiler-sdk] W20260122 20:29:12.017693

139756268912192 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
nonfusion' ::      3.877056 sec

INFO    |-> [rocprofiler-sdk] E20260122 20:29:12.031240

139756268912192 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/379_kernel_trace.csv

INFO    |-> [rocprofiler-sdk] E20260122 20:29:12.077426

139756268912192 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/379_counter_collectio
n.csv

INFO    |-> [rocprofiler-sdk] E20260122 20:29:12.222766

139756268912192 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/379_agent_info.csv

INFO    |-> [rocprofiler-sdk] W20260122 20:29:12.226097

139756268912192 simple_timer.cpp:55] [rocprofv3] output generation ::
0.202996 sec

INFO    |-> [rocprofiler-sdk] W20260122 20:29:12.227081

139756268912192 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.204045 sec

INFO    |-> [rocprofiler-sdk]
WARNING Detected differing Workgroup_Size values while joining
pmc_perf.csv
INFO [roofline] Checking for roofline.csv in
/workspace/user/workloads/nonfusion_matmulbias/MI300X_A1
INFO [roofline] No roofline data found. Generating...

```

Empirical Roofline Calculation

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Total detected GPU devices: 1

GPU Device 0 (gfx942) with 304 CUs: Profiling...

```
99% [||||| ]
HBM BW, GPU ID: 0, workgroupSize:256, workgroups:6225920, experiments:100,
traffic:25501368320 bytes, duration:6.1 ms, mean:4185.1 GB/sec, stdev=18.4
GB/sec
99% [||||| ]
MALL BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:2611340115968 bytes, duration:414.3 ms, mean:6327.4 GB/sec, stdev=19.9
GB/sec
99% [||||| ]
L2 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:1632087572480 bytes, duration:64.1 ms, mean:25504.2 GB/sec, stdev=91.1
GB/sec
99% [||||| ]
L1 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:127506841600 bytes, duration:3.7 ms, mean:34021.6 GB/sec, stdev=159.9
GB/sec
99% [||||| ]
LDS BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:79691776000 bytes, duration:1.3 ms, mean:62144.8 GB/sec, stdev=632.4
GB/sec
99% [||||| ]
Peak FLOPs (FP8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:1081258016768, duration:13.0 ms, mean:82936.7 GFLOPS,
stdev=163.9 GFLOPS
99% [||||| ]
Peak FLOPs (FP16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:530428461056, duration:13.4 ms, mean:39535.4 GFLOPS,
stdev=66.6 GFLOPS
99% [||||| ]
Peak FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:530428461056, duration:92.042 ms, mean:5760.1 GFLOPS,
stdev=9.7 GFLOPS
99% [||||| ]
Peak FLOPs (FP32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:265214230528, duration:2.043 ms, mean:130749.6 GFLOPS,
stdev=1457.2 GFLOPS
99% [||||| ]
Peak FLOPs (FP64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:122406567936, duration:1.2 ms, mean:106125.9 GFLOPS,
stdev=1308.6 GFLOPS
99% [||||| ]
Peak IOPs (INT8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:1081258016768, duration:13.1 ms, mean:82690.7 GOPS,
stdev=143.6 GOPS
```

```

99% [||||| ]
Peak IOPs (INT32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:265214230528, duration:3.7 ms, mean:71761.9 GOPS,
stdev=716.5 GOPS
99% [||||| ]
Peak IOPs (INT64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:122406567936, duration:6.2 ms, mean:19725.4 GOPS,
stdev=56.7 GOPS
100% [||||| ]
Peak MFMA FLOPs (F4), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:1, FLOP:0, duration:0.0 ms, mean:0.0 GFLOPS, stdev=0.0 GFLOPS
100% [||||| ]
Peak MFMA FLOPs (F6), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:1, FLOP:0, duration:0.0 ms, mean:0.0 GFLOPS, stdev=0.0 GFLOPS
99% [||||| ]
Peak MFMA FLOPs (F8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:10200547328000, duration:5.3 ms, mean:1916856.8 GFLOPS,
stdev=17368.4 GFLOPS
99% [||||| ]
Peak MFMA FLOPs (F16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:5100273664000, duration:5.4 ms, mean:940836.6 GFLOPS,
stdev=2812.6 GFLOPS
99% [||||| ]
Peak MFMA FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:2550136832000, duration:4.9 ms, mean:522553.3 GFLOPS,
stdev=2901.8 GFLOPS
99% [||||| ]
Peak MFMA FLOPs (F32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:1275068416000, duration:9.1 ms, mean:140014.0 GFLOPS,
stdev=414.0 GFLOPS
99% [||||| ]
Peak MFMA FLOPs (F64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:637534208000, duration:4.4 ms, mean:145610.5 GFLOPS,
stdev=657.0 GFLOPS
99% [||||| ]
Peak MFMA IOPs (I8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:5100273664000, duration:5.1 ms, mean:1001420.6 GOPS,
stdev=5245.4 GOPS

```

INFO [roofline] Empirical Roofline PDFs saved!

1.4.3 Analyze the Non-Fusion MatMulBias Kernel Profiles

Using the Triton Proton CLI

```
[8]: !proton-viewer -m time/ms -f full profiles/nonfusion_matmulbias.hatchet
```

1004.262 ROOT

875.640 _ZN2at6native29vectorized_elementwise_kernelILi4ENS0_1

0	void at::native::vectorized_elementwise_ 42759.05 42676.75 85.92 kernel<4, at::native::FillFunctor<int>, std::array<char*, 1ul> >(int, at::nat...	1405.00	60076471.91
1	bias_kernel 3554.80 3367.80 8.11	1596.00	5673466.36
2	matmul_kernel 25016.11 24059.46 5.44	152.00	3802448.22
3	__amd_rocclr_copyBuffer 2998.74 3016.84 0.38	89.00	266888.07
4	void at::native::(anonymous namespace):: 6376.64 7507.17 0.03 distribution_elementwise_grid_stride_ker nel<float, 4, at::native::templates::...	3.00	19129.92
5	void at::native::(anonymous namespace):: 2311.88 2215.17 0.02 CatArrayBatchedCopy_contig<at::native::(anonymous namespace)::OpaqueType<2u>,...	7.00	16183.18
6	void at::native::vectorized_elementwise_ 3424.42 3434.50 0.01 kernel<4, at::native::compare_scalar_ker nel<double>(at::TensorIteratorBase&, ...	3.00	10273.25
7	void at::native::vectorized_elementwise_ 3130.50 3130.50 0.01 kernel<8, at::native::UnaryFunctor<c10:	2.00	6261.00

```

:Half, c10::Half, bool, at::native::(...)

8 void at::native::vectorized_elementwise_      2.00      5205.07
2602.53      2602.53      0.01
kernel<8, at::native::BinaryFunctor<c10::

:Half, c10::Half, bool, at::native::(...)

9 void at::native::vectorized_elementwise_      2.00      5004.50
2502.25      2502.25      0.01
kernel<8, at::native::AbsFunctor<c10::Ha

lf>, std::array<char*, 2ul> >(int, at...
```

0.2 Dispatch List

Dispatch_ID	Kernel_Name	GPU_ID
0	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*)::...	0
1	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*)::...	0
2	void at::native::(anonymous	

```

namespace)::distribution_elementwise_grid_stride_ker          0
                        nel<float, 4,
at::native::templates::cuda::normal_and_transform<c10::Half, float
                        , at::CUDAGeneratorImpl*,
at::native::templates::cuda::normal_kernel<at::CUDAGen
                        eratorImpl*>(at::TensorBase const&, double, double,
at::CUDAGeneratorImpl*)::...

```

```

3          3  matmul_kernel
0

```

```

4          4  void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
                        std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
                        l>)

```

```

5          5  matmul_kernel
0

```

```

6          6  void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
                        std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
                        l>)

```

```

7          7  matmul_kernel
0

```

```

8          8  void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
                        std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
                        l>)

```

```

9          9  matmul_kernel
0

```

1. System Info

	Info
workload_name	nonfusion_matmulbias
command	python triton/matmulbias.py nonfusion
ip_blocks	SQ LDS SQC TA TD TCP TCC SPI CPC CPF roofline
timestamp	Thu Jan 22 20:27:26 2026 (UTC)
version	3
hostname	5554ed8fd482
cpu_model	Intel(R) Xeon(R) Platinum 8470
sbios	SeaBIOS1.15.0-1
linux_distro	CentOS Stream 10 (Coughlan)
linux_kernel_version	5.15.0-164-generic
amd_gpu_kernel_version	nan
cpu_memory	230893520
gpu_memory	nan
rocm_version	7.0.3
vbios	113-M3000108-103
compute_partition	SPX
memory_partition	nan
gpu_series	MI300
gpu_model	MI300X_A1
gpu_arch	gfx942

gpu_chip_id	29877
gpu_l1	32
gpu_l2	4096
cu_per_gpu	304
simd_per_cu	4
se_per_gpu	32
wave_size	64
workgroup_max_size	1024
max_waves_per_cu	32
max_sclk	2100
max_mclk	1300
cur_sclk	2100
cur_mclk	1300
total_l2_chan	128
lds_banks_per_cu	32
sqc_per_gpu	160
pipes_per_gpu	4
num_xcd	8
num_hbm_channels	128

2. System Speed-of-Light

2.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit	Peak
Pct of Peak				

2.1.0 0.12	VALU FLOPs	97.15	Gflop/s	81715.2
2.1.1 0.77	VALU IOPs	633.08	Giop/s	81715.2
2.1.2 2614886.4	MFMA FLOPs (F8) 0.0	0.0	Gflop/s	
2.1.3 1307443.2	MFMA FLOPs (BF16) 0.0	0.0	Gflop/s	
2.1.4 1307443.2	MFMA FLOPs (F16) 0.31	4050.42	Gflop/s	
2.1.5 163430.4	MFMA FLOPs (F32) 0.0	0.0	Gflop/s	
2.1.6 163430.4	MFMA FLOPs (F64) 0.0	0.0	Gflop/s	
2.1.7	MFMA FLOPs (F6F4)		Gflop/s	
2.1.8 2614886.4	MFMA IOPs (Int8) 0.0	0.0	Giop/s	
2.1.9 37.5	Active CUs	114.0	Cus	304.0
2.1.10 1.55	SALU Utilization	1.55	Pct	100.0
2.1.11 1.6	VALU Utilization	1.6	Pct	100.0

2.1.12 0.24	MFMA Utilization	0.24	Pct	100.0
2.1.13 0.39	VMEM Utilization	0.39	Pct	100.0
2.1.14 0.28	Branch Utilization	0.28	Pct	100.0
2.1.15 99.38	VALU Active Threads	63.6	Threads	64.0
2.1.16 5.63	IPC	0.28	Instr/cycle	5.0
2.1.17 76.43	Wavefront Occupancy	7434.84	Wavefronts	9728.0
2.1.18 0.31	Theoretical LDS Bandwidth	255.47	Gb/s	81715.2
2.1.19 2.25	LDS Bank Conflicts/Access	0.72	Conflicts/access	32.0
2.1.20 17.27	vL1D Cache Hit Rate	17.27	Pct	100.0
2.1.21 8.35	vL1D Cache BW	6824.66	Gb/s	81715.2
2.1.22 43.03	L2 Cache Hit Rate	43.03	Pct	100.0
2.1.23 18.59	L2 Cache BW	6395.53	Gb/s	34406.4

2.1.24 6.24	L2-Fabric Read BW	332.41	Gb/s	5324.8
2.1.25 56.15	L2-Fabric Write BW	2989.89	Gb/s	5324.8
2.1.26	L2-Fabric Read Latency	415.97	Cycles	
2.1.27	L2-Fabric Write Latency	389.63	Cycles	
2.1.28 98.12	sL1D Cache Hit Rate	98.12	Pct	100.0
2.1.29 0.2	sL1D Cache BW	42.49	Gb/s	21504.0
2.1.30 99.75	L1I Hit Rate	99.75	Pct	100.0
2.1.31 2.21	L1I BW	474.87	Gb/s	21504.0
2.1.32	L1I Fetch Latency		Cycles	

3. Memory Chart

3.1 Memory Chart

(Normalization: per_kernel)

Instr Buff Instr Dispatch Exec
 LDS L2 Cache

SALU: 413307

-----> Active
CUs Req : 9447 Util : 0 %
 xGMI/PCIE

Wave 0 Instr Buf 11
4: 304 <-----

 SMEM:
28157 Lat : 68
cycles Hit : 43
%

>

|

RVGPRseq: 38.0

 | |
Wave N-1 Instr Buf VALU:
371090 Vector L1 Cache
Request |

|

>

v

SGPRs : 37

 MFMA:
6083 Rd : 169439 Hit : 17 % Rd
: 20218 Rd : 23001

```

-----
> <----- <-----
-
Fabric

Wt : 7273757 Lat : N/A Wr : 1821461 Wr :
1821461 Rd :
15181
VMEM: 116319 LDS Alloc: -
-----> -----
> <----- Latency (cycles) Rd:
15183
----->
9 Atomic: 0 Coales: 28 % Atomic: 0 Atomic:
0 Wr : 1816898 <----- HBM
<
-----> <-----
> -----> Rd : 416 Wr:
1816898
LDS:
9447 Stall : 33
cycles Atomic: 0 Wr :
390 ----->
-----> Scratch Alloc:
-----> Atomic: 0
0

Wave Occupancy GWS: 0
Scalar L1D
Cache Latency (cycles)
65 -----
---> Rd :
69

Wave Life
<-----
^
|

8876 BRANCH:
56682 Wavefronts: Rd : 28163 Hit : 91.0 % Wr
: 0 Rd : N/A
|

```

```

----->                                0 <
-----
>                                     |
|
Lat : N/A Atomic: 0 Wr :
N/A |
v   ^
    <-----
>
|                                     Workgroups:
|
|                                     Instr L1 Cache
0                                     GMI
|
|
|
|                                     Fetch :
215281                               Hit : 98 % Req
: 2689
-----
-----<-----

Lat : 1049.0 cycles

```

4. Roofline

4.1 Roofline

140014.0
140014GFLOP/s

36204.4

9361.6

2420.7
@ @

625.9 62145 GB/s

34022 GB/s

161.9

41.9 4185 GB/s

0.01 0.18 3.16 56.23 1000.00

Performance (GFLOP/sec) Arithmetic Intensity (FLOP)s/Byte)

5. Command Processor (CPC/CPF)
5.1 Command Processor Fetcher

Metric_ID	Metric	Avg	Min	Max	Unit
5.1.0	CPF Utilization	100.00	100.00	100.00	Pct
5.1.1	CPF Stall	2.17	0.06	9.59	Pct
5.1.2	CPF-L2 Utilization	23.35	2.92	52.03	Pct
5.1.3	CPF-L2 Stall	0.00	0.00	0.00	Pct
5.1.4	CPF-UTCL1 Stall	2.39	0.06	9.60	Pct

5.2 Packet Processor

Metric_ID	Metric	Avg	Min	Max	Unit
5.2.0	CPC SYNC FIFO Full Rate				Pct
5.2.1	CPC CANE Stall Rate				Pct
5.2.2	CPC ADC Utilization				Pct
5.2.3	CPC Utilization	100.0	100.0	100.0	Pct

5.2.4	CPC Stall Rate	24.16	7.57	56.87	Pct
5.2.5	CPC Packet Decoding Utilization	56.7	23.94	109.93	Pct
5.2.6	CPC-Workgroup Manager Utilization	30.07	0.49	70.58	Pct
5.2.7	CPC-L2 Utilization	0.67	0.17	1.38	Pct
5.2.8	CPC-UTCL1 Stall	0.42	0.11	5.31	Pct
5.2.9	CPC-UTCL2 Utilization	0.41	0.11	4.61	Pct

6. Workgroup Manager (SPI)

6.1 Workgroup Manager Utilizations

Metric_ID	Metric	Avg	Min	Max	Unit
6.1.0	Schedule-Pipe Wave Occupancy				Wave
6.1.1	Accelerator Utilization	100.0	100.0	100.0	Pct
6.1.2	Scheduler-Pipe Utilization	0.0	0.0	0.0	Pct
6.1.3	Scheduler-Pipe Wave Utilization				Pct

6.1.4	Workgroup Manager Utilization	51.7	5.64	83.34	Pct
6.1.5	Shader Engine Utilization	46.52	0.21	93.5	Pct
6.1.6	SIMD Utilization	34.92	0.01	92.65	Pct
6.1.7 Workgroups	Dispatched Workgroups	0.0	0.0	0.0	
6.1.8 Wavefronts	Dispatched Wavefronts	0.0	0.0	0.0	
6.1.9 Cycles/wave	VGPR Writes				
6.1.10 Cycles/wave	SGPR Writes				

6.2 Workgroup Manager - Resource Allocation

Metric_ID Unit	Metric	Avg	Min	Max
6.2.0 Pct	Not-scheduled Rate (Workgroup Manager)	0.0	0.0	0.0
6.2.1 Pct	Not-scheduled Rate (Scheduler-Pipe)	0.0	0.0	0.0
6.2.2 Pct	Scheduler-Pipe FIFO Full Rate			

6.2.3 Pct	Scheduler-Pipe Stall Rate	0.0	0.0	0.0
6.2.4 Pct	Scratch Stall Rate	0.0	0.0	0.0
6.2.5 Pct	Insufficient SIMD Waveslots	0.0	0.0	0.0
6.2.6 Pct	Insufficient SIMD VGPRs	0.0	0.0	0.0
6.2.7 Pct	Insufficient SIMD SGPRs	0.0	0.0	0.0
6.2.8 Pct	Insufficient CU LDS	0.0	0.0	0.0
6.2.9 Pct	Insufficient CU Barriers	0.0	0.0	0.0
6.2.10 Pct	Reached CU Workgroup Limit	0.0	0.0	0.0
6.2.11 Pct	Reached CU Wavefront Limit	0.0	0.0	0.0

7. Wavefront

7.1 Wavefront Launch Stats

Metric_ID	Metric	Avg	Min	Max	Unit
-----------	--------	-----	-----	-----	------

7.1.0 items	Grid Size	1810189.93	32.00	4194304.00	Work
7.1.1 items	Workgroup Size	316.52	32.00	1024.00	Work
7.1.2 Wavefronts	Total Wavefronts	0.00	0.00	0.00	
7.1.3 Wavefronts	Saved Wavefronts	0.00	0.00	0.00	
7.1.4 Wavefronts	Restored Wavefronts	0.00	0.00	0.00	
7.1.5 Registers	VGPRs	38.42	8.00	104.00	
7.1.6 Registers	AGPRs	2.91	0.00	192.00	
7.1.7 Registers	SGPRs	36.80	16.00	112.00	
7.1.8	LDS Allocation	9.38	0.00	29184.00	Bytes
7.1.9 Bytes/workitem	Scratch Allocation	0.00	0.00	0.00	

7.2 Wavefront Runtime Stats

Metric_ID	Metric	Avg	Min
Max Unit			

7.2.0	Kernel Time (Nanosec)	21349.80	1690.51
57988.41	Ns		
7.2.1	Kernel Time (Cycles)	60349.24	18327.75
346390.12	Cycle		
7.2.2	Instructions per wavefront	323.38	19.50
3555.00	Instr/wavefront		
7.2.3	Wave Cycles	287481189.35	4296.00
914262460.00	Cycles per kernel		
7.2.4	Dependency Wait Cycles	65251984.47	3912.00
199297260.00	Cycles per kernel		
7.2.5	Issue Wait Cycles	218339568.19	8.00
713676392.00	Cycles per kernel		
7.2.6	Active Cycles	4128628.97	296.00
16351744.00	Cycles per kernel		
7.2.7	Wavefront Occupancy	7434.84	4.44
35901.38	Wavefronts		

10. Compute Units - Instruction Mix

10.1 Overall Instruction Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.1.0	VALU	365007.10	20.00	2731520.00	Instr per kernel
10.1.1	VMEM	116319.23	2.00	262144.00	Instr per kernel
10.1.2	LDS	9447.42	0.00	212992.00	Instr per kernel
10.1.3	MFMA	6083.34	0.00	131072.00	Instr per kernel

10.1.4	SALU	413307.43	24.00	924160.00	Instr per kernel
10.1.5	SMEM	28157.23	0.00	68096.00	Instr per kernel
10.1.6	Branch	56681.70	0.00	131072.00	Instr per kernel

10.2 VALU Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.2.0	INT32	198085.30	9.00	393216.00	Instr per kernel
10.2.1	INT64	28594.53	0.00	668160.00	Instr per kernel
10.2.2	F16-ADD	3992.19	0.00	8192.00	Instr per kernel
10.2.3	F16-MUL	0.00	0.00	0.00	Instr per kernel
10.2.4	F16-FMA	10.01	0.00	16384.00	Instr per kernel
10.2.5	F16-Trans	0.00	0.00	0.00	Instr per kernel
10.2.6	F32-ADD	23.78	0.00	38912.00	Instr per kernel
10.2.7	F32-MUL	725.30	0.00	123392.00	Instr per kernel
10.2.8	F32-FMA	71.35	0.00	116736.00	Instr per kernel
10.2.9	F32-Trans	689.63	0.00	65024.00	Instr per kernel
10.2.10	F64-ADD	0.00	0.00	0.00	Instr per kernel
10.2.11	F64-MUL	0.00	0.00	1.00	Instr per kernel
10.2.12	F64-FMA	0.00	0.00	6.00	Instr per kernel
10.2.13	F64-Trans	0.00	0.00	1.00	Instr per kernel
10.2.14	Conversion	2095.87	0.00	58368.00	Instr per kernel

10.3 VMEM Instr Mix

Metric_ID	Metric	Avg	Min	Max
Unit				
10.3.0	Global/Generic Instr	112471.92	0.0	262144.0

Instr per kernel

10.3.1	Global/Generic Read	0.07	0.0	6.0
Instr per kernel				
10.3.2	Global/Generic Write	112471.85	0.0	262144.0
Instr per kernel				
10.3.3	Global/Generic Atomic	0.0	0.0	0.0
Instr per kernel				
10.3.4	Spill/Stack Instr	3847.31	0.0	36864.0
Instr per kernel				
10.3.5	Spill/Stack Coalesceable Instr			
Instr per kernel				
10.3.6	Spill/Stack Read	2659.16	0.0	32768.0
Instr per kernel				
10.3.7	Spill/Stack Write	1188.15	0.0	4096.0
Instr per kernel				
10.3.8	Spill/Stack Atomic	0.0	0.0	0.0
Instr per kernel				

10.4 MFMA Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.4.0	MFMA-I8	0.0	0.0	0.0	Instr per kernel
10.4.1	MFMA-F8	0.0	0.0	0.0	Instr per kernel
10.4.2	MFMA-F16	6083.34	0.0	131072.0	Instr per kernel
10.4.3	MFMA-BF16	0.0	0.0	0.0	Instr per kernel
10.4.4	MFMA-F32	0.0	0.0	0.0	Instr per kernel

10.4.5	MFMA-F64	0.0	0.0	0.0	Instr per kernel
10.4.6	MFMA-F6F4				Instr per kernel

11. Compute Units - Compute Pipeline

11.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit	Peak	Pct of Peak
11.1.0	VALU FLOPs	97.15	Gflop	81715.2	0.12
11.1.1	VALU IOPs	633.08	Giop	81715.2	0.77
11.1.2	MFMA FLOPs (F8)	0.0	Gflop	2614886.4	0.0
11.1.3	MFMA FLOPs (BF16)	0.0	Gflop	1307443.2	0.0
11.1.4	MFMA FLOPs (F16)	4050.42	Gflop	1307443.2	0.31
11.1.5	MFMA FLOPs (F32)	0.0	Gflop	163430.4	0.0
11.1.6	MFMA FLOPs (F64)	0.0	Gflop	163430.4	0.0
11.1.7	MFMA FLOPs (F6F4)		Gflop		

11.1.8	MFMA IOPs (INT8)	0.0	Giop	2614886.4	0.0
--------	------------------	-----	------	-----------	-----

11.2 Pipeline Stats

Metric_ID	Metric	Avg	Min	Max	Unit
11.2.0	IPC	0.28	0.04	1.27	Instr/cycle
11.2.1	IPC (Issued)	0.98	0.85	1.0	Instr/cycle
11.2.2	SALU Utilization	1.55	0.0	10.84	Pct
11.2.3	VALU Utilization	1.6	0.0	32.15	Pct
11.2.4	VALU Co-Issue Efficiency				Pct
11.2.5	VMEM Utilization	0.39	0.0	0.85	Pct
11.2.6	Branch Utilization	0.28	0.0	1.49	Pct
11.2.7	VALU Active Threads	63.6	28.46	64.01	Threads
11.2.8	MFMA Utilization	0.24	0.0	5.7	Pct
11.2.9	MFMA Instr Cycles	32.0	32.0	32.0	Cycles/instr

11.2.10	VMEM Latency	Cycles
---------	--------------	--------

11.2.11	SMEM Latency	Cycles
---------	--------------	--------

11.3 Arithmetic Operations

Metric_ID	Metric	Avg	Min	Max	Unit
11.3.0 kernel	FLOPs (Total)	100027462.0	0.0	2147745792.0	Ops per
11.3.1 kernel	IOPs (Total)	14507509.43	960.0	59179008.0	Ops per
11.3.2 kernel	F8 OPs	0.0	0.0	0.0	Ops per
11.3.3 kernel	F16 OPs	99926251.75	0.0	2147483648.0	Ops per
11.3.4 kernel	BF16 OPs	0.0	0.0	0.0	Ops per
11.3.5 kernel	F32 OPs	101209.97	0.0	29491200.0	Ops per
11.3.6 kernel	F64 OPs	0.27	0.0	896.0	Ops per
11.3.7 kernel	F6F4 OPs				Ops per

11.3.8	INT8 OPs	0.0	0.0	0.0	Ops per
--------	----------	-----	-----	-----	---------

kernel

12. Local Data Share (LDS)

12.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
12.1.0	Utilization	0.24	Pct of peak
12.1.1	Access Rate	0.16	Pct of peak
12.1.2	Theoretical Bandwidth (% of Peak)	0.31	Pct of peak
12.1.3	Bank Conflict Rate	2.25	Pct of peak

12.2 LDS Stats

Metric_ID	Metric	Avg	Min	Max
12.2.0	LDS Instrs	9447.42	0.0	212992.0

Unit
Instr per kernel

12.2.1 LDS LOAD
Instr per kernel

12.2.2 LDS STORE
Instr per kernel

12.2.3 LDS ATOMIC
Instr per kernel

12.2.4 LDS LOAD Bandwidth
Gbps

12.2.5	LDS STORE Bandwidth Gbps			
12.2.6	LDS ATOMIC Bandwidth Gbps			
12.2.7	Theoretical Bandwidth Bytes per kernel	6060396.18	0.0	134217728.0
12.2.8	LDS Latency Cycles	67.88	47.8	91.03
12.2.9	Bank Conflicts/Access Conflicts/access	0.72	0.0	1.0
12.2.10	Index Accesses Cycles per kernel	47627.58	0.0	1048576.0
12.2.11	Atomic Return Cycles Cycles per kernel	0.0	0.0	0.0
12.2.12	Bank Conflict Cycles per kernel	280.74	0.0	2304.0
12.2.13	Addr Conflict Cycles per kernel	0.01	0.0	24.0
12.2.14	Unaligned Stall Cycles per kernel	0.0	0.0	0.0
12.2.15	Mem Violations Accesses per kernel	0.0	0.0	0.0
12.2.16	LDS Command FIFO Full Rate Cycles per kernel			

12.2.17 LDS Data FIFO Full Rate
Cycles per kernel

13. Instruction Cache

13.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
13.1.0	Bandwidth	2.21	Pct of peak
13.1.1	Cache Hit Rate	98.25	Pct of peak
13.1.2	L1I-L2 Bandwidth	0.04	Pct of peak

13.2 Instruction Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
13.2.0 per kernel	Req	215280.89	18.0	778240.0	Req
13.2.1 per kernel	Hits	213643.45	14.0	764507.0	Hits
13.2.2 Misses per kernel	Misses - Non Duplicated	97.73	1.0	480.0	
13.2.3 Misses per kernel	Misses - Duplicated	1413.72	1.0	18309.0	
13.2.4	Cache Hit Rate	98.25	77.78	99.65	Pct
13.2.5 Cycles	Instruction Fetch Latency				

13.3 Instruction Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
13.3.0	L1I-L2 Bandwidth	172122.79	1024.00	645120.00	Bytes per kernel

14. Scalar L1 Data Cache

14.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
14.1.0	Bandwidth	0.20	Pct of peak
14.1.1	Cache Hit Rate	91.10	Pct of peak
14.1.2	sL1D-L2 BW	0.00	Pct of peak

14.2 Scalar L1D Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
14.2.0	Req	28163.17	0.00	77824.00	Req per kernel
14.2.1	Hits	26127.38	0.00	67904.00	Req per kernel
14.2.2	Misses - Non Duplicated	68.88	0.00	320.00	Req per kernel
14.2.3	Misses- Duplicated	1967.24	0.00	9600.00	Req per kernel

14.2.4	Cache Hit Rate	91.10	0.00	94.18	Pct
14.2.5 kernel	Read Req (Total)	28157.23	0.00	68096.00	Req per
14.2.6 kernel	Atomic Req	0.00	0.00	0.00	Req per
14.2.7 kernel	Read Req (1 DWord)	17.93	0.00	29184.00	Req per
14.2.8 kernel	Read Req (2 DWord)	11.94	0.00	19456.00	Req per
14.2.9 kernel	Read Req (4 DWord)	28127.36	0.00	65536.00	Req per
14.2.10 kernel	Read Req (8 DWord)	0.01	0.00	8.00	Req per
14.2.11 kernel	Read Req (16 DWord)	0.00	0.00	0.00	Req per

14.3 Scalar L1D Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
14.3.0	sL1D-L2 BW	4408.18	0.00	20480.00	Bytes per kernel
14.3.1	Read Req	68.88	0.00	320.00	Req per kernel
14.3.2	Write Req	0.00	0.00	0.00	Req per kernel
14.3.3	Atomic Req	0.00	0.00	0.00	Req per kernel
14.3.4	Stall Cycles	0.00	0.00	0.00	Cycles per kernel

15. Address Processing Unit and Data Return Path (TA/TD)

15.1 Address Processing Unit

Metric_ID Max	Metric Unit	Avg	Min
15.1.0 90.33	Address Processing Unit Busy Pct	32.88	0.0
15.1.1 5.77	Address Stall Pct	0.34	0.0
15.1.2 67.66	Data Stall Pct	26.22	0.0
15.1.3 0.0	Data-Processor → Address Stall Pct	0.0	0.0
15.1.4	Sequencer → TA Address Stall Cycles per kernel		
15.1.5	Sequencer → TA Command Stall Cycles per kernel		
15.1.6	Sequencer → TA Data Stall Cycles per kernel		
15.1.7 262144.0	Total Instructions Instructions per kernel	116319.23	2.0
15.1.8 262144.0	Global/Generic Instructions Instructions per kernel	112471.92	0.0
15.1.9	Global/Generic Read Instructions	0.07	0.0

6.0 Instructions per kernel

15.1.10 Global/Generic Read Instructions for LDS
Instructions per kernel

15.1.11 Global/Generic Write Instructions 112471.85 0.0
262144.0 Instructions per kernel

15.1.12 Global/Generic Atomic Instructions 0.0 0.0
0.0 Instructions per kernel

15.1.13 Spill/Stack Instructions 3847.31 0.0
36864.0 Instructions per kernel

15.1.14 Spill/Stack Read Instructions 2659.16 0.0
32768.0 Instructions per kernel

15.1.15 Spill/Stack Read Instructions for LDS
Instructions per kernel

15.1.16 Spill/Stack Write Instructions 1188.15 0.0
4096.0 Instructions per kernel

15.1.17 Spill/Stack Atomic Instructions 0.0 0.0
0.0 Instructions per kernel

15.1.18 Spill/Stack Total Cycles 61369.81 0.0
589824.0 Cycles per kernel

15.1.19 Spill/Stack Coalesced Read 62.38 0.0
512.0 Cycles per kernel

15.1.20 Spill/Stack Coalesced Write 0.0 0.0
0.0 Cycles per kernel

15.2 Data-Return Path

Metric_ID Unit	Metric	Avg	Min	Max
15.2.0 78.84	Data-Return Busy Pct	33.69	0.01	
15.2.1 Pct	Cache RAM → Data-Return Stall	32.91	0.01	78.5
15.2.2 Pct	Workgroup manager → Data-Return Stall	0.0	0.0	0.0
15.2.3 16384.0	Coalescable Instructions Instructions per kernel	25.73	0.0	
15.2.4 32768.0	Read Instructions Instructions per kernel	2659.22	0.0	
15.2.5 262144.0	Write Instructions Instructions per kernel	113660.01	1.0	
15.2.6	Atomic Instructions Instructions per kernel	0.0	0.0	0.0
15.2.7	Write Ack Instructions Instructions per kernel			

INFO Not showing table with empty column(s): 16.6 L1D Addr Translation Stalls

16. Vector L1 Data Cache

16.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
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16.1.0	Hit rate	17.27	Pct of peak
16.1.1	Bandwidth	8.35	Pct of peak
16.1.2	Utilization	44.18	Pct of peak
16.1.3	Coalescing	27.62	Pct of peak

16.2 L1D Cache Stalls (%)

Metric_ID	Metric	Min	Q1	Median	Q3	Max
16.2.0 111.15	Stalled on L2 Data	0.02	3.62	5.1	93.12	
16.2.1 82.66	Stalled on L2 Req	0.0	0.0	0.0	76.83	
16.2.2	Stalled on Address					
16.2.3	Stalled on Data					
16.2.4	Stalled on Latency FIFO					
16.2.5	Stalled on Request FIFO					
16.2.6	Stalled on Read Return					
16.2.7	Tag RAM Stall (Read)	0.0	0.0	0.0	0.0	0.0
16.2.8	Tag RAM Stall (Write)	0.0	0.0	0.0	0.0	1.4

16.2.9	Tag RAM Stall (Atomic)	0.0	0.0	0.0	0.0	0.0
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16.3 L1D Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
16.3.0 per kernel	Total Req	7443195.7	32.0	16777216.0	Req
16.3.1 per kernel	Read Req	169438.74	0.0	2097152.0	Req
16.3.2 per kernel	Write Req	7273756.97	16.0	16777216.0	Req
16.3.3 per kernel	Atomic Req	0.0	0.0	0.0	Req
16.3.4 Bytes per kernel	Cache BW	239350203.92	256.0	536870912.0	
16.3.5	Cache Hit Rate	17.27	-29.63	84.09	Pct
16.3.6 per kernel	Cache Accesses	1869923.47	2.0	4194304.0	Req
16.3.7 per kernel	Cache Hits	28245.31	-8.0	393216.0	Req
16.3.8 per kernel	Invalidations	157.07	1.0	304.0	Req

16.3.9	L1-L2 BW	119161323.09	192.0	268435456.0	
Bytes per kernel					
16.3.10	Tag RAM 0 Req				Req
per kernel					
16.3.11	Tag RAM 1 Req				Req
per kernel					
16.3.12	Tag RAM 2 Req				Req
per kernel					
16.3.13	Tag RAM 3 Req				Req
per kernel					
16.3.14	L1-L2 Read	20217.51	0.0	262144.0	Req
per kernel					
16.3.15	L1-L2 Write	1821460.65	1.0	4194304.0	Req
per kernel					
16.3.16	L1-L2 Atomic	0.0	0.0	0.0	Req
per kernel					
16.3.17	L1 Access Latency				
Cycles					
16.3.18	L1-L2 Read Latency				
Cycles					
16.3.19	L1-L2 Write Latency				
Cycles					

16.4 L1D - L2 Transactions

Metric_ID	Metric	Xfer	Coherency	Avg	Min
Max Unit					
16.4.0	NC - Read	Read	NC	0.00	0.00
0.00	Req per kernel				
16.4.1	UC - Read	Read	UC	0.00	0.00
0.00	Req per kernel				
16.4.2	CC - Read	Read	CC	0.00	0.00
0.00	Req per kernel				
16.4.3	RW - Read	Read	RW	20217.51	0.00
262144.00	Req per kernel				
16.4.4	RW - Write	Write	RW	1821460.60	0.00
4194304.00	Req per kernel				
16.4.5	NC - Write	Write	NC	0.00	0.00
0.00	Req per kernel				
16.4.6	UC - Write	Write	UC	0.05	0.00
6.00	Req per kernel				
16.4.7	CC - Write	Write	CC	0.00	0.00
0.00	Req per kernel				
16.4.8	NC - Atomic	Atomic	NC	0.00	0.00
0.00	Req per kernel				
16.4.9	UC - Atomic	Atomic	UC	0.00	0.00
0.00	Req per kernel				
16.4.10	CC - Atomic	Atomic	CC	0.00	0.00
0.00	Req per kernel				

16.4.11	RW - Atomic	Atomic	RW	0.00	0.00
0.00	Req per kernel				

16.5 L1D Addr Translation

Metric_ID	Metric	Avg	Min	Max
Units				
16.5.0	Req	1869923.47	2.0	4194304.0
Req per kernel				
16.5.1	Inflight Req			
Req per kernel				
16.5.2	Hit Ratio	92.74	0.0	100.0
Pct				
16.5.3	Hits	1720416.67	0.0	3863144.0
Req per kernel				
16.5.4	Translation Misses	7093.28	0.0	16384.0
Req per kernel				
16.5.5	Misses under Translation Miss			
Req per kernel				
16.5.6	Permission Misses	0.05	0.0	128.0
Req per kernel				

INFO Not showing table with empty column(s): 17.4 L2 Cache Stalls

17. L2 Cache

17.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
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17.1.0	Utilization	49.92	Pct	
17.1.1	Bandwidth	18.59	Pct	
17.1.2	Hit Rate	43.03	Pct	
17.1.3	L2-Fabric Read BW	332.41	Gb/s	
17.1.4	L2-Fabric Write and Atomic BW	2989.89	Gb/s	
17.1.5	HBM Bandwidth	5324.80	Gb/s	
17.2 L2 - Fabric Transactions				
Metric_ID Unit	Metric	Avg	Min	Max
17.2.0 18953216.0	Read BW Bytes per kernel	1924576.85	1536.0	
17.2.1 116.0	HBM Read Traffic Pct	100.53	60.0	
17.2.2	Remote Read Traffic Pct	0.45	0.0	40.0
17.2.3 161.9	Uncached Read Traffic Pct	3.72	0.0	
17.2.4 268435456.0	Write and Atomic BW Bytes per kernel	116281478.47	32.0	
17.2.5 100.0	HBM Write and Atomic Traffic Pct	97.28	0.0	
17.2.6 100.0	Remote Write and Atomic Traffic Pct	2.72	0.0	

17.2.7 Pct	Atomic Traffic	0.0	0.0	0.0
17.2.8 100.0	Uncached Write and Atomic Traffic Pct	2.72	0.0	
17.2.9 12065.26	Read Latency Cycles	415.97	156.07	
17.2.10 4787.43	Write and Atomic Latency Cycles	389.63	119.5	
17.2.11 Cycles	Atomic Latency			
17.2.12 Pct	Read Stall			
17.2.13 Pct	Write Stall			

17.3 L2 Cache Accesses

Metric_ID Unit	Metric	Avg	Min	Max
17.3.0 Bytes per kernel	Bandwidth	236083804.36	3072.0	537407232.0
17.3.1 Bytes per kernel	Read Bandwidth			
17.3.2 Bytes per kernel	Write Bandwidth			
17.3.3	Atomic Bandwidth			

Bytes per kernel

17.3.4 per kernel	Req	1844404.72	24.0	4198494.0	Req
17.3.5 per kernel	Read Req	23000.71	23.0	270216.0	Req
17.3.6 per kernel	Write Req	1821460.65	1.0	4194304.0	Req
17.3.7 per kernel	Atomic Req	0.0	0.0	0.0	Req
17.3.8 per kernel	Streaming Req	0.0	0.0	3.0	Req
17.3.9 per kernel	Bypassss Req				Req
17.3.10 per kernel	Probe Req	0.01	0.0	8.0	Req
17.3.11 per kernel	Input Buffer Req				Req
17.3.12	Cache Hit	43.03	20.0	61.41	Pct
17.3.13 Hits per kernel	Hits	920964.44	10.0	2101048.0	
17.3.14 Misses per kernel	Misses	923489.15	13.0	2097446.0	
17.3.15	Writeback	908532.02	5.0	2097208.0	

Cachelines per kernel

17.3.16	Writeback (Internal)	843463.32	0.0	1966083.0
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Cachelines per kernel

17.3.17	Writeback (vL1D Req)	64985.73	0.0	131085.0
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Cachelines per kernel

17.3.18	Evict (Internal)	787280.42	0.0	1835120.0
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Cachelines per kernel

17.3.19	Evict (vL1D Req)	0.0	0.0	0.0
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Cachelines per kernel

17.3.20	NC Req	0.0	0.0	0.0	Req
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per kernel

17.3.21	UC Req	6.93	0.0	186.0	Req
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per kernel

17.3.22	CC Req	0.0	0.0	0.0	Req
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per kernel

17.3.23	RW Req	1844435.29	24.0	4198304.0	Req
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per kernel

17.5 L2 - Fabric Interface Stalls

Metric_ID	Metric	Min	Max	Unit	Type
Transaction	Avg				
17.5.0	Read - PCIe Stall Pct				PCIe Stall Read
17.5.1	Read - Infinity Fabric Stall Pct				Infinity Fabric Stall Read

17.5.2	Read - HBM Stall Pct	HBM Stall	Read
17.5.3	Write - PCIe Stall Pct	PCIe Stall	Write
17.5.4	Write - Infinity Fabric Stall Pct	Infinity Fabric Stall	Write
17.5.5	Write - HBM Stall Pct	HBM Stall	Write
17.5.6	Write - Credit Starvation Pct	Credit Starvation	Write
0.0	0.0	0.0	

17.6 L2 - Fabric Detailed Transaction Breakdown

Metric_ID	Metric	Avg	Min	Max
Unit				
17.6.0	Read (32B)	0.0	0.0	0.0
Req per kernel				
17.6.1	Read (64B)	290.38	22.0	
1232.0	Req per kernel			
17.6.2	Read (128B)	14890.57	0.0	
147456.0	Req per kernel			
17.6.3	Read (Uncached)	13.83	0.0	404.0
Req per kernel				
17.6.4	HBM Read	15182.89	23.0	
148688.0	Req per kernel			

17.6.5	Remote Read	1.83	0.0	32.0
Req per kernel				
17.6.6	Read Bandwidth - PCIe			
Bytes per kernel				
17.6.7	Read Bandwidth - Infinity Fabric			
Bytes per kernel				
17.6.8	Read Bandwidth - HBM			
Bytes per kernel				
17.6.9	Write and Atomic (32B)	0.06	0.0	6.0
Req per kernel				
17.6.10	Write and Atomic (Uncached)	0.05	0.0	6.0
Req per kernel				
17.6.11	Write and Atomic (64B)	1816898.07	0.0	
4194304.0	Req per kernel			
17.6.12	HBM Write and Atomic	1816898.08	0.0	
4194304.0	Req per kernel			
17.6.13	Remote Write and Atomic	0.05	0.0	6.0
Req per kernel				
17.6.14	Write Bandwidth - PCIe			
Bytes per kernel				
17.6.15	Write Bandwidth - Infinity Fabric			
Bytes per kernel				
17.6.16	Write Bandwidth - HBM			
Bytes per kernel				

17.6.17	Atomic	0.0	0.0	0.0
Req per kernel				

17.6.18	Atomic - HBM
Req per kernel	

17.6.19	Atomic Bandwidth - PCIe
Bytes per kernel	

17.6.20	Atomic Bandwidth - Infinity Fabric
Bytes per kernel	

17.6.21	Atomic Bandwidth - HBM
Bytes per kernel	

INFO Not showing table with empty column(s): 18.9 L2-Fabric Read Stall (Cycles per normUnit)

INFO Not showing table with empty column(s): 18.10 L2-Fabric Write and Atomic Stall (Cycles per normUnit)

18. L2 Cache (per Channel)

18.1 Aggregate Stats (All channels)

Metric_ID	Metric	Avg	Std Dev	Min	Max	Unit
18.1.0	L2 Cache Hit Rate	42.70	8.59	0.00	61.41	Pct

18.2 L2 Cache Hit Rate (pct)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.2.0	0.00	0.00	36.59	37.98	50.12	61.31
18.2.1	1.00	35.06	36.59	37.98	50.00	61.31
18.2.2	2.00	0.00	36.14	37.98	50.00	61.31
18.2.3	3.00	0.00	36.14	37.98	50.00	61.31
18.2.4	4.00	0.00	35.32	37.98	50.12	100.00

18.2.5	5.00	0.00	35.32	37.98	50.12	61.13
18.2.6	6.00	0.00	35.32	37.98	50.12	61.31
18.2.7	7.00	35.24	35.32	37.98	50.12	61.31
18.2.8	8.00	0.00	33.33	35.50	50.00	61.31
18.2.9	9.00	0.00	33.33	35.50	50.00	61.31
18.2.10	10.00	0.00	33.33	35.50	50.00	63.38
18.2.11	11.00	33.07	33.33	35.50	50.00	65.33
18.2.12	12.00	0.00	33.33	35.50	50.00	66.67
18.2.13	13.00	33.07	33.33	35.50	50.00	66.67
18.2.14	14.00	0.00	33.33	35.50	50.00	61.31
18.2.15	15.00	0.00	33.33	35.50	50.12	61.31
18.2.16	16.00	0.00	33.33	35.50	50.00	61.31
18.2.17	17.00	0.00	33.33	35.50	50.00	61.31
18.2.18	18.00	0.00	33.33	35.50	50.00	63.38
18.2.19	19.00	0.00	33.33	35.50	50.00	65.33
18.2.20	20.00	0.00	33.33	35.50	50.00	61.31
18.2.21	21.00	33.07	33.33	35.50	50.00	61.31
18.2.22	22.00	0.00	33.33	35.50	50.00	66.67
18.2.23	23.00	0.00	33.33	35.50	50.12	66.67
18.2.24	24.00	0.00	36.59	37.98	50.12	61.31
18.2.25	25.00	35.06	36.59	37.98	50.00	61.31
18.2.26	26.00	0.00	36.14	37.98	50.00	61.31
18.2.27	27.00	0.00	36.14	37.98	50.00	61.31
18.2.28	28.00	0.00	35.32	37.98	50.12	100.00

18.2.29	29.00	0.00	35.32	37.98	50.12	61.13
18.2.30	30.00	0.00	35.32	37.98	50.12	61.31
18.2.31	31.00	33.33	35.32	37.98	50.12	61.31
18.2.32	32.00	0.00	36.59	37.98	50.12	61.31
18.2.33	33.00	0.00	36.59	37.98	50.00	61.31
18.2.34	34.00	0.00	36.14	37.98	50.00	61.31
18.2.35	35.00	0.00	36.14	37.98	50.00	61.31
18.2.36	36.00	0.00	35.32	37.98	50.12	100.00
18.2.37	37.00	0.00	35.32	37.98	50.12	61.13
18.2.38	38.00	0.00	35.32	37.98	50.12	61.31
18.2.39	39.00	35.24	35.32	37.98	50.12	61.31
18.2.40	40.00	0.00	33.33	35.50	50.00	61.31
18.2.41	41.00	0.00	33.33	35.50	50.00	61.31
18.2.42	42.00	0.00	33.33	35.50	50.00	63.38
18.2.43	43.00	33.07	33.33	35.50	50.00	65.33
18.2.44	44.00	0.00	33.33	35.50	50.00	61.31
18.2.45	45.00	33.07	33.33	35.50	50.00	61.31
18.2.46	46.00	0.00	33.33	35.50	50.00	61.31
18.2.47	47.00	0.00	33.33	35.50	50.12	61.31
18.2.48	48.00	0.00	33.33	35.50	50.00	61.31
18.2.49	49.00	0.00	33.33	35.50	50.00	61.31
18.2.50	50.00	0.00	33.33	35.50	50.00	63.38
18.2.51	51.00	33.07	33.33	35.50	50.00	65.33
18.2.52	52.00	33.07	33.33	35.50	50.00	61.31

18.2.53	53.00	33.07	33.33	35.50	50.00	61.31
18.2.54	54.00	0.00	33.33	35.50	50.00	61.31
18.2.55	55.00	0.00	33.33	35.50	50.12	61.31
18.2.56	56.00	0.00	36.14	37.98	50.12	61.31
18.2.57	57.00	35.06	36.14	37.98	50.00	61.31
18.2.58	58.00	0.00	35.96	37.98	50.00	61.31
18.2.59	59.00	0.00	35.96	37.98	50.00	61.31
18.2.60	60.00	0.00	35.32	37.98	50.12	100.00
18.2.61	61.00	0.00	35.32	37.98	50.12	61.13
18.2.62	62.00	0.00	35.32	37.98	50.12	61.31
18.2.63	63.00	35.24	35.32	37.98	50.12	61.31
18.2.64	64.00	0.00	33.33	35.50	50.00	61.31
18.2.65	65.00	0.00	33.33	35.50	50.00	61.31
18.2.66	66.00	0.00	33.33	35.50	50.00	63.38
18.2.67	67.00	33.07	33.33	35.50	50.00	65.33
18.2.68	68.00	33.07	33.33	35.50	50.00	61.31
18.2.69	69.00	33.07	33.33	35.50	50.00	61.31
18.2.70	70.00	33.07	33.33	35.50	50.00	61.31
18.2.71	71.00	0.00	33.33	35.50	50.12	61.31
18.2.72	72.00	0.00	36.59	37.98	50.12	61.31
18.2.73	73.00	35.06	36.59	37.98	50.00	61.31
18.2.74	74.00	0.00	36.05	37.98	50.00	61.31
18.2.75	75.00	0.00	36.14	37.98	50.00	61.31
18.2.76	76.00	0.00	35.32	37.98	50.12	61.31

18.2.77	77.00	0.00	35.32	37.98	50.12	61.13
18.2.78	78.00	0.00	35.32	37.98	50.12	61.31
18.2.79	79.00	35.24	35.32	37.98	50.12	61.31
18.2.80	80.00	0.00	36.59	37.98	50.12	61.31
18.2.81	81.00	35.06	36.59	37.98	50.00	61.31
18.2.82	82.00	0.00	36.05	37.98	50.00	61.31
18.2.83	83.00	0.00	36.05	37.98	50.00	61.31
18.2.84	84.00	0.00	35.32	37.98	50.12	61.31
18.2.85	85.00	0.00	35.32	37.98	50.12	61.13
18.2.86	86.00	0.00	35.32	37.98	50.12	61.31
18.2.87	87.00	35.24	35.32	37.98	50.12	61.31
18.2.88	88.00	0.00	33.33	35.50	50.00	61.31
18.2.89	89.00	0.00	33.33	35.50	50.00	61.31
18.2.90	90.00	0.00	33.33	35.50	50.00	63.38
18.2.91	91.00	33.07	33.33	35.50	50.00	65.33
18.2.92	92.00	33.07	33.33	35.50	50.00	61.31
18.2.93	93.00	33.07	33.33	35.50	50.00	61.31
18.2.94	94.00	33.07	33.33	35.50	50.00	61.31
18.2.95	95.00	0.00	33.33	35.50	50.12	61.31
18.2.96	96.00	0.00	33.33	35.50	50.00	100.00
18.2.97	97.00	0.00	33.33	50.00	50.00	100.00
18.2.98	98.00	0.00	33.33	50.00	50.00	100.00
18.2.99	99.00	0.00	33.33	50.00	50.00	100.00
18.2.100	100.00	0.00	33.33	50.00	50.00	100.00

18.2.101	101.00	0.00	33.33	40.00	50.00	66.67
18.2.102	102.00	0.00	33.33	35.50	50.00	100.00
18.2.103	103.00	0.00	33.33	35.50	50.12	66.67
18.2.104	104.00	0.00	36.59	37.98	50.12	100.00
18.2.105	105.00	0.00	36.59	37.98	50.00	100.00
18.2.106	106.00	0.00	36.14	37.98	50.00	100.00
18.2.107	107.00	0.00	36.14	37.98	50.00	100.00
18.2.108	108.00	0.00	35.32	37.98	50.12	100.00
18.2.109	109.00	0.00	35.32	37.98	50.12	100.00
18.2.110	110.00	0.00	35.32	37.98	50.12	100.00
18.2.111	111.00	0.00	35.32	37.98	50.12	100.00
18.2.112	112.00	0.00	36.59	37.98	50.12	100.00
18.2.113	113.00	0.00	36.59	37.98	50.00	61.31
18.2.114	114.00	0.00	36.05	37.98	50.00	61.31
18.2.115	115.00	0.00	36.14	37.98	50.00	61.31
18.2.116	116.00	0.00	35.32	37.98	50.12	100.00
18.2.117	117.00	0.00	35.32	37.98	50.12	61.13
18.2.118	118.00	0.00	35.32	37.89	50.12	61.31
18.2.119	119.00	0.00	35.32	37.98	50.12	61.31
18.2.120	120.00	0.00	33.33	35.50	50.00	61.31
18.2.121	121.00	0.00	33.33	35.50	50.00	61.31
18.2.122	122.00	0.00	33.33	35.50	50.00	63.38
18.2.123	123.00	0.00	33.33	35.50	50.00	66.67
18.2.124	124.00	0.00	33.33	35.50	50.00	100.00

18.2.125	125.00	0.00	33.33	35.50	50.00	61.31
18.2.126	126.00	0.00	33.33	35.50	50.00	61.31
18.2.127	127.00	0.00	33.33	35.50	50.12	61.31

18.3 L2 Requests (per normUnit)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.3.0	0.00	0.00	388.00	414.00	32848.00	32869.00
18.3.1	1.00	0.00	388.00	414.00	32848.00	32869.00
18.3.2	2.00	0.00	388.00	414.00	32848.00	32870.00
18.3.3	3.00	0.00	388.00	414.00	32848.00	32869.00
18.3.4	4.00	0.00	388.00	414.00	32768.00	32789.00
18.3.5	5.00	0.00	388.00	414.00	32848.00	32869.00
18.3.6	6.00	0.00	388.00	414.00	32768.00	32790.00
18.3.7	7.00	0.00	388.00	414.00	32768.00	32789.00
18.3.8	8.00	0.00	400.00	412.00	32768.00	32789.00
18.3.9	9.00	0.00	400.00	412.00	32768.00	32789.00
18.3.10	10.00	0.00	400.00	412.00	32848.00	32869.00
18.3.11	11.00	0.00	400.00	416.00	32768.00	32789.00
18.3.12	12.00	0.00	400.00	412.00	32768.00	32789.00
18.3.13	13.00	0.00	400.00	412.00	32768.00	32789.00
18.3.14	14.00	0.00	400.00	416.00	32768.00	32789.00
18.3.15	15.00	0.00	400.00	416.00	32768.00	32789.00
18.3.16	16.00	0.00	400.00	412.00	32768.00	32789.00
18.3.17	17.00	0.00	400.00	412.00	32768.00	32789.00
18.3.18	18.00	0.00	400.00	412.00	32848.00	32869.00

18.3.19	19.00	0.00	400.00	416.00	32768.00	32789.00
18.3.20	20.00	0.00	400.00	412.00	32768.00	32789.00
18.3.21	21.00	0.00	400.00	412.00	32768.00	32789.00
18.3.22	22.00	0.00	400.00	416.00	32768.00	32789.00
18.3.23	23.00	0.00	400.00	416.00	32768.00	32789.00
18.3.24	24.00	0.00	388.00	414.00	32848.00	32869.00
18.3.25	25.00	0.00	388.00	414.00	32848.00	32869.00
18.3.26	26.00	0.00	388.00	414.00	32848.00	32870.00
18.3.27	27.00	0.00	388.00	414.00	32848.00	32869.00
18.3.28	28.00	0.00	388.00	414.00	32768.00	32789.00
18.3.29	29.00	0.00	388.00	414.00	32848.00	32869.00
18.3.30	30.00	0.00	388.00	414.00	32768.00	32790.00
18.3.31	31.00	0.00	388.00	414.00	32768.00	32789.00
18.3.32	32.00	0.00	388.00	414.00	32848.00	32869.00
18.3.33	33.00	0.00	388.00	414.00	32848.00	32869.00
18.3.34	34.00	0.00	388.00	414.00	32848.00	32870.00
18.3.35	35.00	0.00	388.00	414.00	32848.00	32869.00
18.3.36	36.00	0.00	388.00	414.00	32768.00	32789.00
18.3.37	37.00	0.00	388.00	414.00	32848.00	32869.00
18.3.38	38.00	0.00	388.00	414.00	32768.00	32790.00
18.3.39	39.00	0.00	388.00	414.00	32768.00	32789.00
18.3.40	40.00	0.00	400.00	412.00	32768.00	32789.00
18.3.41	41.00	0.00	400.00	412.00	32768.00	32789.00
18.3.42	42.00	0.00	400.00	412.00	32848.00	32869.00

18.3.43	43.00	0.00	400.00	416.00	32768.00	32789.00
18.3.44	44.00	0.00	400.00	412.00	32768.00	32789.00
18.3.45	45.00	0.00	400.00	412.00	32768.00	32789.00
18.3.46	46.00	0.00	400.00	416.00	32768.00	32789.00
18.3.47	47.00	0.00	400.00	416.00	32768.00	32789.00
18.3.48	48.00	0.00	400.00	412.00	32768.00	32789.00
18.3.49	49.00	0.00	400.00	412.00	32768.00	32789.00
18.3.50	50.00	0.00	400.00	412.00	32848.00	32869.00
18.3.51	51.00	0.00	400.00	416.00	32768.00	32789.00
18.3.52	52.00	0.00	400.00	412.00	32768.00	32789.00
18.3.53	53.00	0.00	400.00	412.00	32768.00	32789.00
18.3.54	54.00	0.00	400.00	416.00	32768.00	32789.00
18.3.55	55.00	0.00	400.00	416.00	32768.00	32789.00
18.3.56	56.00	0.00	388.00	414.00	32848.00	32869.00
18.3.57	57.00	0.00	388.00	414.00	32848.00	32869.00
18.3.58	58.00	0.00	388.00	414.00	32848.00	32870.00
18.3.59	59.00	0.00	388.00	414.00	32848.00	32869.00
18.3.60	60.00	0.00	388.00	414.00	32768.00	32789.00
18.3.61	61.00	0.00	388.00	414.00	32848.00	32869.00
18.3.62	62.00	0.00	388.00	414.00	32768.00	32790.00
18.3.63	63.00	0.00	388.00	414.00	32768.00	32789.00
18.3.64	64.00	0.00	400.00	412.00	32768.00	32789.00
18.3.65	65.00	0.00	400.00	412.00	32768.00	32789.00
18.3.66	66.00	0.00	400.00	412.00	32848.00	32869.00

18.3.67	67.00	0.00	400.00	416.00	32768.00	32789.00
18.3.68	68.00	0.00	400.00	412.00	32768.00	32789.00
18.3.69	69.00	0.00	400.00	412.00	32768.00	32789.00
18.3.70	70.00	0.00	400.00	416.00	32768.00	32789.00
18.3.71	71.00	0.00	400.00	416.00	32768.00	32789.00
18.3.72	72.00	0.00	388.00	414.00	32848.00	32869.00
18.3.73	73.00	0.00	388.00	414.00	32848.00	32869.00
18.3.74	74.00	0.00	388.00	414.00	32848.00	32870.00
18.3.75	75.00	0.00	388.00	414.00	32848.00	32869.00
18.3.76	76.00	0.00	388.00	414.00	32768.00	32789.00
18.3.77	77.00	0.00	388.00	414.00	32848.00	32869.00
18.3.78	78.00	0.00	388.00	414.00	32768.00	32790.00
18.3.79	79.00	0.00	388.00	414.00	32768.00	32789.00
18.3.80	80.00	0.00	388.00	414.00	32848.00	32869.00
18.3.81	81.00	0.00	388.00	414.00	32848.00	32869.00
18.3.82	82.00	0.00	388.00	414.00	32848.00	32870.00
18.3.83	83.00	0.00	388.00	414.00	32848.00	32869.00
18.3.84	84.00	0.00	388.00	414.00	32768.00	32789.00
18.3.85	85.00	0.00	388.00	414.00	32848.00	32869.00
18.3.86	86.00	0.00	388.00	414.00	32768.00	32790.00
18.3.87	87.00	0.00	388.00	414.00	32768.00	32789.00
18.3.88	88.00	0.00	400.00	412.00	32768.00	32789.00
18.3.89	89.00	0.00	400.00	412.00	32768.00	32789.00
18.3.90	90.00	0.00	400.00	412.00	32848.00	32869.00

18.3.91	91.00	0.00	400.00	416.00	32768.00	32789.00
18.3.92	92.00	0.00	400.00	412.00	32768.00	32789.00
18.3.93	93.00	0.00	400.00	412.00	32768.00	32789.00
18.3.94	94.00	0.00	400.00	416.00	32768.00	32789.00
18.3.95	95.00	0.00	400.00	416.00	32768.00	32789.00
18.3.96	96.00	0.00	400.00	412.00	32768.00	32789.00
18.3.97	97.00	0.00	400.00	412.00	32768.00	32789.00
18.3.98	98.00	0.00	400.00	412.00	32848.00	32869.00
18.3.99	99.00	0.00	400.00	416.00	32768.00	32789.00
18.3.100	100.00	0.00	400.00	412.00	32768.00	32789.00
18.3.101	101.00	0.00	400.00	412.00	32768.00	32789.00
18.3.102	102.00	0.00	400.00	416.00	32768.00	32789.00
18.3.103	103.00	0.00	400.00	416.00	32768.00	32789.00
18.3.104	104.00	0.00	388.00	414.00	32848.00	32869.00
18.3.105	105.00	0.00	388.00	414.00	32848.00	32869.00
18.3.106	106.00	0.00	388.00	415.00	32848.00	32870.00
18.3.107	107.00	0.00	388.00	414.00	32848.00	32869.00
18.3.108	108.00	0.00	388.00	414.00	32768.00	32789.00
18.3.109	109.00	0.00	388.00	414.00	32848.00	32869.00
18.3.110	110.00	0.00	388.00	414.00	32768.00	32790.00
18.3.111	111.00	0.00	388.00	414.00	32768.00	32789.00
18.3.112	112.00	0.00	388.00	414.00	32848.00	32869.00
18.3.113	113.00	0.00	388.00	414.00	32848.00	32869.00
18.3.114	114.00	0.00	388.00	415.00	32848.00	32870.00

18.3.115	115.00	0.00	388.00	414.00	32848.00	32869.00
18.3.116	116.00	0.00	388.00	414.00	32768.00	32789.00
18.3.117	117.00	0.00	388.00	414.00	32848.00	32869.00
18.3.118	118.00	0.00	388.00	414.00	32768.00	32790.00
18.3.119	119.00	0.00	388.00	414.00	32768.00	32789.00
18.3.120	120.00	0.00	400.00	412.00	32768.00	32789.00
18.3.121	121.00	0.00	400.00	412.00	32768.00	32789.00
18.3.122	122.00	0.00	400.00	412.00	32848.00	32869.00
18.3.123	123.00	0.00	400.00	416.00	32768.00	32789.00
18.3.124	124.00	0.00	400.00	412.00	32768.00	32789.00
18.3.125	125.00	0.00	400.00	412.00	32768.00	32789.00
18.3.126	126.00	0.00	400.00	416.00	32768.00	32789.00
18.3.127	127.00	0.00	400.00	416.00	32768.00	32789.00

18.4 L2 Requests (per normUnit)

Metric_ID	Channel	L2 Read	L2 Write	L2 Atomic
18.4.0	0.00	202.70	14230.16	0.00
18.4.1	1.00	202.67	14230.16	0.00
18.4.2	2.00	202.41	14230.16	0.00
18.4.3	3.00	202.17	14230.16	0.00
18.4.4	4.00	166.30	14230.16	0.00
18.4.5	5.00	200.57	14230.16	0.00
18.4.6	6.00	167.12	14230.16	0.00
18.4.7	7.00	166.29	14230.16	0.00
18.4.8	8.00	165.61	14230.16	0.00

18.4.9	9.00	165.32	14230.16	0.00
18.4.10	10.00	199.61	14230.16	0.00
18.4.11	11.00	165.30	14230.16	0.00
18.4.12	12.00	165.28	14230.16	0.00
18.4.13	13.00	165.28	14230.16	0.00
18.4.14	14.00	165.29	14230.16	0.00
18.4.15	15.00	165.29	14230.16	0.00
18.4.16	16.00	165.60	14230.16	0.00
18.4.17	17.00	165.31	14230.16	0.00
18.4.18	18.00	199.60	14230.16	0.00
18.4.19	19.00	165.29	14230.16	0.00
18.4.20	20.00	165.27	14230.16	0.00
18.4.21	21.00	165.27	14230.16	0.00
18.4.22	22.00	165.29	14230.16	0.00
18.4.23	23.00	165.28	14230.16	0.00
18.4.24	24.00	202.62	14230.16	0.00
18.4.25	25.00	202.60	14230.16	0.00
18.4.26	26.00	202.32	14230.16	0.00
18.4.27	27.00	202.09	14230.16	0.00
18.4.28	28.00	166.26	14230.16	0.00
18.4.29	29.00	200.55	14230.16	0.00
18.4.30	30.00	167.04	14230.16	0.00
18.4.31	31.00	166.26	14230.16	0.00
18.4.32	32.00	202.71	14230.16	0.00

18.4.33	33.00	202.68	14230.16	0.00
18.4.34	34.00	202.42	14230.16	0.00
18.4.35	35.00	202.19	14230.16	0.00
18.4.36	36.00	166.31	14230.16	0.00
18.4.37	37.00	200.58	14230.16	0.00
18.4.38	38.00	167.13	14230.16	0.00
18.4.39	39.00	166.30	14230.16	0.00
18.4.40	40.00	165.66	14230.16	0.00
18.4.41	41.00	165.37	14230.16	0.00
18.4.42	42.00	199.67	14230.16	0.00
18.4.43	43.00	165.36	14230.16	0.00
18.4.44	44.00	165.34	14230.16	0.00
18.4.45	45.00	165.34	14230.16	0.00
18.4.46	46.00	165.35	14230.16	0.00
18.4.47	47.00	165.35	14230.16	0.00
18.4.48	48.00	165.57	14230.16	0.00
18.4.49	49.00	165.28	14230.16	0.00
18.4.50	50.00	199.57	14230.16	0.00
18.4.51	51.00	165.26	14230.16	0.00
18.4.52	52.00	165.24	14230.16	0.00
18.4.53	53.00	165.24	14230.16	0.00
18.4.54	54.00	165.26	14230.16	0.00
18.4.55	55.00	165.25	14230.16	0.00
18.4.56	56.00	202.65	14230.16	0.00

18.4.57	57.00	202.63	14230.16	0.00
18.4.58	58.00	202.37	14230.16	0.00
18.4.59	59.00	202.13	14230.16	0.00
18.4.60	60.00	166.27	14230.16	0.00
18.4.61	61.00	200.56	14230.16	0.00
18.4.62	62.00	167.08	14230.16	0.00
18.4.63	63.00	166.26	14230.16	0.00
18.4.64	64.00	165.53	14230.16	0.00
18.4.65	65.00	165.24	14230.16	0.00
18.4.66	66.00	199.53	14230.16	0.00
18.4.67	67.00	165.22	14230.16	0.00
18.4.68	68.00	165.20	14230.16	0.00
18.4.69	69.00	165.20	14230.16	0.00
18.4.70	70.00	165.22	14230.16	0.00
18.4.71	71.00	165.21	14230.16	0.00
18.4.72	72.00	202.66	14230.16	0.00
18.4.73	73.00	202.63	14230.16	0.00
18.4.74	74.00	202.37	14230.16	0.00
18.4.75	75.00	202.14	14230.16	0.00
18.4.76	76.00	166.31	14230.16	0.00
18.4.77	77.00	200.60	14230.16	0.00
18.4.78	78.00	167.10	14230.16	0.00
18.4.79	79.00	166.30	14230.16	0.00
18.4.80	80.00	202.64	14230.16	0.00

18.4.81	81.00	202.62	14230.16	0.00
18.4.82	82.00	202.35	14230.16	0.00
18.4.83	83.00	202.12	14230.16	0.00
18.4.84	84.00	166.28	14230.16	0.00
18.4.85	85.00	200.57	14230.16	0.00
18.4.86	86.00	167.09	14230.16	0.00
18.4.87	87.00	166.28	14230.16	0.00
18.4.88	88.00	165.57	14230.16	0.00
18.4.89	89.00	165.28	14230.16	0.00
18.4.90	90.00	199.57	14230.16	0.00
18.4.91	91.00	165.27	14230.16	0.00
18.4.92	92.00	165.25	14230.16	0.00
18.4.93	93.00	165.25	14230.16	0.00
18.4.94	94.00	165.26	14230.16	0.00
18.4.95	95.00	165.25	14230.16	0.00
18.4.96	96.00	165.80	14230.17	0.00
18.4.97	97.00	165.50	14230.16	0.00
18.4.98	98.00	199.69	14230.16	0.00
18.4.99	99.00	165.38	14230.16	0.00
18.4.100	100.00	165.47	14230.16	0.00
18.4.101	101.00	165.46	14230.16	0.00
18.4.102	102.00	165.48	14230.17	0.00
18.4.103	103.00	165.48	14230.16	0.00
18.4.104	104.00	202.70	14230.19	0.00

18.4.105	105.00	202.68	14230.16	0.00
18.4.106	106.00	202.42	14230.16	0.00
18.4.107	107.00	202.19	14230.16	0.00
18.4.108	108.00	166.33	14230.16	0.00
18.4.109	109.00	200.62	14230.16	0.00
18.4.110	110.00	167.14	14230.17	0.00
18.4.111	111.00	166.32	14230.16	0.00
18.4.112	112.00	202.61	14230.16	0.00
18.4.113	113.00	202.59	14230.16	0.00
18.4.114	114.00	202.34	14230.16	0.00
18.4.115	115.00	202.10	14230.16	0.00
18.4.116	116.00	166.30	14230.16	0.00
18.4.117	117.00	200.58	14230.16	0.00
18.4.118	118.00	167.07	14230.16	0.00
18.4.119	119.00	166.29	14230.16	0.00
18.4.120	120.00	165.65	14230.16	0.00
18.4.121	121.00	165.36	14230.16	0.00
18.4.122	122.00	199.65	14230.16	0.00
18.4.123	123.00	165.34	14230.16	0.00
18.4.124	124.00	165.32	14230.16	0.00
18.4.125	125.00	165.32	14230.16	0.00
18.4.126	126.00	165.34	14230.16	0.00
18.4.127	127.00	165.33	14230.16	0.00

18.5 L2-Fabric Requests (per normUnit)

Metric_ID L2-Fabric Atomic	Channel	L2-Fabric Read	L2-Fabric Write and Atomic
18.5.0 0.00	0.00	119.97	14194.52
18.5.1 0.00	1.00	118.26	14194.52
18.5.2 0.00	2.00	118.29	14194.52
18.5.3 0.00	3.00	118.53	14194.52
18.5.4 0.00	4.00	119.80	14194.52
18.5.5 0.00	5.00	119.80	14194.52
18.5.6 0.00	6.00	119.94	14194.52
18.5.7 0.00	7.00	120.40	14194.52
18.5.8 0.00	8.00	117.61	14194.52
18.5.9 0.00	9.00	117.60	14194.52
18.5.10 0.00	10.00	117.94	14194.52

18.5.11 0.00	11.00	117.89	14194.52
18.5.12 0.00	12.00	117.60	14194.52
18.5.13 0.00	13.00	117.61	14194.52
18.5.14 0.00	14.00	117.75	14194.52
18.5.15 0.00	15.00	119.44	14194.52
18.5.16 0.00	16.00	117.61	14194.52
18.5.17 0.00	17.00	117.60	14194.52
18.5.18 0.00	18.00	117.94	14194.52
18.5.19 0.00	19.00	117.89	14194.52
18.5.20 0.00	20.00	117.60	14194.52
18.5.21 0.00	21.00	117.61	14194.52
18.5.22 0.00	22.00	117.75	14194.52

18.5.23 0.00	23.00	119.44	14194.52
18.5.24 0.00	24.00	119.97	14194.52
18.5.25 0.00	25.00	118.26	14194.52
18.5.26 0.00	26.00	118.29	14194.52
18.5.27 0.00	27.00	118.53	14194.52
18.5.28 0.00	28.00	119.80	14194.52
18.5.29 0.00	29.00	119.80	14194.52
18.5.30 0.00	30.00	119.94	14194.52
18.5.31 0.00	31.00	120.40	14194.52
18.5.32 0.00	32.00	119.97	14194.52
18.5.33 0.00	33.00	118.26	14194.52
18.5.34 0.00	34.00	118.29	14194.52

18.5.35 0.00	35.00	118.53	14194.52
18.5.36 0.00	36.00	119.80	14194.52
18.5.37 0.00	37.00	119.80	14194.52
18.5.38 0.00	38.00	119.94	14194.52
18.5.39 0.00	39.00	120.40	14194.52
18.5.40 0.00	40.00	117.61	14194.52
18.5.41 0.00	41.00	117.60	14194.52
18.5.42 0.00	42.00	117.94	14194.52
18.5.43 0.00	43.00	117.89	14194.52
18.5.44 0.00	44.00	117.59	14194.52
18.5.45 0.00	45.00	117.61	14194.52
18.5.46 0.00	46.00	117.75	14194.52

18.5.47 0.00	47.00	119.43	14194.52
18.5.48 0.00	48.00	117.61	14194.52
18.5.49 0.00	49.00	117.60	14194.52
18.5.50 0.00	50.00	117.94	14194.52
18.5.51 0.00	51.00	117.89	14194.52
18.5.52 0.00	52.00	117.59	14194.52
18.5.53 0.00	53.00	117.61	14194.52
18.5.54 0.00	54.00	117.75	14194.52
18.5.55 0.00	55.00	119.43	14194.52
18.5.56 0.00	56.00	119.97	14194.52
18.5.57 0.00	57.00	118.26	14194.52
18.5.58 0.00	58.00	118.29	14194.52

18.5.59 0.00	59.00	118.53	14194.52
18.5.60 0.00	60.00	119.80	14194.52
18.5.61 0.00	61.00	119.80	14194.52
18.5.62 0.00	62.00	119.94	14194.52
18.5.63 0.00	63.00	120.40	14194.52
18.5.64 0.00	64.00	117.61	14194.52
18.5.65 0.00	65.00	117.60	14194.52
18.5.66 0.00	66.00	117.94	14194.52
18.5.67 0.00	67.00	117.89	14194.52
18.5.68 0.00	68.00	117.59	14194.52
18.5.69 0.00	69.00	117.61	14194.52
18.5.70 0.00	70.00	117.75	14194.52

18.5.71 0.00	71.00	119.43	14194.52
18.5.72 0.00	72.00	119.97	14194.52
18.5.73 0.00	73.00	118.26	14194.52
18.5.74 0.00	74.00	118.29	14194.52
18.5.75 0.00	75.00	118.53	14194.52
18.5.76 0.00	76.00	119.80	14194.52
18.5.77 0.00	77.00	119.80	14194.52
18.5.78 0.00	78.00	119.94	14194.52
18.5.79 0.00	79.00	120.40	14194.52
18.5.80 0.00	80.00	119.97	14194.52
18.5.81 0.00	81.00	118.26	14194.52
18.5.82 0.00	82.00	118.29	14194.52

18.5.83 0.00	83.00	118.53	14194.52
18.5.84 0.00	84.00	119.80	14194.52
18.5.85 0.00	85.00	119.80	14194.52
18.5.86 0.00	86.00	119.94	14194.52
18.5.87 0.00	87.00	120.40	14194.52
18.5.88 0.00	88.00	117.61	14194.52
18.5.89 0.00	89.00	117.60	14194.52
18.5.90 0.00	90.00	117.94	14194.52
18.5.91 0.00	91.00	117.89	14194.52
18.5.92 0.00	92.00	117.59	14194.52
18.5.93 0.00	93.00	117.61	14194.52
18.5.94 0.00	94.00	117.75	14194.52

18.5.95 0.00	95.00	119.43	14194.52
18.5.96 0.00	96.00	117.64	14194.52
18.5.97 0.00	97.00	117.65	14194.52
18.5.98 0.00	98.00	118.08	14194.52
18.5.99 0.00	99.00	118.04	14194.52
18.5.100 0.00	100.00	117.73	14194.52
18.5.101 0.00	101.00	117.74	14194.52
18.5.102 0.00	102.00	117.86	14194.53
18.5.103 0.00	103.00	119.57	14194.52
18.5.104 0.00	104.00	120.01	14194.54
18.5.105 0.00	105.00	118.30	14194.52
18.5.106 0.00	106.00	118.33	14194.52

18.5.107 0.00	107.00	118.57	14194.52
18.5.108 0.00	108.00	119.83	14194.52
18.5.109 0.00	109.00	119.84	14194.52
18.5.110 0.00	110.00	119.98	14194.52
18.5.111 0.00	111.00	120.44	14194.52
18.5.112 0.00	112.00	119.97	14194.52
18.5.113 0.00	113.00	118.26	14194.52
18.5.114 0.00	114.00	118.29	14194.52
18.5.115 0.00	115.00	118.53	14194.52
18.5.116 0.00	116.00	119.80	14194.52
18.5.117 0.00	117.00	119.80	14194.52
18.5.118 0.00	118.00	119.95	14194.52

18.5.119 0.00	119.00	120.41	14194.52
18.5.120 0.00	120.00	117.61	14194.52
18.5.121 0.00	121.00	117.61	14194.52
18.5.122 0.00	122.00	117.94	14194.52
18.5.123 0.00	123.00	117.89	14194.52
18.5.124 0.00	124.00	117.60	14194.52
18.5.125 0.00	125.00	117.61	14194.52
18.5.126 0.00	126.00	117.75	14194.52
18.5.127 0.00	127.00	119.44	14194.52

18.6 L2-Fabric Read Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.6.0	0.00	0.00	230.25	472.30	485.60	1179.75
18.6.1	1.00	0.00	492.51	499.04	529.77	1939.00
18.6.2	2.00	0.00	502.80	509.40	526.45	1541.00

18.6.3	3.00	0.00	498.69	507.48	554.94	1821.00
18.6.4	4.00	0.00	0.00	493.20	510.39	708.93
18.6.5	5.00	0.00	248.75	489.86	508.07	951.50
18.6.6	6.00	0.00	0.00	524.73	542.55	741.76
18.6.7	7.00	0.00	0.00	504.54	521.72	727.91
18.6.8	8.00	0.00	471.05	485.49	507.74	1145.00
18.6.9	9.00	0.00	492.01	509.17	526.63	2355.00
18.6.10	10.00	0.00	487.02	504.77	541.76	1777.00
18.6.11	11.00	0.00	490.90	508.26	529.62	714.39
18.6.12	12.00	0.00	494.72	500.98	516.58	734.03
18.6.13	13.00	0.00	491.15	497.16	512.68	719.88
18.6.14	14.00	0.00	513.21	524.20	537.90	741.26
18.6.15	15.00	0.00	0.00	491.12	508.22	718.71
18.6.16	16.00	0.00	471.89	488.81	509.00	1374.00
18.6.17	17.00	0.00	490.22	508.79	529.33	1634.00
18.6.18	18.00	0.00	485.06	502.96	543.67	1680.00
18.6.19	19.00	0.00	487.63	506.65	528.20	708.32
18.6.20	20.00	0.00	490.62	499.84	521.89	767.33
18.6.21	21.00	0.00	486.93	495.28	513.00	707.75
18.6.22	22.00	0.00	509.04	521.82	537.53	737.27
18.6.23	23.00	0.00	0.00	486.89	508.44	712.05
18.6.24	24.00	0.00	229.50	469.19	482.30	1478.75
18.6.25	25.00	0.00	490.83	498.34	530.84	1899.00
18.6.26	26.00	0.00	500.04	506.48	531.32	1734.00

18.6.27	27.00	0.00	494.62	504.12	553.12	2014.00
18.6.28	28.00	0.00	0.00	491.20	509.70	708.89
18.6.29	29.00	0.00	269.25	487.64	507.68	709.52
18.6.30	30.00	0.00	0.00	522.57	542.72	749.19
18.6.31	31.00	0.00	0.00	501.31	520.38	734.16
18.6.32	32.00	0.00	129.50	475.72	490.58	971.00
18.6.33	33.00	275.00	500.08	507.50	532.22	1328.00
18.6.34	34.00	0.00	495.23	501.63	526.00	1279.00
18.6.35	35.00	0.00	499.84	508.25	544.00	1514.00
18.6.36	36.00	0.00	0.00	493.59	511.12	694.16
18.6.37	37.00	0.00	154.75	490.96	510.79	717.60
18.6.38	38.00	0.00	0.00	516.15	537.90	744.05
18.6.39	39.00	0.00	0.00	494.04	514.82	722.85
18.6.40	40.00	0.00	465.36	480.95	505.38	1130.00
18.6.41	41.00	0.00	482.59	503.90	537.84	1198.00
18.6.42	42.00	0.00	491.75	508.87	538.32	1391.00
18.6.43	43.00	0.00	487.02	508.08	536.19	707.26
18.6.44	44.00	0.00	491.50	499.36	525.30	768.00
18.6.45	45.00	0.00	488.54	496.32	519.57	715.03
18.6.46	46.00	0.00	519.99	531.23	550.92	751.20
18.6.47	47.00	0.00	0.00	499.01	519.04	738.19
18.6.48	48.00	0.00	464.94	482.56	509.05	1074.00
18.6.49	49.00	0.00	480.35	497.07	521.82	1119.00
18.6.50	50.00	0.00	487.47	504.84	537.18	1447.00

18.6.51	51.00	0.00	482.99	503.77	532.72	705.01
18.6.52	52.00	0.00	488.65	496.33	519.73	717.94
18.6.53	53.00	0.00	485.71	494.17	516.74	683.19
18.6.54	54.00	0.00	516.04	529.95	549.43	761.59
18.6.55	55.00	0.00	0.00	494.91	515.66	732.87
18.6.56	56.00	0.00	130.50	470.55	485.66	698.17
18.6.57	57.00	0.00	497.13	504.33	526.40	1403.00
18.6.58	58.00	0.00	493.47	499.62	529.71	1250.00
18.6.59	59.00	0.00	497.59	507.46	542.00	1508.00
18.6.60	60.00	0.00	0.00	490.94	509.48	725.85
18.6.61	61.00	0.00	155.75	488.04	505.05	693.31
18.6.62	62.00	0.00	0.00	513.41	531.44	739.12
18.6.63	63.00	0.00	0.00	491.11	514.17	714.88
18.6.64	64.00	0.00	468.95	484.95	512.21	785.00
18.6.65	65.00	0.00	483.20	503.87	532.48	777.00
18.6.66	66.00	0.00	492.16	510.31	556.67	1826.00
18.6.67	67.00	0.00	490.43	509.29	536.19	745.01
18.6.68	68.00	0.00	489.05	495.25	519.73	723.17
18.6.69	69.00	0.00	486.26	492.29	512.57	727.41
18.6.70	70.00	0.00	518.87	526.80	546.64	759.90
18.6.71	71.00	0.00	0.00	497.69	513.22	768.20
18.6.72	72.00	0.00	281.31	477.87	495.23	688.05
18.6.73	73.00	146.75	506.29	514.84	552.53	2309.00
18.6.74	74.00	0.00	502.80	510.29	538.30	1985.00

18.6.75	75.00	0.00	508.38	517.66	573.38	2041.00
18.6.76	76.00	0.00	0.00	499.53	516.95	715.12
18.6.77	77.00	0.00	292.69	496.35	517.19	731.06
18.6.78	78.00	0.00	0.00	522.29	541.92	751.44
18.6.79	79.00	0.00	0.00	499.57	517.99	741.34
18.6.80	80.00	0.00	289.75	481.28	496.91	761.50
18.6.81	81.00	146.25	507.24	515.12	550.04	2350.00
18.6.82	82.00	0.00	502.02	509.17	536.87	1928.00
18.6.83	83.00	0.00	508.72	518.62	578.50	2044.00
18.6.84	84.00	0.00	0.00	500.45	517.80	722.63
18.6.85	85.00	0.00	309.00	497.19	514.96	709.85
18.6.86	86.00	0.00	0.00	523.54	543.64	732.87
18.6.87	87.00	0.00	0.00	501.09	520.42	716.77
18.6.88	88.00	0.00	467.93	485.59	517.66	842.00
18.6.89	89.00	0.00	485.31	506.55	535.15	786.00
18.6.90	90.00	0.00	492.92	512.72	558.67	1849.00
18.6.91	91.00	0.00	491.13	511.42	536.30	723.48
18.6.92	92.00	0.00	489.58	495.55	519.44	710.74
18.6.93	93.00	0.00	486.61	493.30	517.42	706.25
18.6.94	94.00	0.00	520.77	528.78	551.38	746.39
18.6.95	95.00	0.00	0.00	497.21	515.07	738.18
18.6.96	96.00	0.00	473.62	489.50	514.47	1442.00
18.6.97	97.00	0.00	492.67	512.31	539.22	2398.00
18.6.98	98.00	0.00	483.65	503.17	545.34	2193.00

18.6.99	99.00	0.00	486.40	507.38	530.32	1020.50
18.6.100	100.00	0.00	488.64	497.42	517.35	693.57
18.6.101	101.00	0.00	485.45	495.21	514.24	1063.00
18.6.102	102.00	0.00	511.00	519.50	533.84	738.30
18.6.103	103.00	0.00	0.00	486.80	507.95	1093.00
18.6.104	104.00	0.00	384.00	479.93	494.15	1010.50
18.6.105	105.00	0.00	501.84	510.19	543.89	2881.00
18.6.106	106.00	0.00	509.96	517.82	549.32	2364.00
18.6.107	107.00	0.00	507.51	518.11	580.37	2430.00
18.6.108	108.00	0.00	0.00	500.40	517.65	1361.00
18.6.109	109.00	0.00	416.50	498.71	519.36	745.72
18.6.110	110.00	0.00	0.00	532.55	556.73	773.25
18.6.111	111.00	0.00	0.00	510.47	530.59	1566.50
18.6.112	112.00	0.00	382.75	483.66	497.45	960.00
18.6.113	113.00	0.00	500.98	508.64	540.72	2851.00
18.6.114	114.00	0.00	509.55	516.58	541.94	2198.00
18.6.115	115.00	0.00	507.66	518.09	566.22	2557.00
18.6.116	116.00	0.00	0.00	500.34	520.49	718.99
18.6.117	117.00	0.00	414.75	499.67	522.62	724.41
18.6.118	118.00	0.00	0.00	534.34	553.20	758.34
18.6.119	119.00	0.00	0.00	511.19	529.16	737.36
18.6.120	120.00	0.00	468.13	487.19	507.65	1499.00
18.6.121	121.00	0.00	490.14	508.70	528.90	955.00
18.6.122	122.00	0.00	485.20	503.54	540.87	2368.00

18.6.123	123.00	0.00	490.85	508.86	536.52	724.44
18.6.124	124.00	0.00	491.40	499.00	520.12	728.10
18.6.125	125.00	0.00	487.95	496.03	514.98	719.86
18.6.126	126.00	0.00	511.09	521.27	538.81	740.73
18.6.127	127.00	0.00	0.00	487.34	506.31	728.18

18.7 L2-Fabric Write and Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.7.0	0.00	302.43	329.97	343.44	353.97	513.23
18.7.1	1.00	308.50	328.87	338.00	351.74	540.30
18.7.2	2.00	312.93	343.80	353.91	364.20	545.40
18.7.3	3.00	314.76	342.93	352.08	362.66	544.76
18.7.4	4.00	310.01	343.62	370.67	381.21	534.32
18.7.5	5.00	311.34	344.12	362.85	372.62	531.12
18.7.6	6.00	324.02	360.97	385.19	394.54	565.30
18.7.7	7.00	319.98	358.18	372.91	383.46	571.98
18.7.8	8.00	302.98	338.77	376.89	459.02	663.10
18.7.9	9.00	311.66	354.77	398.96	469.56	660.46
18.7.10	10.00	308.75	345.60	381.88	450.79	641.75
18.7.11	11.00	316.08	356.58	397.06	443.58	636.75
18.7.12	12.00	0.00	347.59	388.78	413.84	605.48
18.7.13	13.00	0.00	349.62	388.36	406.51	603.00
18.7.14	14.00	319.41	353.77	389.71	435.13	642.78
18.7.15	15.00	313.83	353.92	396.33	454.85	652.47
18.7.16	16.00	302.87	348.69	376.25	452.76	640.24

18.7.17	17.00	312.06	368.59	398.70	465.05	655.35
18.7.18	18.00	307.43	355.38	381.05	446.08	621.92
18.7.19	19.00	316.11	369.07	396.15	437.59	631.40
18.7.20	20.00	0.00	359.55	387.80	409.30	594.68
18.7.21	21.00	312.24	362.04	387.47	402.43	592.29
18.7.22	22.00	0.00	362.95	389.05	430.95	616.95
18.7.23	23.00	0.00	366.77	395.18	450.38	640.22
18.7.24	24.00	303.38	333.49	344.44	355.57	508.62
18.7.25	25.00	309.17	328.30	337.76	367.01	532.93
18.7.26	26.00	315.75	344.71	354.16	372.79	540.88
18.7.27	27.00	314.81	342.57	351.55	370.80	541.39
18.7.28	28.00	311.93	356.41	370.37	379.80	527.24
18.7.29	29.00	313.67	351.39	363.07	372.75	525.31
18.7.30	30.00	325.19	373.08	384.78	394.28	557.79
18.7.31	31.00	321.68	362.26	372.74	388.95	561.61
18.7.32	32.00	304.87	346.58	379.71	466.17	655.85
18.7.33	33.00	314.61	363.87	402.42	468.27	653.41
18.7.34	34.00	309.24	351.33	383.17	464.92	661.64
18.7.35	35.00	317.81	366.28	402.66	458.96	662.78
18.7.36	36.00	312.36	357.23	393.13	422.63	613.91
18.7.37	37.00	314.71	359.05	392.96	415.09	598.09
18.7.38	38.00	320.13	368.17	404.75	426.85	622.76
18.7.39	39.00	314.33	355.90	388.43	451.49	649.62
18.7.40	40.00	304.95	338.35	350.23	359.95	504.62

18.7.41	41.00	309.91	333.81	342.32	368.77	477.95
18.7.42	42.00	317.94	347.37	357.36	372.43	508.47
18.7.43	43.00	317.05	345.94	355.70	371.68	505.23
18.7.44	44.00	0.00	359.59	377.71	388.20	561.75
18.7.45	45.00	315.23	356.58	370.32	379.91	541.67
18.7.46	46.00	327.43	376.34	392.54	403.09	574.95
18.7.47	47.00	322.14	362.85	373.42	388.64	528.56
18.7.48	48.00	302.45	335.34	346.93	357.04	497.23
18.7.49	49.00	309.15	330.52	338.91	366.64	472.46
18.7.50	50.00	313.58	343.73	353.65	369.78	499.27
18.7.51	51.00	315.46	343.16	352.35	369.49	496.51
18.7.52	52.00	0.00	357.12	374.44	384.60	560.01
18.7.53	53.00	311.46	353.99	367.23	376.80	536.43
18.7.54	54.00	324.33	374.09	389.09	398.95	571.88
18.7.55	55.00	319.14	359.84	370.04	385.90	526.15
18.7.56	56.00	303.21	348.58	377.49	460.82	651.11
18.7.57	57.00	312.67	366.67	399.19	463.33	647.19
18.7.58	58.00	308.50	353.57	380.80	459.26	658.71
18.7.59	59.00	316.50	368.57	399.81	453.81	658.67
18.7.60	60.00	310.79	359.65	390.49	418.17	601.57
18.7.61	61.00	313.11	362.04	390.47	410.08	590.92
18.7.62	62.00	318.77	371.83	401.69	422.98	608.25
18.7.63	63.00	313.43	358.90	385.75	447.28	646.66
18.7.64	64.00	301.90	332.62	342.12	357.27	518.64

18.7.65	65.00	308.67	326.74	334.78	371.10	541.43
18.7.66	66.00	313.33	340.41	348.84	374.11	545.98
18.7.67	67.00	314.87	339.18	347.75	373.60	543.12
18.7.68	68.00	309.54	355.97	370.02	380.43	535.21
18.7.69	69.00	310.82	351.03	361.63	374.00	534.24
18.7.70	70.00	323.40	372.61	384.81	394.64	559.24
18.7.71	71.00	319.41	357.55	365.93	388.93	567.64
18.7.72	72.00	301.80	350.92	382.45	450.48	658.25
18.7.73	73.00	311.72	368.57	404.88	463.79	659.73
18.7.74	74.00	308.59	355.66	384.91	440.41	629.81
18.7.75	75.00	316.10	370.36	403.27	436.07	641.27
18.7.76	76.00	309.82	362.09	392.23	410.93	576.94
18.7.77	77.00	312.62	363.91	391.82	405.22	563.85
18.7.78	78.00	320.00	373.79	402.69	416.48	568.19
18.7.79	79.00	313.57	359.89	389.78	434.39	596.49
18.7.80	80.00	302.08	352.33	382.56	454.40	660.06
18.7.81	81.00	312.90	372.10	405.91	465.38	658.98
18.7.82	82.00	307.69	357.87	385.60	445.69	629.36
18.7.83	83.00	317.54	373.34	404.90	441.29	642.95
18.7.84	84.00	310.37	363.88	393.96	413.12	585.16
18.7.85	85.00	313.13	365.94	393.44	407.91	567.36
18.7.86	86.00	319.40	377.08	404.36	418.92	574.62
18.7.87	87.00	313.36	361.99	390.25	436.44	593.15
18.7.88	88.00	303.89	336.07	345.67	360.66	495.87

18.7.89	89.00	308.92	329.84	338.74	375.23	518.55
18.7.90	90.00	317.09	344.18	352.97	377.98	521.00
18.7.91	91.00	316.26	342.32	351.19	376.59	519.22
18.7.92	92.00	311.84	360.15	373.23	383.12	519.93
18.7.93	93.00	313.55	354.72	364.85	377.17	511.96
18.7.94	94.00	325.62	376.71	388.30	397.71	544.30
18.7.95	95.00	322.49	360.21	368.95	393.54	541.11
18.7.96	96.00	0.00	352.97	381.89	458.26	656.60
18.7.97	97.00	0.00	374.78	405.22	462.88	651.94
18.7.98	98.00	0.00	359.81	385.96	446.29	658.57
18.7.99	99.00	0.00	375.22	400.96	434.06	658.92
18.7.100	100.00	0.00	364.41	391.12	409.28	608.41
18.7.101	101.00	0.00	367.17	390.06	402.09	602.36
18.7.102	102.00	0.00	365.99	393.27	427.26	612.75
18.7.103	103.00	0.00	373.57	401.76	449.71	652.42
18.7.104	104.00	0.00	332.36	341.98	359.34	508.73
18.7.105	105.00	0.00	325.22	333.46	374.39	476.28
18.7.106	106.00	0.00	342.47	351.38	379.17	508.19
18.7.107	107.00	316.47	340.39	349.29	377.29	505.57
18.7.108	108.00	0.00	358.13	370.81	382.65	545.40
18.7.109	109.00	0.00	351.97	361.97	376.43	530.74
18.7.110	110.00	0.00	375.71	386.97	398.38	560.13
18.7.111	111.00	0.00	359.03	367.50	396.75	525.38
18.7.112	112.00	0.00	331.90	341.38	355.91	498.49

18.7.113	113.00	309.13	325.48	333.62	368.37	484.09
18.7.114	114.00	0.00	342.40	351.24	373.93	503.69
18.7.115	115.00	314.81	340.85	349.62	373.16	500.27
18.7.116	116.00	0.00	355.08	370.25	380.68	542.19
18.7.117	117.00	311.39	350.90	361.43	373.51	527.79
18.7.118	118.00	0.00	373.78	386.55	396.66	558.26
18.7.119	119.00	319.92	359.16	368.00	390.39	526.38
18.7.120	120.00	0.00	350.73	380.45	461.28	654.74
18.7.121	121.00	312.78	368.69	402.78	463.35	651.58
18.7.122	122.00	0.00	357.66	384.93	449.86	656.43
18.7.123	123.00	0.00	370.05	400.76	439.41	655.33
18.7.124	124.00	0.00	360.23	390.95	410.83	607.15
18.7.125	125.00	312.81	363.45	390.20	404.11	600.15
18.7.126	126.00	318.90	365.46	392.88	429.60	621.91
18.7.127	127.00	314.35	368.02	401.00	452.52	656.60

18.8 L2-Fabric Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.8.0	0.00	0.00	0.00	0.00	0.00	0.00
18.8.1	1.00	0.00	0.00	0.00	0.00	0.00
18.8.2	2.00	0.00	0.00	0.00	0.00	0.00
18.8.3	3.00	0.00	0.00	0.00	0.00	0.00
18.8.4	4.00	0.00	0.00	0.00	0.00	0.00
18.8.5	5.00	0.00	0.00	0.00	0.00	0.00
18.8.6	6.00	0.00	0.00	0.00	0.00	0.00

18.8.7	7.00	0.00	0.00	0.00	0.00	0.00
18.8.8	8.00	0.00	0.00	0.00	0.00	0.00
18.8.9	9.00	0.00	0.00	0.00	0.00	0.00
18.8.10	10.00	0.00	0.00	0.00	0.00	0.00
18.8.11	11.00	0.00	0.00	0.00	0.00	0.00
18.8.12	12.00	0.00	0.00	0.00	0.00	0.00
18.8.13	13.00	0.00	0.00	0.00	0.00	0.00
18.8.14	14.00	0.00	0.00	0.00	0.00	0.00
18.8.15	15.00	0.00	0.00	0.00	0.00	0.00
18.8.16	16.00	0.00	0.00	0.00	0.00	0.00
18.8.17	17.00	0.00	0.00	0.00	0.00	0.00
18.8.18	18.00	0.00	0.00	0.00	0.00	0.00
18.8.19	19.00	0.00	0.00	0.00	0.00	0.00
18.8.20	20.00	0.00	0.00	0.00	0.00	0.00
18.8.21	21.00	0.00	0.00	0.00	0.00	0.00
18.8.22	22.00	0.00	0.00	0.00	0.00	0.00
18.8.23	23.00	0.00	0.00	0.00	0.00	0.00
18.8.24	24.00	0.00	0.00	0.00	0.00	0.00
18.8.25	25.00	0.00	0.00	0.00	0.00	0.00
18.8.26	26.00	0.00	0.00	0.00	0.00	0.00
18.8.27	27.00	0.00	0.00	0.00	0.00	0.00
18.8.28	28.00	0.00	0.00	0.00	0.00	0.00
18.8.29	29.00	0.00	0.00	0.00	0.00	0.00
18.8.30	30.00	0.00	0.00	0.00	0.00	0.00

18.8.31	31.00	0.00	0.00	0.00	0.00	0.00
18.8.32	32.00	0.00	0.00	0.00	0.00	0.00
18.8.33	33.00	0.00	0.00	0.00	0.00	0.00
18.8.34	34.00	0.00	0.00	0.00	0.00	0.00
18.8.35	35.00	0.00	0.00	0.00	0.00	0.00
18.8.36	36.00	0.00	0.00	0.00	0.00	0.00
18.8.37	37.00	0.00	0.00	0.00	0.00	0.00
18.8.38	38.00	0.00	0.00	0.00	0.00	0.00
18.8.39	39.00	0.00	0.00	0.00	0.00	0.00
18.8.40	40.00	0.00	0.00	0.00	0.00	0.00
18.8.41	41.00	0.00	0.00	0.00	0.00	0.00
18.8.42	42.00	0.00	0.00	0.00	0.00	0.00
18.8.43	43.00	0.00	0.00	0.00	0.00	0.00
18.8.44	44.00	0.00	0.00	0.00	0.00	0.00
18.8.45	45.00	0.00	0.00	0.00	0.00	0.00
18.8.46	46.00	0.00	0.00	0.00	0.00	0.00
18.8.47	47.00	0.00	0.00	0.00	0.00	0.00
18.8.48	48.00	0.00	0.00	0.00	0.00	0.00
18.8.49	49.00	0.00	0.00	0.00	0.00	0.00
18.8.50	50.00	0.00	0.00	0.00	0.00	0.00
18.8.51	51.00	0.00	0.00	0.00	0.00	0.00
18.8.52	52.00	0.00	0.00	0.00	0.00	0.00
18.8.53	53.00	0.00	0.00	0.00	0.00	0.00
18.8.54	54.00	0.00	0.00	0.00	0.00	0.00

18.8.55	55.00	0.00	0.00	0.00	0.00	0.00
18.8.56	56.00	0.00	0.00	0.00	0.00	0.00
18.8.57	57.00	0.00	0.00	0.00	0.00	0.00
18.8.58	58.00	0.00	0.00	0.00	0.00	0.00
18.8.59	59.00	0.00	0.00	0.00	0.00	0.00
18.8.60	60.00	0.00	0.00	0.00	0.00	0.00
18.8.61	61.00	0.00	0.00	0.00	0.00	0.00
18.8.62	62.00	0.00	0.00	0.00	0.00	0.00
18.8.63	63.00	0.00	0.00	0.00	0.00	0.00
18.8.64	64.00	0.00	0.00	0.00	0.00	0.00
18.8.65	65.00	0.00	0.00	0.00	0.00	0.00
18.8.66	66.00	0.00	0.00	0.00	0.00	0.00
18.8.67	67.00	0.00	0.00	0.00	0.00	0.00
18.8.68	68.00	0.00	0.00	0.00	0.00	0.00
18.8.69	69.00	0.00	0.00	0.00	0.00	0.00
18.8.70	70.00	0.00	0.00	0.00	0.00	0.00
18.8.71	71.00	0.00	0.00	0.00	0.00	0.00
18.8.72	72.00	0.00	0.00	0.00	0.00	0.00
18.8.73	73.00	0.00	0.00	0.00	0.00	0.00
18.8.74	74.00	0.00	0.00	0.00	0.00	0.00
18.8.75	75.00	0.00	0.00	0.00	0.00	0.00
18.8.76	76.00	0.00	0.00	0.00	0.00	0.00
18.8.77	77.00	0.00	0.00	0.00	0.00	0.00
18.8.78	78.00	0.00	0.00	0.00	0.00	0.00

18.8.79	79.00	0.00	0.00	0.00	0.00	0.00
18.8.80	80.00	0.00	0.00	0.00	0.00	0.00
18.8.81	81.00	0.00	0.00	0.00	0.00	0.00
18.8.82	82.00	0.00	0.00	0.00	0.00	0.00
18.8.83	83.00	0.00	0.00	0.00	0.00	0.00
18.8.84	84.00	0.00	0.00	0.00	0.00	0.00
18.8.85	85.00	0.00	0.00	0.00	0.00	0.00
18.8.86	86.00	0.00	0.00	0.00	0.00	0.00
18.8.87	87.00	0.00	0.00	0.00	0.00	0.00
18.8.88	88.00	0.00	0.00	0.00	0.00	0.00
18.8.89	89.00	0.00	0.00	0.00	0.00	0.00
18.8.90	90.00	0.00	0.00	0.00	0.00	0.00
18.8.91	91.00	0.00	0.00	0.00	0.00	0.00
18.8.92	92.00	0.00	0.00	0.00	0.00	0.00
18.8.93	93.00	0.00	0.00	0.00	0.00	0.00
18.8.94	94.00	0.00	0.00	0.00	0.00	0.00
18.8.95	95.00	0.00	0.00	0.00	0.00	0.00
18.8.96	96.00	0.00	0.00	0.00	0.00	0.00
18.8.97	97.00	0.00	0.00	0.00	0.00	0.00
18.8.98	98.00	0.00	0.00	0.00	0.00	0.00
18.8.99	99.00	0.00	0.00	0.00	0.00	0.00
18.8.100	100.00	0.00	0.00	0.00	0.00	0.00
18.8.101	101.00	0.00	0.00	0.00	0.00	0.00
18.8.102	102.00	0.00	0.00	0.00	0.00	0.00

18.8.103	103.00	0.00	0.00	0.00	0.00	0.00
18.8.104	104.00	0.00	0.00	0.00	0.00	0.00
18.8.105	105.00	0.00	0.00	0.00	0.00	0.00
18.8.106	106.00	0.00	0.00	0.00	0.00	0.00
18.8.107	107.00	0.00	0.00	0.00	0.00	0.00
18.8.108	108.00	0.00	0.00	0.00	0.00	0.00
18.8.109	109.00	0.00	0.00	0.00	0.00	0.00
18.8.110	110.00	0.00	0.00	0.00	0.00	0.00
18.8.111	111.00	0.00	0.00	0.00	0.00	0.00
18.8.112	112.00	0.00	0.00	0.00	0.00	0.00
18.8.113	113.00	0.00	0.00	0.00	0.00	0.00
18.8.114	114.00	0.00	0.00	0.00	0.00	0.00
18.8.115	115.00	0.00	0.00	0.00	0.00	0.00
18.8.116	116.00	0.00	0.00	0.00	0.00	0.00
18.8.117	117.00	0.00	0.00	0.00	0.00	0.00
18.8.118	118.00	0.00	0.00	0.00	0.00	0.00
18.8.119	119.00	0.00	0.00	0.00	0.00	0.00
18.8.120	120.00	0.00	0.00	0.00	0.00	0.00
18.8.121	121.00	0.00	0.00	0.00	0.00	0.00
18.8.122	122.00	0.00	0.00	0.00	0.00	0.00
18.8.123	123.00	0.00	0.00	0.00	0.00	0.00
18.8.124	124.00	0.00	0.00	0.00	0.00	0.00
18.8.125	125.00	0.00	0.00	0.00	0.00	0.00
18.8.126	126.00	0.00	0.00	0.00	0.00	0.00

18.8.127	127.00	0.00	0.00	0.00	0.00	0.00
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18.12 L2-Fabric (128B read requests per normUnit)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.12.0	0.00	0.00	0.00	130.00	130.00	1152.00
18.12.1	1.00	0.00	0.00	130.00	130.00	1152.00
18.12.2	2.00	0.00	0.00	130.00	130.00	1152.00
18.12.3	3.00	0.00	0.00	130.00	130.00	1152.00
18.12.4	4.00	0.00	0.00	130.00	130.00	1152.00
18.12.5	5.00	0.00	0.00	130.00	130.00	1152.00
18.12.6	6.00	0.00	0.00	130.00	130.00	1152.00
18.12.7	7.00	0.00	0.00	130.00	130.00	1152.00
18.12.8	8.00	0.00	0.00	128.00	128.00	1152.00
18.12.9	9.00	0.00	0.00	128.00	128.00	1152.00
18.12.10	10.00	0.00	0.00	128.00	128.00	1152.00
18.12.11	11.00	0.00	0.00	128.00	128.00	1152.00
18.12.12	12.00	0.00	0.00	128.00	128.00	1152.00
18.12.13	13.00	0.00	0.00	128.00	128.00	1152.00
18.12.14	14.00	0.00	0.00	128.00	128.00	1152.00
18.12.15	15.00	0.00	0.00	128.00	128.00	1152.00
18.12.16	16.00	0.00	0.00	128.00	128.00	1152.00
18.12.17	17.00	0.00	0.00	128.00	128.00	1152.00
18.12.18	18.00	0.00	0.00	128.00	128.00	1152.00
18.12.19	19.00	0.00	0.00	128.00	128.00	1152.00
18.12.20	20.00	0.00	0.00	128.00	128.00	1152.00

18.12.21	21.00	0.00	0.00	128.00	128.00	1152.00
18.12.22	22.00	0.00	0.00	128.00	128.00	1152.00
18.12.23	23.00	0.00	0.00	128.00	128.00	1152.00
18.12.24	24.00	0.00	0.00	130.00	130.00	1152.00
18.12.25	25.00	0.00	0.00	130.00	130.00	1152.00
18.12.26	26.00	0.00	0.00	130.00	130.00	1152.00
18.12.27	27.00	0.00	0.00	130.00	130.00	1152.00
18.12.28	28.00	0.00	0.00	130.00	130.00	1152.00
18.12.29	29.00	0.00	0.00	130.00	130.00	1152.00
18.12.30	30.00	0.00	0.00	130.00	130.00	1152.00
18.12.31	31.00	0.00	0.00	130.00	130.00	1152.00
18.12.32	32.00	0.00	0.00	130.00	130.00	1152.00
18.12.33	33.00	0.00	0.00	130.00	130.00	1152.00
18.12.34	34.00	0.00	0.00	130.00	130.00	1152.00
18.12.35	35.00	0.00	0.00	130.00	130.00	1152.00
18.12.36	36.00	0.00	0.00	130.00	130.00	1152.00
18.12.37	37.00	0.00	0.00	130.00	130.00	1152.00
18.12.38	38.00	0.00	0.00	130.00	130.00	1152.00
18.12.39	39.00	0.00	0.00	130.00	130.00	1152.00
18.12.40	40.00	0.00	0.00	128.00	128.00	1152.00
18.12.41	41.00	0.00	0.00	128.00	128.00	1152.00
18.12.42	42.00	0.00	0.00	128.00	128.00	1152.00
18.12.43	43.00	0.00	0.00	128.00	128.00	1152.00
18.12.44	44.00	0.00	0.00	128.00	128.00	1152.00

18.12.45	45.00	0.00	0.00	128.00	128.00	1152.00
18.12.46	46.00	0.00	0.00	128.00	128.00	1152.00
18.12.47	47.00	0.00	0.00	128.00	128.00	1152.00
18.12.48	48.00	0.00	0.00	128.00	128.00	1152.00
18.12.49	49.00	0.00	0.00	128.00	128.00	1152.00
18.12.50	50.00	0.00	0.00	128.00	128.00	1152.00
18.12.51	51.00	0.00	0.00	128.00	128.00	1152.00
18.12.52	52.00	0.00	0.00	128.00	128.00	1152.00
18.12.53	53.00	0.00	0.00	128.00	128.00	1152.00
18.12.54	54.00	0.00	0.00	128.00	128.00	1152.00
18.12.55	55.00	0.00	0.00	128.00	128.00	1152.00
18.12.56	56.00	0.00	0.00	130.00	130.00	1152.00
18.12.57	57.00	0.00	0.00	130.00	130.00	1152.00
18.12.58	58.00	0.00	0.00	130.00	130.00	1152.00
18.12.59	59.00	0.00	0.00	130.00	130.00	1152.00
18.12.60	60.00	0.00	0.00	130.00	130.00	1152.00
18.12.61	61.00	0.00	0.00	130.00	130.00	1152.00
18.12.62	62.00	0.00	0.00	130.00	130.00	1152.00
18.12.63	63.00	0.00	0.00	130.00	130.00	1152.00
18.12.64	64.00	0.00	0.00	128.00	128.00	1152.00
18.12.65	65.00	0.00	0.00	128.00	128.00	1152.00
18.12.66	66.00	0.00	0.00	128.00	128.00	1152.00
18.12.67	67.00	0.00	0.00	128.00	128.00	1152.00
18.12.68	68.00	0.00	0.00	128.00	128.00	1152.00

18.12.69	69.00	0.00	0.00	128.00	128.00	1152.00
18.12.70	70.00	0.00	0.00	128.00	128.00	1152.00
18.12.71	71.00	0.00	0.00	128.00	128.00	1152.00
18.12.72	72.00	0.00	0.00	130.00	130.00	1152.00
18.12.73	73.00	0.00	0.00	130.00	130.00	1152.00
18.12.74	74.00	0.00	0.00	130.00	130.00	1152.00
18.12.75	75.00	0.00	0.00	130.00	130.00	1152.00
18.12.76	76.00	0.00	0.00	130.00	130.00	1152.00
18.12.77	77.00	0.00	0.00	130.00	130.00	1152.00
18.12.78	78.00	0.00	0.00	130.00	130.00	1152.00
18.12.79	79.00	0.00	0.00	130.00	130.00	1152.00
18.12.80	80.00	0.00	0.00	130.00	130.00	1152.00
18.12.81	81.00	0.00	0.00	130.00	130.00	1152.00
18.12.82	82.00	0.00	0.00	130.00	130.00	1152.00
18.12.83	83.00	0.00	0.00	130.00	130.00	1152.00
18.12.84	84.00	0.00	0.00	130.00	130.00	1152.00
18.12.85	85.00	0.00	0.00	130.00	130.00	1152.00
18.12.86	86.00	0.00	0.00	130.00	130.00	1152.00
18.12.87	87.00	0.00	0.00	130.00	130.00	1152.00
18.12.88	88.00	0.00	0.00	128.00	128.00	1152.00
18.12.89	89.00	0.00	0.00	128.00	128.00	1152.00
18.12.90	90.00	0.00	0.00	128.00	128.00	1152.00
18.12.91	91.00	0.00	0.00	128.00	128.00	1152.00
18.12.92	92.00	0.00	0.00	128.00	128.00	1152.00

18.12.93	93.00	0.00	0.00	128.00	128.00	1152.00
18.12.94	94.00	0.00	0.00	128.00	128.00	1152.00
18.12.95	95.00	0.00	0.00	128.00	128.00	1152.00
18.12.96	96.00	0.00	0.00	128.00	128.00	1152.00
18.12.97	97.00	0.00	0.00	128.00	128.00	1152.00
18.12.98	98.00	0.00	0.00	128.00	128.00	1152.00
18.12.99	99.00	0.00	0.00	128.00	128.00	1152.00
18.12.100	100.00	0.00	0.00	128.00	128.00	1152.00
18.12.101	101.00	0.00	0.00	128.00	128.00	1152.00
18.12.102	102.00	0.00	0.00	128.00	128.00	1152.00
18.12.103	103.00	0.00	0.00	128.00	128.00	1152.00
18.12.104	104.00	0.00	0.00	130.00	130.00	1152.00
18.12.105	105.00	0.00	0.00	130.00	130.00	1152.00
18.12.106	106.00	0.00	0.00	130.00	130.00	1152.00
18.12.107	107.00	0.00	0.00	130.00	130.00	1152.00
18.12.108	108.00	0.00	0.00	130.00	130.00	1152.00
18.12.109	109.00	0.00	0.00	130.00	130.00	1152.00
18.12.110	110.00	0.00	0.00	130.00	130.00	1152.00
18.12.111	111.00	0.00	0.00	130.00	130.00	1152.00
18.12.112	112.00	0.00	0.00	130.00	130.00	1152.00
18.12.113	113.00	0.00	0.00	130.00	130.00	1152.00
18.12.114	114.00	0.00	0.00	130.00	130.00	1152.00
18.12.115	115.00	0.00	0.00	130.00	130.00	1152.00
18.12.116	116.00	0.00	0.00	130.00	130.00	1152.00

18.12.117	117.00	0.00	0.00	130.00	130.00	1152.00
18.12.118	118.00	0.00	0.00	130.00	130.00	1152.00
18.12.119	119.00	0.00	0.00	130.00	130.00	1152.00
18.12.120	120.00	0.00	0.00	128.00	128.00	1152.00
18.12.121	121.00	0.00	0.00	128.00	128.00	1152.00
18.12.122	122.00	0.00	0.00	128.00	128.00	1152.00
18.12.123	123.00	0.00	0.00	128.00	128.00	1152.00
18.12.124	124.00	0.00	0.00	128.00	128.00	1152.00
18.12.125	125.00	0.00	0.00	128.00	128.00	1152.00
18.12.126	126.00	0.00	0.00	128.00	128.00	1152.00
18.12.127	127.00	0.00	0.00	128.00	128.00	1152.00

1.5 Fusion MatMulBias Kernel

1.5.1 Verify the Fusion MatMulBias Kernel

This also ensures the kernel has been built and cached.

```
[10]: !python triton/matmulbias.py --verify fusion
```

Running the fusion matmulbias kernel ...

```
triton_output: tensor([[ 2.7480, 43.6250, 30.9531, ..., 26.2500,  8.8203,
 6.4609],
```

```
    [ 7.1562, -21.3906, 17.7969, ..., -16.8125, 29.6562, -48.3125],
```

```
    [ 13.8125, -6.4609,  0.5146, ..., 48.0000, 25.1562, -27.5156],
```

```
    ...,
```

```
    [-54.4375, -1.4375, -0.4214, ...,  5.4297, 17.2812, -39.0625],
```

```
    [ 9.6094, -52.2500, 43.0000, ..., -45.8438, -49.3125, 32.6875],
```

```
    [-12.8516,  9.5391, 53.3750, ..., -9.3438,  7.6797, 54.1250]],
```

```
device='cuda:0', dtype=torch.float16)
```

Running the torch matmulbias kernel ...

```
torch_output: tensor([[ 2.7480, 43.6562, 30.9531, ..., 26.2500,  8.8203,
 6.4609],
```

```
    [ 7.1602, -21.3906, 17.7969, ..., -16.8125, 29.6406, -48.3125],
```

```
    [ 13.8125, -6.4609,  0.5142, ..., 48.0312, 25.1406, -27.5156],
```

```
    ...,
```

```
    [-54.4375, -1.4375, -0.4214, ...,  5.4297, 17.2812, -39.0625],
```

```

        [ 9.6094, -52.2500, 43.0000, ..., -45.8125, -49.3125, 32.7188],
        [-12.8516, 9.5391, 53.3750, ..., -9.3438, 7.6797, 54.1562]],
        device='cuda:0', dtype=torch.float16)
Verifying triton results with torch ...
Verified

```

1.5.2 Profile the Fusion MatMulBias Kernel

Using the Triton Proton CLI Generates a hatchet file, fusion_matmulbias.hatchet, under the profiles directory.

```
[11]: !proton -n profiles/fusion_matmulbias -k triton triton/matmulbias.py fusion
```

```

Running the fusion matmulbias kernel ...
triton_output: tensor([[ 2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203,
6.4609],
        [ 7.1562, -21.3906, 17.7969, ..., -16.8125, 29.6562, -48.3125],
        [ 13.8125, -6.4609, 0.5146, ..., 48.0000, 25.1562, -27.5156],
        ...,
        [-54.4375, -1.4375, -0.4214, ..., 5.4297, 17.2812, -39.0625],
        [ 9.6094, -52.2500, 43.0000, ..., -45.8438, -49.3125, 32.6875],
        [-12.8516, 9.5391, 53.3750, ..., -9.3438, 7.6797, 54.1250]],
        device='cuda:0', dtype=torch.float16)

```

Using the Triton Proton API Generates a hatchet file, fusion_matmulbias.hatchet, under the profiles directory and outputs the profile.

```
[12]: !python triton/matmulbias.py --profile fusion
```

```

Profiling the fusion matmulbias kernel ...
triton_output: tensor([[ 2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203,
6.4609],
        [ 7.1562, -21.3906, 17.7969, ..., -16.8125, 29.6562, -48.3125],
        [ 13.8125, -6.4609, 0.5146, ..., 48.0000, 25.1562, -27.5156],
        ...,
        [-54.4375, -1.4375, -0.4214, ..., 5.4297, 17.2812, -39.0625],
        [ 9.6094, -52.2500, 43.0000, ..., -45.8438, -49.3125, 32.6875],
        [-12.8516, 9.5391, 53.3750, ..., -9.3438, 7.6797, 54.1250]],
        device='cuda:0', dtype=torch.float16)

```

Proton profile results for fusion_matmulbias

```

95.951 38.025 ROOT
57.152 nan _ZN2at6native29vectorized_elementwise_kernel
ILi4ENS0_11FillFunctorIiEES5arrayIPcLm1EEEEviT0_T1_
38.799 94.038 matmulbias_kernel [M=1024, N=1024,
K=1024] [BM=128, BN=128 BK=64]

```

Legend (Metric: time/ms (inc) Min: 38.80 Max: 95.95)

```

90.24 - 95.95
78.81 - 90.24

```

```
name User code      Only in left graph
Only in right graph
```

```
[13]: !ROCPROF=rocprofiler-sdk rocprof-compute profile -n fusion_matmulbias -- python_
      ↪ triton/matmulbias.py fusion
```

WARNING DEPRECATION WARNING: rocm-smi is deprecated in ROCm 7.0 and will be removed from rocprof-compute in ROCm 7.1. Please migrate to amd-smi for compute partition parsing. For migration help, see <https://github.com/ROCm/amdsmi>

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```

n_matmulbias/MI300X_A1/perfmon/SQ_IFETCH_LEVEL.txt
INFO      |-> [rocprofiler-sdk] W20260122 20:38:13.898392
140133751795264 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.177593 sec
INFO      |-> [rocprofiler-sdk] W20260122 20:38:13.898493
140133751795264 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec
INFO      |-> [rocprofiler-sdk] W20260122 20:38:15.160795
140133751795264 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)
INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906,  17.7969, ...,
-16.8125,  29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000,  25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500,  43.0000, ...,
-45.8438, -49.3125,  32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391,  53.3750, ...,
-9.3438,  7.6797,  54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260122 20:38:16.434699
140133751795264 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.536205 sec

INFO      |-> [rocprofiler-sdk] E20260122 20:38:16.447654
140133751795264 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/625_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:38:16.461380
140133751795264 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/625_counter_collection.c
sv

INFO      |-> [rocprofiler-sdk] E20260122 20:38:16.779406
140133751795264 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/625_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260122 20:38:16.782336
140133751795264 simple_timer.cpp:55] [rocprofv3] output generation ::
0.342456 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:16.783473
140133751795264 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.343663 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 2/12] [Approximate profiling time left: 53 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias/MI300X_A1/perfmon/SQ_INST_LEVEL_LDS.txt
INFO      |-> [rocprofiler-sdk] W20260122 20:38:18.954827
139739090050624 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128179 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:18.954926
139739090050624 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:20.218464
139739090050624 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,

```



```

43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO    |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO    |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO    |-> [rocprofiler-sdk] ...,
INFO    |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO    |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO    |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO    |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO    |-> [rocprofiler-sdk] W20260122 20:38:21.472985

139739090050624 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.518060 sec

INFO    |-> [rocprofiler-sdk] E20260122 20:38:21.486034

139739090050624 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/650_kernel_trace.csv

INFO    |-> [rocprofiler-sdk] E20260122 20:38:21.499953

139739090050624 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/650_counter_collection.c
sv

INFO    |-> [rocprofiler-sdk] E20260122 20:38:21.521561

139739090050624 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/650_agent_info.csv

INFO    |-> [rocprofiler-sdk] W20260122 20:38:21.524644

139739090050624 simple_timer.cpp:55] [rocprofv3] output generation ::
0.046412 sec

INFO    |-> [rocprofiler-sdk] W20260122 20:38:21.525624

139739090050624 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.047464 sec

INFO    |-> [rocprofiler-sdk]
INFO [Run 3/12][Approximate profiling time left: 41 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias/MI300X_A1/perfmon/SQ_INST_LEVEL_SMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260122 20:38:23.197306
139677527932480 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.180110 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:23.197415
139677527932480 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000001 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:24.463593
139677527932480 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260122 20:38:25.732680
139677527932480 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.535265 sec

INFO      |-> [rocprofiler-sdk] E20260122 20:38:25.745642
139677527932480 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/670_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:38:25.759524
139677527932480 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/670_counter_collection.c
sv

INFO      |-> [rocprofiler-sdk] E20260122 20:38:26.065935
139677527932480 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/670_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260122 20:38:26.068874
139677527932480 simple_timer.cpp:55] [rocprofv3] output generation ::
0.331005 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:26.069976
139677527932480 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.332174 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 4/12] [Approximate profiling time left: 38 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias/MI300X_A1/perfmon/SQ_INST_LEVEL_VMEM.txt
INFO      |-> [rocprofiler-sdk] W20260122 20:38:28.389564
140547139194432 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.165892 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:28.389673
140547139194432 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000001 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:29.654622
140547139194432 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,

```

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43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO    |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO    |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO    |-> [rocprofiler-sdk] ...,
INFO    |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO    |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO    |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO    |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO    |-> [rocprofiler-sdk] W20260122 20:38:30.924104

140547139194432 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.534432 sec

INFO    |-> [rocprofiler-sdk] E20260122 20:38:30.937123

140547139194432 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/691_kernel_trace.csv

INFO    |-> [rocprofiler-sdk] E20260122 20:38:30.951007

140547139194432 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/691_counter_collection.c
sv

INFO    |-> [rocprofiler-sdk] E20260122 20:38:31.187268

140547139194432 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/691_agent_info.csv

INFO    |-> [rocprofiler-sdk] W20260122 20:38:31.190274

140547139194432 simple_timer.cpp:55] [rocprofv3] output generation ::
0.260979 sec

INFO    |-> [rocprofiler-sdk] W20260122 20:38:31.191326

140547139194432 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.262099 sec

INFO    |-> [rocprofiler-sdk]
INFO [Run 5/12][Approximate profiling time left: 34 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/fusion_matmulbias/MI300X_A1/perfmon/SQ_LEVEL_WAVES.txt

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```

INFO      |-> [rocprofiler-sdk] W20260122 20:38:33.355617
140202194579008 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.165899 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:33.355722
140202194579008 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:34.616728
140202194579008 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

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INFO      |-> [rocprofiler-sdk] W20260122 20:38:35.880206
140202194579008 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.524485 sec

INFO      |-> [rocprofiler-sdk] E20260122 20:38:35.893209
140202194579008 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/711_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:38:35.907139
140202194579008 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/711_counter_collection.c
sv

INFO      |-> [rocprofiler-sdk] E20260122 20:38:36.162151
140202194579008 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/711_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260122 20:38:36.165084
140202194579008 simple_timer.cpp:55] [rocprofv3] output generation ::
0.279691 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:36.166166
140202194579008 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.280842 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 6/12] [Approximate profiling time left: 29 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/fusion_matmulbias/MI300X_A1/perfmon/pmc_perf_0.txt
INFO      |-> [rocprofiler-sdk] W20260122 20:38:38.145354
140004028415552 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128258 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:38.145477
140004028415552 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:39.406490
140004028415552 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,

```

```

43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO    |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO    |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO    |-> [rocprofiler-sdk] ...,
INFO    |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO    |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO    |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO    |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO    |-> [rocprofiler-sdk] W20260122 20:38:40.657758

140004028415552 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.512281 sec

INFO    |-> [rocprofiler-sdk] E20260122 20:38:40.670916

140004028415552 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/732_kernel_trace.csv

INFO    |-> [rocprofiler-sdk] E20260122 20:38:40.684901

140004028415552 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/732_counter_collection.c
sv

INFO    |-> [rocprofiler-sdk] E20260122 20:38:40.705018

140004028415552 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/732_agent_info.csv

INFO    |-> [rocprofiler-sdk] W20260122 20:38:40.708111

140004028415552 simple_timer.cpp:55] [rocprofv3] output generation ::
0.045304 sec

INFO    |-> [rocprofiler-sdk] W20260122 20:38:40.709051

140004028415552 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.046312 sec

INFO    |-> [rocprofiler-sdk]
INFO [Run 7/12][Approximate profiling time left: 23 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/fusion_matmulbias/MI300X_A1/perfmon/pmc_perf_1.txt

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INFO      |-> [rocprofiler-sdk] W20260122 20:38:42.015380
140695764520512 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128236 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:42.015485
140695764520512 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:43.276193
140695764520512 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```



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INFO      |-> [rocprofiler-sdk] W20260122 20:38:44.528694
140695764520512 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.513209 sec

INFO      |-> [rocprofiler-sdk] E20260122 20:38:44.541665
140695764520512 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/752_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:38:44.555566
140695764520512 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/752_counter_collection.c
sv

INFO      |-> [rocprofiler-sdk] E20260122 20:38:44.575425
140695764520512 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/752_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260122 20:38:44.578545
140695764520512 simple_timer.cpp:55] [rocprofv3] output generation ::
0.044672 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:44.579513
140695764520512 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.045704 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 8/12] [Approximate profiling time left: 18 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/fusion_matmulbias/MI300X_A1/perfmon/pmc_perf_2.txt
INFO      |-> [rocprofiler-sdk] W20260122 20:38:45.893231
140250941754944 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128618 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:45.893330
140250941754944 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:47.157245
140250941754944 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,

```

```

43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO    |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO    |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO    |-> [rocprofiler-sdk] ...,
INFO    |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO    |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO    |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO    |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO    |-> [rocprofiler-sdk] W20260122 20:38:48.409285

140250941754944 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.515956 sec

INFO    |-> [rocprofiler-sdk] E20260122 20:38:48.422214

140250941754944 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/772_kernel_trace.csv

INFO    |-> [rocprofiler-sdk] E20260122 20:38:48.436181

140250941754944 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/772_counter_collection.c
sv

INFO    |-> [rocprofiler-sdk] E20260122 20:38:48.456042

140250941754944 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/772_agent_info.csv

INFO    |-> [rocprofiler-sdk] W20260122 20:38:48.459185

140250941754944 simple_timer.cpp:55] [rocprofv3] output generation ::
0.044737 sec

INFO    |-> [rocprofiler-sdk] W20260122 20:38:48.460124

140250941754944 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.045740 sec

INFO    |-> [rocprofiler-sdk]
INFO [Run 9/12][Approximate profiling time left: 13 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/fusion_matmulbias/MI300X_A1/perfmon/pmc_perf_3.txt

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INFO      |-> [rocprofiler-sdk] W20260122 20:38:49.776539
140630524247616 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128351 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:49.776642
140630524247616 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:51.038690
140630524247616 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

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INFO      |-> [rocprofiler-sdk] W20260122 20:38:52.276800
140630524247616 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.500158 sec

INFO      |-> [rocprofiler-sdk] E20260122 20:38:52.290179
140630524247616 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/793_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:38:52.304400
140630524247616 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/793_counter_collection.c
sv

INFO      |-> [rocprofiler-sdk] E20260122 20:38:52.325302
140630524247616 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/793_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260122 20:38:52.328410
140630524247616 simple_timer.cpp:55] [rocprofv3] output generation ::
0.046016 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:52.329370
140630524247616 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.047035 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 10/12] [Approximate profiling time left: 8 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/fusion_matmulbias/MI300X_A1/perfmon/pmc_perf_4.txt
INFO      |-> [rocprofiler-sdk] W20260122 20:38:53.647757
140424691277376 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128317 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:53.647861
140424691277376 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:54.908000
140424691277376 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,

```

```

43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO    |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO    |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO    |-> [rocprofiler-sdk] ...,
INFO    |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO    |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO    |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO    |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO    |-> [rocprofiler-sdk] W20260122 20:38:56.128044

140424691277376 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.480183 sec

INFO    |-> [rocprofiler-sdk] E20260122 20:38:56.141176

140424691277376 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/813_kernel_trace.csv

INFO    |-> [rocprofiler-sdk] E20260122 20:38:56.156233

140424691277376 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/813_counter_collection.c
sv

INFO    |-> [rocprofiler-sdk] E20260122 20:38:56.180721

140424691277376 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/813_agent_info.csv

INFO    |-> [rocprofiler-sdk] W20260122 20:38:56.183789

140424691277376 simple_timer.cpp:55] [rocprofv3] output generation ::
0.050401 sec

INFO    |-> [rocprofiler-sdk] W20260122 20:38:56.184730

140424691277376 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.051412 sec

INFO    |-> [rocprofiler-sdk]
INFO [Run 11/12] [Approximate profiling time left: 4 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/fusion_matmulbias/MI300X_A1/perfmon/pmc_perf_5.txt

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INFO      |-> [rocprofiler-sdk] W20260122 20:38:57.514901
139885200830016 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.127768 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:57.515005
139885200830016 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:38:58.773395
139885200830016 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,
43.6250, 30.9531, ..., 26.2500,  8.8203,  6.4609],
INFO      |-> [rocprofiler-sdk] [  7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609,  0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [  9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516,  9.5391, 53.3750, ...,
-9.3438,  7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260122 20:38:59.994375
139885200830016 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.479371 sec

INFO      |-> [rocprofiler-sdk] E20260122 20:39:00.007455
139885200830016 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/834_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260122 20:39:00.022558
139885200830016 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/834_counter_collection.c
sv

INFO      |-> [rocprofiler-sdk] E20260122 20:39:00.047424
139885200830016 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/834_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260122 20:39:00.050472
139885200830016 simple_timer.cpp:55] [rocprofv3] output generation ::
0.050828 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:39:00.051445
139885200830016 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.051872 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 12/12] [Approximate profiling time left: 0 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/fusion_matmulbias/MI300X_A1/perfmon/pmc_perf_6.txt
INFO      |-> [rocprofiler-sdk] W20260122 20:39:01.389153
140668741766720 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.129133 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:39:01.389256
140668741766720 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260122 20:39:02.650756
140668741766720 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] triton_output: tensor([[ 2.7480,

```

```

43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO    |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO    |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO    |-> [rocprofiler-sdk] ...,
INFO    |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO    |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO    |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO    |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO    |-> [rocprofiler-sdk] W20260122 20:39:03.852275

140668741766720 simple_timer.cpp:55] [rocprofv3] 'python triton/matmulbias.py
fusion' ::      2.463019 sec

INFO    |-> [rocprofiler-sdk] E20260122 20:39:03.865221

140668741766720 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/854_kernel_trace.csv

INFO    |-> [rocprofiler-sdk] E20260122 20:39:03.880905

140668741766720 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/854_counter_collection.c
sv

INFO    |-> [rocprofiler-sdk] E20260122 20:39:03.898839

140668741766720 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/5554ed8fd482/854_agent_info.csv

INFO    |-> [rocprofiler-sdk] W20260122 20:39:03.901898

140668741766720 simple_timer.cpp:55] [rocprofv3] output generation ::
0.044440 sec

INFO    |-> [rocprofiler-sdk] W20260122 20:39:03.902954

140668741766720 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.045576 sec

INFO    |-> [rocprofiler-sdk]
INFO [roofline] Checking for roofline.csv in
/workspace/user/workloads/fusion_matmulbias/MI300X_A1
INFO [roofline] No roofline data found. Generating...
Empirical Roofline Calculation
Copyright © 2025 Advanced Micro Devices, Inc. All rights reserved.

```



```

Total detected GPU devices: 1
GPU Device 0 (gfx942) with 304 CUs: Profiling...
  99% [||||| ]
HBM BW, GPU ID: 0, workgroupSize:256, workgroups:6225920, experiments:100,
traffic:25501368320 bytes, duration:6.0 ms, mean:4210.4 GB/sec, stdev=8.4 GB/sec
  99% [||||| ]
MALL BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:2611340115968 bytes, duration:412.6 ms, mean:6334.9 GB/sec, stdev=15.8
GB/sec
  99% [||||| ]
L2 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:1632087572480 bytes, duration:63.8 ms, mean:25601.0 GB/sec, stdev=48.5
GB/sec
  99% [||||| ]
L1 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:127506841600 bytes, duration:3.8 ms, mean:33573.5 GB/sec, stdev=294.1
GB/sec
  99% [||||| ]
LDS BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:79691776000 bytes, duration:1.3 ms, mean:63298.1 GB/sec, stdev=201.2
GB/sec
  99% [||||| ]
Peak FLOPs (FP8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:1081258016768, duration:13.0 ms, mean:82966.6 GFLOPS,
stdev=151.2 GFLOPS
  99% [||||| ]
Peak FLOPs (FP16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:530428461056, duration:13.4 ms, mean:39545.6 GFLOPS,
stdev=55.5 GFLOPS
  99% [||||| ]
Peak FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:530428461056, duration:91.763 ms, mean:5763.9 GFLOPS,
stdev=9.9 GFLOPS
  99% [||||| ]
Peak FLOPs (FP32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:265214230528, duration:2.038 ms, mean:130797.2 GFLOPS,
stdev=1509.9 GFLOPS
  99% [||||| ]
Peak FLOPs (FP64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:122406567936, duration:1.1 ms, mean:106166.8 GFLOPS,
stdev=1557.0 GFLOPS
  99% [||||| ]
Peak IOPs (INT8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:1081258016768, duration:13.1 ms, mean:82699.1 GOPS,
stdev=175.3 GOPS
  99% [||||| ]
Peak IOPs (INT32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:265214230528, duration:3.7 ms, mean:71769.6 GOPS,

```

```

stdev=718.9 GOPS
 99% [||||| ]
Peak IOPs (INT64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:122406567936, duration:6.2 ms, mean:19721.1 GOPS,
stdev=59.2 GOPS
100% [||||| ]
Peak MFMA FLOPs (F4), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:1, FLOP:0, duration:0.0 ms, mean:0.0 GFLOPS, stdev=0.0 GFLOPS
100% [||||| ]
Peak MFMA FLOPs (F6), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:1, FLOP:0, duration:0.0 ms, mean:0.0 GFLOPS, stdev=0.0 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (F8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:10200547328000, duration:5.3 ms, mean:1917747.2 GFLOPS,
stdev=16459.0 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (F16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:5100273664000, duration:5.4 ms, mean:940750.0 GFLOPS,
stdev=2868.8 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:2550136832000, duration:4.9 ms, mean:522404.8 GFLOPS,
stdev=3304.2 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (F32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:1275068416000, duration:9.1 ms, mean:139962.7 GFLOPS,
stdev=390.9 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (F64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:637534208000, duration:4.4 ms, mean:145494.9 GFLOPS,
stdev=598.7 GFLOPS
 99% [||||| ]
Peak MFMA IOPs (I8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:5100273664000, duration:5.1 ms, mean:1001899.7 GOPS,
stdev=4451.8 GOPS

```

INFO [roofline] Empirical Roofline PDFs saved!

1.5.3 Analyze the Fusion MatMulBias Kernel Profiles

Using the Triton Proton

```
[14]: !proton-viewer -m time/ms -f full profiles/fusion_matmulbias.hatchet
```

```

95.951 ROOT
 57.152 _ZN2at6native29vectorized_elementwise_kernelILi4ENS0_11F
illFunctorIiEEST5arrayIPcLm1EEEEviT0_T1_
 38.799 matmulbias_kernel [M=1024, N=1024, K=1024] [BM=128,
BN=128 BK=64]

```

90.24 - 95.95
78.81 - 90.24
67.37 - 78.81
55.94 - 67.37
44.51 - 55.94
38.80 - 44.51

Using the ROCm Systems Compute analyzer

----- /-| ----- -| |
 |' _/ _ \ / _| ' \ | ' _/ _ \ | | _ _ _ / _/ _ \ | ' _ \ \ | ' \ | | | _/ _
 \
 | | | () | (_ | |) | | | () | _| _ _ | (_ () | | | | | | | | | | | | | | |
 _/
 | _ \ _ _/ \ _ _ | . _/ | _ \ _ _/ | _ \ _ _ \ _ _/ | _ | | _ | _ | _ _/
 \ _ _ _ | \ _ _ \ _ _ |
 | _ | | _ |

0. Top Stats

0.1 Top Kernels

147

1	matmulbias_kernel	172.00	4524976.79
26308.00	24453.88 41.55		
2	__amd_roccclr_copyBuffer	89.00	269701.62
3030.36	3033.75 2.48		
3	void at::native::(anonymous namespace)::	3.00	18926.58
6308.86	7363.58 0.17		
	distribution_elementwise_grid_stride_ker		
	nel<float, 4, at::native::templates::...		
4	void at::native::(anonymous namespace)::	7.00	16096.67
2299.52	2178.50 0.15		
	CatArrayBatchedCopy_contig<at::native::(		
	anonymous namespace)::OpaqueType<2u>,...		
5	void at::native::vectorized_elementwise_	3.00	10176.34
3392.11	3384.50 0.09		
	kernel<4, at::native::compare_scalar_ker		
	nel<double>(at::TensorIteratorBase&, ...		
6	void at::native::vectorized_elementwise_	2.00	6174.33
3087.17	3087.17 0.06		
	kernel<8, at::native::AUnaryFunctor<c10:		
	:Half, c10::Half, bool, at::native::(...		
7	void at::native::vectorized_elementwise_	2.00	4971.49
2485.75	2485.75 0.05		
	kernel<8, at::native::BinaryFunctor<c10:		
	:Half, c10::Half, bool, at::native::(...		

8	void at::native::vectorized_elementwise_	2.00	4914.50
2457.25	2457.25 0.05		
	kernel<8, at::native::AbsFunctor<c10::Ha		
	lf>, std::array<char*, 2ul> >(int, at...		
9	void rocprim::ROCPRIM_400001_NS::detail:	1.00	4236.25
4236.25	4236.25 0.04		
	:partition_kernel<(rocprim::ROCPRIM_4000		
	01_NS::detail::select_method)0, true,...		

0.2 Dispatch List

	Dispatch_ID	Kernel_Name	
	GPU_ID		
0	0	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*):...	0
1	1	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*):...	0
2	2	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4,	0

```

at::native::templates::cuda::normal_and_transform<c10::Half, float
    , at::CUDAGeneratorImpl*,
at::native::templates::cuda::normal_kernel<at::CUDAGen
    eratorImpl*>(at::TensorBase const&, double, double,
at::CUDAGeneratorImpl*):...

```

```

3          3  matmulbias_kernel
0

```

```

4          4  void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
    std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
    l>)

```

```

5          5  matmulbias_kernel
0

```

```

6          6  void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
    std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
    l>)

```

```

7          7  matmulbias_kernel
0

```

```

8          8  void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
    std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
    l>)

```

```

9          9  matmulbias_kernel
0

```

1. System Info

	Info
workload_name	fusion_matmulbias
command	python triton/matmulbias.py fusion
ip_blocks	SQ LDS SQC TA TD TCP TCC SPI CPC CPF roofline
timestamp	Thu Jan 22 20:38:10 2026 (UTC)
version	3
hostname	5554ed8fd482
cpu_model	Intel(R) Xeon(R) Platinum 8470
sbios	SeaBIOS1.15.0-1
linux_distro	CentOS Stream 10 (Coughlan)
linux_kernel_version	5.15.0-164-generic
amd_gpu_kernel_version	nan
cpu_memory	230893520
gpu_memory	nan
rocm_version	7.0.3
vbios	113-M3000108-103
compute_partition	SPX
memory_partition	nan
gpu_series	MI300
gpu_model	MI300X_A1
gpu_arch	gfx942
gpu_chip_id	29877

gpu_l1	32
gpu_l2	4096
cu_per_gpu	304
simd_per_cu	4
se_per_gpu	32
wave_size	64
workgroup_max_size	1024
max_waves_per_cu	32
max_sclk	2100
max_mclk	1300
cur_sclk	2100
cur_mclk	1300
total_l2_chan	128
lds_banks_per_cu	32
sqc_per_gpu	160
pipes_per_gpu	4
num_xcd	8
num_hbm_channels	128

2. System Speed-of-Light

2.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit	Peak
Pct of Peak				
2.1.0	VALU FLOPs	34.92	Gflop/s	

81715.2	0.04			
2.1.1	VALU IOPs	542.72	Giop/s	
81715.2	0.66			
2.1.2	MFMA FLOPs (F8)	0.0	Gflop/s	
2614886.4	0.0			
2.1.3	MFMA FLOPs (BF16)	0.0	Gflop/s	
1307443.2	0.0			
2.1.4	MFMA FLOPs (F16)	33195.12	Gflop/s	
1307443.2	2.54			
2.1.5	MFMA FLOPs (F32)	0.0	Gflop/s	
163430.4	0.0			
2.1.6	MFMA FLOPs (F64)	0.0	Gflop/s	
163430.4	0.0			
2.1.7	MFMA FLOPs (F6F4)		Gflop/s	
2.1.8	MFMA IOPs (Int8)	0.0	Giop/s	
2614886.4	0.0			
2.1.9	Active CUs	103.0	Cus	304.0
33.88				
2.1.10	SALU Utilization	1.3	Pct	100.0
1.3				
2.1.11	VALU Utilization	2.14	Pct	100.0
2.14				
2.1.12	MFMA Utilization	1.95	Pct	100.0

1.95					
2.1.13 0.34	VMEM Utilization	0.34	Pct	100.0	
2.1.14 0.25	Branch Utilization	0.25	Pct	100.0	
2.1.15 98.12	VALU Active Threads	62.8	Threads	64.0	
2.1.16 4.66	IPC	0.23	Instr/cycle	5.0	
2.1.17 36.64	Wavefront Occupancy	3564.05	Wavefronts	9728.0	
2.1.18 81715.2	Theoretical LDS Bandwidth 2.43	1982.13	Gb/s		
2.1.19 0.0	LDS Bank Conflicts/Access	0.0	Conflicts/access	32.0	
2.1.20 26.03	vL1D Cache Hit Rate	26.03	Pct	100.0	
2.1.21 81715.2	vL1D Cache BW 6.84	5590.05	Gb/s		
2.1.22 52.07	L2 Cache Hit Rate	52.07	Pct	100.0	
2.1.23 34406.4	L2 Cache BW 13.83	4760.05	Gb/s		
2.1.24	L2-Fabric Read BW	293.2	Gb/s	5324.8	

5.51

2.1.25 37.86	L2-Fabric Write BW	2015.73	Gb/s	5324.8
2.1.26	L2-Fabric Read Latency	389.41	Cycles	
2.1.27	L2-Fabric Write Latency	454.42	Cycles	
2.1.28 86.0	sL1D Cache Hit Rate	86.0	Pct	100.0
2.1.29 21504.0	sL1D Cache BW	34.01	Gb/s	
	0.16			
2.1.30 99.36	L1I Hit Rate	99.36	Pct	100.0
2.1.31 21504.0	L1I BW	464.61	Gb/s	
	2.16			
2.1.32	L1I Fetch Latency		Cycles	

3. Memory Chart

3.1 Memory Chart

(Normalization: per_kernel)

Instr Buff	Instr Dispatch	Exec
LDS	L2 Cache	

SALU: 350676

		----->	Active
CUs	Req : 76033	Util : 2 %	
	xGMI/PCIE		

Wave	0	Instr Buf	10
3: 304	<-----		

	SMEM:	
21252	Lat :	
N/A	Hit : 52	
%		

>		
	~	
	RVGPRseq:	44.0

Wave N-1	Instr Buf	VALU:	
509268		Vector L1 Cache	
Request			

>			
v			
	SGPRs :	40	

	MFMA:	
52186	Rd : 887173	Hit : 26 %
: 104423	Rd : 107922	Rd

```

> <-----<-----
-
Fabric

Wt      : 5464985 Lat      :   N/A      Wr      : 1392186 Wr      :
1392186      Rd      :
59193

VMEM: 99310      LDS Alloc:      -
----->
> <----- Latency (cycles)      Rd:
59195

----->
71 Atomic:      0 Coales:      44 %      Atomic:      0 Atomic:
0 Wr      : 1353046 <----- HBM
<
-----> <-----
> -----> Rd      :      389      Wr:
1353046

LDS:
76033 Stall :      25
cycles Atomic:      0 Wr      :
454 ----->
-----> Scratch Alloc:

-----> Atomic:      0
0

Wave Occupancy GWS: 0
Scalar L1D
Cache Latency (cycles)

35 -----
---> Rd      :
53

Wave Life
<-----
^

|
23554
BRANCH: 45537 Wavefronts: Rd      : 21297 Hit      : 79.0
% Wr      : 0 Rd      : N/A |
|
-----> 0 <
-----
> |
|

```

Lat : N/A Atomic: 0 Wr :
N/A |

v

^

<-----

>

|

Workgroups:

|

0

Instr L1 Cache
GMI

|

|

|

Fetch :

236908

Hit : 96 % Req

: 3449

<-----

Lat : 2258.0 cycles

4. Roofline

4.1 Roofline

Roofline (FP32) -
/workspace/user/workloads/fusion_matmulbias/MI300X_A1

139962.7

139963GFLOP/s

36229.6

9378.1

2427.5

628.4 63298 GB/s

33574 GB/s

162.7

42.1 4210 GB/s

0.01

0.18

3.16

56.23

1000.00

Performance (GFLOP/sec) Arithmetic Intensity (FLOP)s/Byte)

5. Command Processor (CPC/CPF)

5.1 Command Processor Fetcher

Metric_ID	Metric	Avg	Min	Max	Unit
5.1.0	CPF Utilization	100.00	100.00	100.00	Pct
5.1.1	CPF Stall	0.62	0.06	5.27	Pct
5.1.2	CPF-L2 Utilization	12.01	2.68	52.31	Pct
5.1.3	CPF-L2 Stall	0.00	0.00	0.00	Pct
5.1.4	CPF-UTCL1 Stall	0.77	0.06	5.49	Pct

5.2 Packet Processor

Metric_ID	Metric	Avg	Min	Max	Unit
5.2.0	CPC SYNC FIFO Full Rate				Pct
5.2.1	CPC CANE Stall Rate				Pct
5.2.2	CPC ADC Utilization				Pct
5.2.3	CPC Utilization	100.0	100.0	100.0	Pct
5.2.4	CPC Stall Rate	23.94	7.2	44.66	Pct

5.2.5	CPC Packet Decoding Utilization	67.67	24.04	107.87	Pct
5.2.6	CPC-Workgroup Manager Utilization	22.09	0.51	70.95	Pct
5.2.7	CPC-L2 Utilization	0.47	0.16	1.39	Pct
5.2.8	CPC-UTCL1 Stall	0.34	0.11	5.07	Pct
5.2.9	CPC-UTCL2 Utilization	0.33	0.04	5.07	Pct

6. Workgroup Manager (SPI)

6.1 Workgroup Manager Utilizations

Metric_ID	Metric	Avg	Min	Max	Unit
6.1.0	Schedule-Pipe Wave Occupancy				Wave
6.1.1	Accelerator Utilization	100.73	80.84	390.66	Pct
6.1.2	Scheduler-Pipe Utilization	0.0	0.0	0.0	Pct
6.1.3	Scheduler-Pipe Wave Utilization				Pct
6.1.4	Workgroup Manager Utilization	57.82	6.25	83.04	Pct

6.1.5	Shader Engine Utilization	53.82	0.25	80.52	Pct
6.1.6	SIMD Utilization	31.49	0.01	319.43	Pct
6.1.7 Workgroups	Dispatched Workgroups	0.0	0.0	0.0	
6.1.8 Wavefronts	Dispatched Wavefronts	0.0	0.0	0.0	
6.1.9 Cycles/wave	VGPR Writes				
6.1.10 Cycles/wave	SGPR Writes				

6.2 Workgroup Manager - Resource Allocation

Metric_ID Unit	Metric	Avg	Min	Max
6.2.0 Pct	Not-scheduled Rate (Workgroup Manager)	0.0	0.0	0.0
6.2.1 Pct	Not-scheduled Rate (Scheduler-Pipe)	0.0	0.0	0.0
6.2.2 Pct	Scheduler-Pipe FIFO Full Rate			
6.2.3 Pct	Scheduler-Pipe Stall Rate	0.0	0.0	0.0

6.2.4 Pct	Scratch Stall Rate	0.0	0.0	0.0
6.2.5 Pct	Insufficient SIMD Waveslots	0.0	0.0	0.0
6.2.6 Pct	Insufficient SIMD VGPRs	0.0	0.0	0.0
6.2.7 Pct	Insufficient SIMD SGPRs	0.0	0.0	0.0
6.2.8 Pct	Insufficient CU LDS	0.0	0.0	0.0
6.2.9 Pct	Insufficient CU Barriers	0.0	0.0	0.0
6.2.10 Pct	Reached CU Workgroup Limit	0.0	0.0	0.0
6.2.11 Pct	Reached CU Wavefront Limit	0.0	0.0	0.0

7. Wavefront

7.1 Wavefront Launch Stats

Metric_ID	Metric	Avg	Min	Max	Unit
7.1.0 items	Grid Size	1354433.93	32.00	4194304.00	Work

7.1.1 items	Workgroup Size	429.78	32.00	1024.00	Work
7.1.2 Wavefronts	Total Wavefronts	0.00	0.00	0.00	
7.1.3 Wavefronts	Saved Wavefronts	0.00	0.00	0.00	
7.1.4 Wavefronts	Restored Wavefronts	0.00	0.00	0.00	
7.1.5 Registers	VGPRs	43.68	8.00	108.00	
7.1.6 Registers	AGPRs	20.45	0.00	192.00	
7.1.7 Registers	SGPRs	39.56	16.00	112.00	
7.1.8	LDS Allocation	71.11	0.00	29184.00	Bytes
7.1.9 Bytes/workitem	Scratch Allocation	0.00	0.00	0.00	

7.2 Wavefront Runtime Stats

Metric_ID	Metric	Avg	Min
Max Unit			
7.2.0 65816.16 Ns	Kernel Time (Nanosec)	25210.32	1710.41

7.2.1	Kernel Time (Cycles)	70130.45	17532.75
329610.50	Cycle		
7.2.2	Instructions per wavefront	1091.01	19.50
3797.00	Instr/wavefront		
7.2.3	Wave Cycles	223986349.06	5668.00
916594400.00	Cycles per kernel		
7.2.4	Dependency Wait Cycles	52832719.77	4356.00
200490988.00	Cycles per kernel		
7.2.5	Issue Wait Cycles	166837944.58	8.00
715126636.00	Cycles per kernel		
7.2.6	Active Cycles	4738371.07	296.00
16351744.00	Cycles per kernel		
7.2.7	Wavefront Occupancy	3564.05	0.17
38005.81	Wavefronts		

10. Compute Units - Instruction Mix

10.1 Overall Instruction Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.1.0	VALU	457082.06	20.00	2731520.00	Instr per kernel
10.1.1	VMEM	99310.09	2.00	262144.00	Instr per kernel
10.1.2	LDS	76032.60	0.00	212992.00	Instr per kernel
10.1.3	MFMA	52186.07	0.00	131072.00	Instr per kernel
10.1.4	SALU	350675.87	24.00	924160.00	Instr per kernel
10.1.5	SMEM	21251.87	0.00	68096.00	Instr per kernel

10.1.6	Branch	45536.73	10.00	131072.00	Instr per kernel
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10.2 VALU Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.2.0	INT32	240579.02	9.00	393216.00	Instr per kernel
10.2.1	INT64	24626.34	0.00	668160.00	Instr per kernel
10.2.2	F16-ADD	4892.44	0.00	12288.00	Instr per kernel
10.2.3	F16-MUL	0.00	0.00	0.00	Instr per kernel
10.2.4	F16-FMA	75.89	0.00	16384.00	Instr per kernel
10.2.5	F16-Trans	0.00	0.00	0.00	Instr per kernel
10.2.6	F32-ADD	180.30	0.00	38912.00	Instr per kernel
10.2.7	F32-MUL	1292.30	0.00	123392.00	Instr per kernel
10.2.8	F32-FMA	540.89	0.00	116736.00	Instr per kernel
10.2.9	F32-Trans	1021.85	0.00	65024.00	Instr per kernel
10.2.10	F64-ADD	0.00	0.00	0.00	Instr per kernel
10.2.11	F64-MUL	0.00	0.00	1.00	Instr per kernel
10.2.12	F64-FMA	0.01	0.00	6.00	Instr per kernel
10.2.13	F64-Trans	0.00	0.00	1.00	Instr per kernel
10.2.14	Conversion	8234.89	0.00	58368.00	Instr per kernel

10.3 VMEM Instr Mix

Metric_ID	Metric	Avg	Min	Max
Unit				
10.3.0	Global/Generic Instr	83817.35	0.0	262144.0
Instr per kernel				
10.3.1	Global/Generic Read	0.51	0.0	6.0

Instr per kernel

10.3.2	Global/Generic Write	83816.84	0.0	262144.0
Instr per kernel				

10.3.3	Global/Generic Atomic	0.0	0.0	0.0
Instr per kernel				

10.3.4	Spill/Stack Instr	15492.74	0.0	38912.0
Instr per kernel				

10.3.5	Spill/Stack Coalesceable Instr			
Instr per kernel				

10.3.6	Spill/Stack Read	13861.93	0.0	34816.0
Instr per kernel				

10.3.7	Spill/Stack Write	1630.81	0.0	4096.0
Instr per kernel				

10.3.8	Spill/Stack Atomic	0.0	0.0	0.0
Instr per kernel				

10.4 MFMA Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.4.0	MFMA-I8	0.0	0.0	0.0	Instr per kernel
10.4.1	MFMA-F8	0.0	0.0	0.0	Instr per kernel
10.4.2	MFMA-F16	52186.07	0.0	131072.0	Instr per kernel
10.4.3	MFMA-BF16	0.0	0.0	0.0	Instr per kernel
10.4.4	MFMA-F32	0.0	0.0	0.0	Instr per kernel
10.4.5	MFMA-F64	0.0	0.0	0.0	Instr per kernel
10.4.6	MFMA-F6F4				Instr per kernel

11. Compute Units - Compute Pipeline

11.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit	Peak	Pct of Peak
11.1.0	VALU FLOPs	34.92	Gflop	81715.2	0.04
11.1.1	VALU IOPs	542.72	Giop	81715.2	0.66
11.1.2	MFMA FLOPs (F8)	0.0	Gflop	2614886.4	0.0
11.1.3	MFMA FLOPs (BF16)	0.0	Gflop	1307443.2	0.0
11.1.4	MFMA FLOPs (F16)	33195.12	Gflop	1307443.2	2.54
11.1.5	MFMA FLOPs (F32)	0.0	Gflop	163430.4	0.0
11.1.6	MFMA FLOPs (F64)	0.0	Gflop	163430.4	0.0
11.1.7	MFMA FLOPs (F6F4)		Gflop		
11.1.8	MFMA IOPs (INT8)	0.0	Giop	2614886.4	0.0

11.2 Pipeline Stats

Metric_ID	Metric	Avg	Min	Max	Unit
11.2.0	IPC	0.23	0.02	1.27	Instr/cycle
11.2.1	IPC (Issued)	0.95	0.86	1.0	Instr/cycle
11.2.2	SALU Utilization	1.3	0.0	11.26	Pct
11.2.3	VALU Utilization	2.14	0.0	33.4	Pct
11.2.4	VALU Co-Issue Efficiency				Pct
11.2.5	VMEM Utilization	0.34	0.0	0.85	Pct
11.2.6	Branch Utilization	0.25	0.0	1.55	Pct
11.2.7	VALU Active Threads	62.8	28.46	64.0	Threads
11.2.8	MFMA Utilization	1.95	0.0	5.68	Pct
11.2.9	MFMA Instr Cycles	32.0	32.0	32.0	Cycles/instr
11.2.10	VMEM Latency				Cycles

11.2.11	SMEM Latency	Cycles
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11.3 Arithmetic Operations

Metric_ID	Metric	Avg	Min	Max	Unit
11.3.0 kernel	FLOPs (Total)	855568348.15	0.0	2148532224.0	Ops per
11.3.1 kernel	IOPs (Total)	16973142.96	960.0	59179008.0	Ops per
11.3.2 kernel	F8 OPs	0.0	0.0	0.0	Ops per
11.3.3 kernel	F16 OPs	855339467.85	0.0	2148270080.0	Ops per
11.3.4 kernel	BF16 OPs	0.0	0.0	0.0	Ops per
11.3.5 kernel	F32 OPs	228878.22	0.0	29491200.0	Ops per
11.3.6 kernel	F64 OPs	2.07	0.0	896.0	Ops per
11.3.7 kernel	F6F4 OPs				Ops per
11.3.8 kernel	INT8 OPs	0.0	0.0	0.0	Ops per

12. Local Data Share (LDS)

12.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
12.1.0	Utilization	1.86	Pct of peak
12.1.1	Access Rate	1.22	Pct of peak
12.1.2	Theoretical Bandwidth (% of Peak)	2.43	Pct of peak
12.1.3	Bank Conflict Rate	0.00	Pct of peak

12.2 LDS Stats

Metric_ID	Metric	Avg	Min	Max
Unit				
12.2.0	LDS Instrs	76032.6	0.0	212992.0
Instr per kernel				
12.2.1	LDS LOAD			
Instr per kernel				
12.2.2	LDS STORE			
Instr per kernel				
12.2.3	LDS ATOMIC			
Instr per kernel				
12.2.4	LDS LOAD Bandwidth			
Gbps				
12.2.5	LDS STORE Bandwidth			
Gbps				

12.2.6	LDS ATOMIC Bandwidth Gbps			
12.2.7	Theoretical Bandwidth Bytes per kernel	50331929.48	0.0	134217728.0
12.2.8	LDS Latency Cycles			
12.2.9	Bank Conflicts/Access Conflicts/access	0.0	0.0	0.15
12.2.10	Index Accesses Cycles per kernel	393218.46	0.0	1048576.0
12.2.11	Atomic Return Cycles Cycles per kernel	0.0	0.0	0.0
12.2.12	Bank Conflict Cycles per kernel	0.26	0.0	112.0
12.2.13	Addr Conflict Cycles per kernel	0.06	0.0	24.0
12.2.14	Unaligned Stall Cycles per kernel	0.0	0.0	0.0
12.2.15	Mem Violations Accesses per kernel	0.0	0.0	0.0
12.2.16	LDS Command FIFO Full Rate Cycles per kernel			
12.2.17	LDS Data FIFO Full Rate Cycles per kernel			

13. Instruction Cache

13.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
13.1.0	Bandwidth	2.16	Pct of peak
13.1.1	Cache Hit Rate	95.63	Pct of peak
13.1.2	L1I-L2 Bandwidth	0.02	Pct of peak

13.2 Instruction Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
13.2.0 per kernel	Req	236908.24	18.0	778240.0	Req
13.2.1 per kernel	Hits	235477.27	14.0	763511.0	Hits
13.2.2 Misses per kernel	Misses - Non Duplicated	101.49	1.0	480.0	
13.2.3 Misses per kernel	Misses - Duplicated	1332.25	1.0	18053.0	
13.2.4	Cache Hit Rate	95.63	77.78	99.66	Pct
13.2.5 Cycles	Instruction Fetch Latency				

13.3 Instruction Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
13.3.0	L1I-L2 Bandwidth	220706.67	1024.00	645120.00	Bytes per kernel

14. Scalar L1 Data Cache

14.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
14.1.0	Bandwidth	0.16	Pct of peak
14.1.1	Cache Hit Rate	78.52	Pct of peak
14.1.2	sL1D-L2 BW	0.00	Pct of peak

14.2 Scalar L1D Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
14.2.0	Req	21296.96	0.00	77824.00	Req per kernel
14.2.1	Hits	19738.62	0.00	67914.00	Req per kernel
14.2.2	Misses - Non Duplicated	52.91	0.00	320.00	Req per kernel
14.2.3	Misses- Duplicated	1508.34	0.00	9600.00	Req per kernel
14.2.4	Cache Hit Rate	78.52	0.00	93.28	Pct

14.2.5 kernel	Read Req (Total)	21251.87	0.00	68096.00	Req per
14.2.6 kernel	Atomic Req	0.00	0.00	0.00	Req per
14.2.7 kernel	Read Req (1 DWord)	135.92	0.00	29184.00	Req per
14.2.8 kernel	Read Req (2 DWord)	90.49	0.00	19456.00	Req per
14.2.9 kernel	Read Req (4 DWord)	21025.41	0.00	65536.00	Req per
14.2.10 kernel	Read Req (8 DWord)	0.04	0.00	8.00	Req per
14.2.11 kernel	Read Req (16 DWord)	0.00	0.00	0.00	Req per

14.3 Scalar L1D Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
14.3.0	sL1D-L2 BW	3385.93	0.00	20480.00	Bytes per kernel
14.3.1	Read Req	52.91	0.00	320.00	Req per kernel
14.3.2	Write Req	0.00	0.00	0.00	Req per kernel
14.3.3	Atomic Req	0.00	0.00	0.00	Req per kernel
14.3.4	Stall Cycles	0.00	0.00	0.00	Cycles per kernel

15. Address Processing Unit and Data Return Path (TA/TD)

15.1 Address Processing Unit

Metric_ID Max	Metric Unit	Avg	Min
15.1.0 90.5	Address Processing Unit Busy Pct	27.07	0.0
15.1.1 6.58	Address Stall Pct	1.85	0.0
15.1.2 67.87	Data Stall Pct	20.34	0.0
15.1.3 0.0	Data-Processor → Address Stall Pct	0.0	0.0
15.1.4	Sequencer → TA Address Stall Cycles per kernel		
15.1.5	Sequencer → TA Command Stall Cycles per kernel		
15.1.6	Sequencer → TA Data Stall Cycles per kernel		
15.1.7 262144.0	Total Instructions Instructions per kernel	99310.09	2.0
15.1.8 262144.0	Global/Generic Instructions Instructions per kernel	83817.35	0.0
15.1.9 6.0	Global/Generic Read Instructions Instructions per kernel	0.51	0.0
15.1.10	Global/Generic Read Instructions for LDS		

Instructions per kernel

15.1.11	Global/Generic Write Instructions	83816.84	0.0
262144.0	Instructions per kernel		
15.1.12	Global/Generic Atomic Instructions	0.0	0.0
0.0	Instructions per kernel		
15.1.13	Spill/Stack Instructions	15492.74	0.0
38912.0	Instructions per kernel		
15.1.14	Spill/Stack Read Instructions	13861.93	0.0
34816.0	Instructions per kernel		
15.1.15	Spill/Stack Read Instructions for LDS		
	Instructions per kernel		
15.1.16	Spill/Stack Write Instructions	1630.81	0.0
4096.0	Instructions per kernel		
15.1.17	Spill/Stack Atomic Instructions	0.0	0.0
0.0	Instructions per kernel		
15.1.18	Spill/Stack Total Cycles	247883.85	0.0
622592.0	Cycles per kernel		
15.1.19	Spill/Stack Coalesced Read	0.0	0.0
0.0	Cycles per kernel		
15.1.20	Spill/Stack Coalesced Write	0.0	0.0
0.0	Cycles per kernel		

15.2 Data-Return Path

Metric_ID	Metric	Avg	Min	Max
Unit				

15.2.0 Pct	Data-Return Busy	29.16	0.01	79.07
15.2.1 Pct	Cache RAM → Data-Return Stall	27.82	0.0	78.43
15.2.2 Pct	Workgroup manager → Data-Return Stall	0.0	0.0	0.0
15.2.3 16384.0	Coalescable Instructions Instructions per kernel	76.85	0.0	
15.2.4 34816.0	Read Instructions Instructions per kernel	13862.44	0.0	
15.2.5 262144.0	Write Instructions Instructions per kernel	85447.66	1.0	
15.2.6	Atomic Instructions Instructions per kernel	0.0	0.0	0.0
15.2.7	Write Ack Instructions Instructions per kernel			

INFO Not showing table with empty column(s): 16.6 L1D Addr Translation Stalls

16. Vector L1 Data Cache

16.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
16.1.0	Hit rate	26.03	Pct of peak
16.1.1	Bandwidth	6.84	Pct of peak

16.1.2	Utilization	37.07	Pct of peak
16.1.3	Coalescing	44.33	Pct of peak

16.2 L1D Cache Stalls (%)

Metric_ID	Metric	Min	Q1	Median	Q3	Max
16.2.0 98.16	Stalled on L2 Data	0.02	0.28	13.46	92.92	
16.2.1 82.82	Stalled on L2 Req	0.0	0.0	0.0	76.68	
16.2.2	Stalled on Address					
16.2.3	Stalled on Data					
16.2.4	Stalled on Latency FIFO					
16.2.5	Stalled on Request FIFO					
16.2.6	Stalled on Read Return					
16.2.7	Tag RAM Stall (Read)	0.0	0.0	0.0	0.0	0.0
16.2.8	Tag RAM Stall (Write)	0.0	0.0	0.0	1.26	1.36
16.2.9	Tag RAM Stall (Atomic)	0.0	0.0	0.0	0.0	0.0

16.3 L1D Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
16.3.0 per kernel	Total Req	6352157.65	32.0	16777216.0	Req
16.3.1 per kernel	Read Req	887173.02	0.0	2228224.0	Req
16.3.2 per kernel	Write Req	5464984.63	16.0	16777216.0	Req
16.3.3 per kernel	Atomic Req	0.0	0.0	0.0	Req
16.3.4 Bytes per kernel	Cache BW	213288314.67	256.0	536870912.0	
16.3.5	Cache Hit Rate	26.03	-29.63	84.09	Pct
16.3.6 per kernel	Cache Accesses	1666314.96	2.0	4194304.0	Req
16.3.7 per kernel	Cache Hits	169705.96	-8.0	425856.0	Req
16.3.8 per kernel	Invalidations	124.31	1.0	304.0	Req
16.3.9 Bytes per kernel	L1-L2 BW	102466076.15	192.0	268435456.0	

16.3.10 per kernel	Tag RAM 0 Req					Req
16.3.11 per kernel	Tag RAM 1 Req					Req
16.3.12 per kernel	Tag RAM 2 Req					Req
16.3.13 per kernel	Tag RAM 3 Req					Req
16.3.14 per kernel	L1-L2 Read	104423.44	0.0	262272.0		Req
16.3.15 per kernel	L1-L2 Write	1392185.56	1.0	4194304.0		Req
16.3.16 per kernel	L1-L2 Atomic	0.0	0.0	0.0		Req
16.3.17 Cycles	L1 Access Latency					
16.3.18 Cycles	L1-L2 Read Latency					
16.3.19 Cycles	L1-L2 Write Latency					

16.4 L1D - L2 Transactions

Metric_ID	Metric	Xfer	Coherency	Avg	Min
Max Unit					

16.4.0	NC - Read	Read	NC	0.00	0.00
0.00	Req per kernel				
16.4.1	UC - Read	Read	UC	0.00	0.00
0.00	Req per kernel				
16.4.2	CC - Read	Read	CC	0.00	0.00
0.00	Req per kernel				
16.4.3	RW - Read	Read	RW	104423.44	0.00
262272.00	Req per kernel				
16.4.4	RW - Write	Write	RW	1392185.17	0.00
4194304.00	Req per kernel				
16.4.5	NC - Write	Write	NC	0.00	0.00
0.00	Req per kernel				
16.4.6	UC - Write	Write	UC	0.39	0.00
6.00	Req per kernel				
16.4.7	CC - Write	Write	CC	0.00	0.00
0.00	Req per kernel				
16.4.8	NC - Atomic	Atomic	NC	0.00	0.00
0.00	Req per kernel				
16.4.9	UC - Atomic	Atomic	UC	0.00	0.00
0.00	Req per kernel				
16.4.10	CC - Atomic	Atomic	CC	0.00	0.00
0.00	Req per kernel				
16.4.11	RW - Atomic	Atomic	RW	0.00	0.00
0.00	Req per kernel				

16.5 L1D Addr Translation

Metric_ID Units	Metric	Avg	Min	Max
16.5.0 Req per kernel	Req	1666314.94	2.0	4194304.0
16.5.1 Req per kernel	Inflight Req			
16.5.2 Pct	Hit Ratio	82.05	0.0	100.0
16.5.3 Req per kernel	Hits	1554645.25	0.0	3863915.0
16.5.4 Req per kernel	Translation Misses	5316.5	0.0	16383.0
16.5.5 Req per kernel	Misses under Translation Miss			
16.5.6 Req per kernel	Permission Misses	0.36	0.0	128.0

INFO Not showing table with empty column(s): 17.4 L2 Cache Stalls

17. L2 Cache

17.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
17.1.0	Utilization	48.60	Pct
17.1.1	Bandwidth	13.83	Pct

17.1.2	Hit Rate	52.07	Pct	
17.1.3	L2-Fabric Read BW	293.20	Gb/s	
17.1.4	L2-Fabric Write and Atomic BW	2015.73	Gb/s	
17.1.5	HBM Bandwidth	5324.80	Gb/s	
17.2 L2 - Fabric Transactions				
Metric_ID Unit	Metric	Avg	Min	Max
17.2.0 18984704.0	Read BW Bytes per kernel	7549017.93	1536.0	
17.2.1 Pct	HBM Read Traffic	100.65	72.41	108.5
17.2.2 Pct	Remote Read Traffic	0.37	0.0	27.59
17.2.3 102.86	Uncached Read Traffic Pct	4.17	0.0	
17.2.4 268435456.0	Write and Atomic BW Bytes per kernel	86594925.19	32.0	
17.2.5 Pct	HBM Write and Atomic Traffic	79.4	0.0	100.0
17.2.6 Pct	Remote Write and Atomic Traffic	20.6	0.0	100.0
17.2.7 Pct	Atomic Traffic	0.0	0.0	0.0

17.2.8	Uncached Write and Atomic Traffic	20.63	0.0	100.0	
Pct					
17.2.9	Read Latency	389.41	136.47	690.9	
Cycles					
17.2.10	Write and Atomic Latency	454.42	111.0	842.0	
Cycles					
17.2.11	Atomic Latency				
Cycles					
17.2.12	Read Stall				
Pct					
17.2.13	Write Stall				
Pct					
17.3 L2 Cache Accesses					
Metric_ID	Metric	Avg	Min	Max	Unit
17.3.0	Bandwidth	192009912.0	3072.0	537406080.0	
Bytes per kernel					
17.3.1	Read Bandwidth				
Bytes per kernel					
17.3.2	Write Bandwidth				
Bytes per kernel					
17.3.3	Atomic Bandwidth				
Bytes per kernel					
17.3.4	Req	1500077.44	24.0	4198485.0	Req

per kernel

17.3.5 per kernel	Read Req	107922.39	23.0	272008.0	Req
17.3.6 per kernel	Write Req	1392185.56	1.0	4194304.0	Req
17.3.7 per kernel	Atomic Req	0.0	0.0	0.0	Req
17.3.8 per kernel	Streaming Req	0.01	0.0	3.0	Req
17.3.9 per kernel	Bypassss Req				Req
17.3.10 per kernel	Probe Req	0.05	0.0	4.0	Req
17.3.11 per kernel	Input Buffer Req				Req
17.3.12	Cache Hit	52.07	16.67	61.41	Pct
17.3.13 per kernel	Hits	764626.6	10.0	2101048.0	Hits
17.3.14 Misses per kernel	Misses	735506.0	13.0	2097792.0	
17.3.15 Cachelines per kernel	Writeback	676587.5	6.0	2097208.0	
17.3.16	Writeback (Internal)	628053.31	0.0	1966080.0	

Cachelines per kernel

17.3.17	Writeback (vL1D Req)	48469.48	0.0	131072.0
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Cachelines per kernel

17.3.18	Evict (Internal)	586218.57	0.0	1835123.0
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Cachelines per kernel

17.3.19	Evict (vL1D Req)	0.0	0.0	0.0
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Cachelines per kernel

17.3.20	NC Req	0.0	0.0	0.0	Req
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per kernel

17.3.21	UC Req	3.42	0.0	185.0	Req
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per kernel

17.3.22	CC Req	0.0	0.0	0.0	Req
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per kernel

17.3.23	RW Req	1500102.14	24.0	4198304.0	Req
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per kernel

17.5 L2 - Fabric Interface Stalls

Metric_ID	Metric	Min	Max	Unit	Type
Transaction	Avg				
17.5.0	Read - PCIe Stall Pct				PCIe Stall Read
17.5.1	Read - Infinity Fabric Stall Pct				Infinity Fabric Stall Read
17.5.2	Read - HBM Stall Pct				HBM Stall Read

17.5.3	Write - PCIe Stall Pct	PCIe Stall	Write
17.5.4	Write - Infinity Fabric Stall Pct	Infinity Fabric Stall	Write
17.5.5	Write - HBM Stall Pct	HBM Stall	Write
17.5.6	Write - Credit Starvation 0.0 0.0 0.0 Pct	Credit Starvation	Write

17.6 L2 - Fabric Detailed Transaction Breakdown

Metric_ID Unit	Metric	Avg	Min	Max
17.6.0 Req per kernel	Read (32B)	0.0	0.0	0.0
17.6.1 1468.0 Req per kernel	Read (64B)	432.16	22.0	
17.6.2 147584.0 Req per kernel	Read (128B)	58760.62	0.0	
17.6.3 Req per kernel	Read (Uncached)	11.45	0.0	366.0
17.6.4 149044.0 Req per kernel	HBM Read	59195.47	23.0	
17.6.5 Req per kernel	Remote Read	1.04	0.0	16.0

17.6.6	Read Bandwidth - PCIe			
	Bytes per kernel			
17.6.7	Read Bandwidth - Infinity Fabric			
	Bytes per kernel			
17.6.8	Read Bandwidth - HBM			
	Bytes per kernel			
17.6.9	Write and Atomic (32B)	0.48	0.0	6.0
	Req per kernel			
17.6.10	Write and Atomic (Uncached)	0.4	0.0	6.0
	Req per kernel			
17.6.11	Write and Atomic (64B)	1353045.47	0.0	
4194304.0	Req per kernel			
17.6.12	HBM Write and Atomic	1353045.55	0.0	
4194304.0	Req per kernel			
17.6.13	Remote Write and Atomic	0.39	0.0	6.0
	Req per kernel			
17.6.14	Write Bandwidth - PCIe			
	Bytes per kernel			
17.6.15	Write Bandwidth - Infinity Fabric			
	Bytes per kernel			
17.6.16	Write Bandwidth - HBM			
	Bytes per kernel			
17.6.17	Atomic	0.0	0.0	0.0
	Req per kernel			

17.6.18 Atomic - HBM
Req per kernel

17.6.19 Atomic Bandwidth - PCIe
Bytes per kernel

17.6.20 Atomic Bandwidth - Infinity Fabric
Bytes per kernel

17.6.21 Atomic Bandwidth - HBM
Bytes per kernel

INFO Not showing table with empty column(s): 18.9 L2-Fabric Read
Stall (Cycles per normUnit)

INFO Not showing table with empty column(s): 18.10 L2-Fabric Write
and Atomic Stall (Cycles per normUnit)

18. L2 Cache (per Channel)

18.1 Aggregate Stats (All channels)

Metric_ID	Metric	Avg	Std Dev	Min	Max	Unit
18.1.0	L2 Cache Hit Rate	52.00	12.92	0.00	61.41	Pct

18.2 L2 Cache Hit Rate (pct)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.2.0	0.00	0.00	50.00	58.74	58.85	61.31
18.2.1	1.00	0.00	50.00	58.48	58.59	61.31
18.2.2	2.00	0.00	50.00	58.74	58.85	61.31
18.2.3	3.00	49.94	50.00	58.72	58.85	61.31
18.2.4	4.00	0.00	50.12	58.74	58.85	61.31
18.2.5	5.00	0.00	50.00	58.63	58.85	61.31
18.2.6	6.00	0.00	50.00	58.56	58.72	61.31

18.2.7	7.00	0.00	50.00	58.72	58.85	61.31
18.2.8	8.00	0.00	50.00	58.45	58.55	65.16
18.2.9	9.00	50.00	50.00	58.50	58.55	65.16
18.2.10	10.00	0.00	50.00	58.48	58.55	59.37
18.2.11	11.00	40.00	50.12	58.48	58.55	61.31
18.2.12	12.00	0.00	50.12	58.50	58.55	61.31
18.2.13	13.00	0.00	50.12	58.50	58.55	61.13
18.2.14	14.00	0.00	50.12	58.49	58.55	61.31
18.2.15	15.00	50.00	50.12	58.50	58.55	61.31
18.2.16	16.00	0.00	50.00	58.50	58.61	65.16
18.2.17	17.00	50.00	50.00	58.50	58.61	65.16
18.2.18	18.00	0.00	50.00	58.50	58.61	59.37
18.2.19	19.00	40.00	50.12	58.50	58.61	61.31
18.2.20	20.00	0.00	50.12	58.50	58.61	61.31
18.2.21	21.00	0.00	50.12	58.50	58.61	61.13
18.2.22	22.00	0.00	50.12	58.50	58.61	61.31
18.2.23	23.00	50.00	50.12	58.50	58.61	61.31
18.2.24	24.00	0.00	50.00	58.74	58.94	61.31
18.2.25	25.00	0.00	50.00	58.52	58.63	61.31
18.2.26	26.00	0.00	50.00	58.74	58.94	61.31
18.2.27	27.00	0.00	50.00	58.74	58.85	61.31
18.2.28	28.00	0.00	50.12	58.74	58.94	61.31
18.2.29	29.00	0.00	50.00	58.72	58.93	61.31
18.2.30	30.00	0.00	50.00	58.64	58.80	61.31

18.2.31	31.00	0.00	50.00	58.74	58.94	61.31
18.2.32	32.00	0.00	50.00	58.74	58.95	61.31
18.2.33	33.00	0.00	50.00	58.53	58.65	61.31
18.2.34	34.00	0.00	50.00	58.74	58.95	61.31
18.2.35	35.00	49.95	50.00	58.74	58.85	61.31
18.2.36	36.00	0.00	50.12	58.74	58.95	61.31
18.2.37	37.00	0.00	50.00	58.74	58.95	61.31
18.2.38	38.00	0.00	50.00	58.64	58.80	61.31
18.2.39	39.00	0.00	50.00	58.74	58.95	61.31
18.2.40	40.00	0.00	50.00	58.50	58.61	65.16
18.2.41	41.00	0.00	50.00	58.50	58.61	65.16
18.2.42	42.00	0.00	50.00	58.50	58.61	59.37
18.2.43	43.00	0.00	50.12	58.50	58.61	61.31
18.2.44	44.00	0.00	50.12	58.50	58.61	61.31
18.2.45	45.00	0.00	50.12	58.50	58.61	61.13
18.2.46	46.00	0.00	50.12	58.50	58.61	61.31
18.2.47	47.00	50.11	50.12	58.50	58.61	61.31
18.2.48	48.00	0.00	50.00	58.50	58.61	65.16
18.2.49	49.00	50.00	50.00	58.50	58.61	65.16
18.2.50	50.00	0.00	50.00	58.50	58.61	59.37
18.2.51	51.00	0.00	50.12	58.50	58.61	61.31
18.2.52	52.00	0.00	50.12	58.50	58.61	61.31
18.2.53	53.00	0.00	50.12	58.50	58.61	61.13
18.2.54	54.00	0.00	50.12	58.50	58.61	61.31

18.2.55	55.00	50.11	50.12	58.50	58.61	61.31
18.2.56	56.00	0.00	50.00	58.74	58.94	61.31
18.2.57	57.00	0.00	50.00	58.54	58.63	61.31
18.2.58	58.00	0.00	50.00	58.74	58.94	61.31
18.2.59	59.00	49.95	50.00	58.74	58.85	61.31
18.2.60	60.00	0.00	50.12	58.74	58.94	61.31
18.2.61	61.00	0.00	50.00	58.72	58.93	61.31
18.2.62	62.00	0.00	50.00	58.64	58.80	61.31
18.2.63	63.00	0.00	50.00	58.74	58.94	61.31
18.2.64	64.00	0.00	50.00	58.50	58.61	65.16
18.2.65	65.00	50.00	50.00	58.50	58.61	65.16
18.2.66	66.00	0.00	50.00	58.50	58.61	59.21
18.2.67	67.00	0.00	50.12	58.50	58.61	61.31
18.2.68	68.00	0.00	50.12	58.50	58.61	61.31
18.2.69	69.00	0.00	50.12	58.50	58.61	61.13
18.2.70	70.00	0.00	50.12	58.50	58.61	61.31
18.2.71	71.00	50.11	50.12	58.50	58.61	61.31
18.2.72	72.00	0.00	50.00	58.74	58.95	61.31
18.2.73	73.00	0.00	50.00	58.54	58.63	61.31
18.2.74	74.00	0.00	50.00	58.74	58.95	61.31
18.2.75	75.00	49.95	50.00	58.74	58.85	61.31
18.2.76	76.00	0.00	50.12	58.74	58.95	61.31
18.2.77	77.00	0.00	50.00	58.74	58.93	61.31
18.2.78	78.00	0.00	50.00	58.64	58.80	61.31

18.2.79	79.00	0.00	50.00	58.74	58.95	61.31
18.2.80	80.00	0.00	50.00	58.74	58.94	61.31
18.2.81	81.00	0.00	50.00	58.54	58.63	61.31
18.2.82	82.00	0.00	50.00	58.74	58.94	61.31
18.2.83	83.00	49.94	50.00	58.74	58.85	61.31
18.2.84	84.00	0.00	50.12	58.74	58.94	61.31
18.2.85	85.00	0.00	50.00	58.74	58.93	61.31
18.2.86	86.00	0.00	50.00	58.64	58.80	61.31
18.2.87	87.00	0.00	50.00	58.74	58.94	61.31
18.2.88	88.00	0.00	50.00	58.50	58.61	65.16
18.2.89	89.00	50.00	50.00	58.50	58.61	65.16
18.2.90	90.00	0.00	50.00	58.50	58.61	59.37
18.2.91	91.00	0.00	50.12	58.50	58.61	61.31
18.2.92	92.00	0.00	50.12	58.50	58.61	61.31
18.2.93	93.00	0.00	50.12	58.50	58.61	61.13
18.2.94	94.00	0.00	50.12	58.50	58.61	61.31
18.2.95	95.00	50.11	50.12	58.50	58.61	61.31
18.2.96	96.00	0.00	50.00	50.00	58.55	65.16
18.2.97	97.00	0.00	50.00	50.00	58.55	65.16
18.2.98	98.00	0.00	50.00	58.45	58.55	59.37
18.2.99	99.00	0.00	50.12	58.44	58.55	61.31
18.2.100	100.00	0.00	50.00	50.12	58.55	61.31
18.2.101	101.00	0.00	50.00	50.12	58.55	61.13
18.2.102	102.00	25.00	50.00	50.12	58.55	61.31

18.2.103	103.00	0.00	50.00	50.12	58.55	61.31
18.2.104	104.00	0.00	50.00	58.63	58.85	61.31
18.2.105	105.00	0.00	50.00	58.48	58.59	61.31
18.2.106	106.00	0.00	50.00	58.63	58.85	61.31
18.2.107	107.00	0.00	50.00	58.63	58.85	61.31
18.2.108	108.00	0.00	50.12	58.63	58.85	61.31
18.2.109	109.00	0.00	50.00	50.00	58.85	61.31
18.2.110	110.00	0.00	50.00	58.56	58.72	61.31
18.2.111	111.00	0.00	50.00	58.63	58.85	83.33
18.2.112	112.00	0.00	50.00	58.74	58.94	61.31
18.2.113	113.00	0.00	50.00	58.52	58.63	61.31
18.2.114	114.00	0.00	50.00	58.74	58.94	61.31
18.2.115	115.00	0.00	50.00	58.74	58.85	61.31
18.2.116	116.00	0.00	50.12	58.74	58.94	61.31
18.2.117	117.00	0.00	50.00	58.74	58.93	61.31
18.2.118	118.00	0.00	50.00	58.64	58.80	61.31
18.2.119	119.00	0.00	50.00	58.74	58.94	61.31
18.2.120	120.00	0.00	50.00	58.50	58.61	65.16
18.2.121	121.00	0.00	50.00	58.50	58.61	65.16
18.2.122	122.00	0.00	50.00	58.50	58.61	59.37
18.2.123	123.00	40.00	50.12	58.50	58.61	61.31
18.2.124	124.00	0.00	50.12	58.50	58.61	61.31
18.2.125	125.00	0.00	50.12	58.50	58.61	61.13
18.2.126	126.00	0.00	50.12	58.50	58.61	61.31

18.2.127	127.00	50.00	50.12	58.50	58.61	61.31
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18.3 L2 Requests (per normUnit)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.3.0	0.00	0.00	0.00	3128.00	32768.00	32789.00
18.3.1	1.00	0.00	0.00	3128.00	32768.00	32788.00
18.3.2	2.00	0.00	0.00	3128.00	32768.00	32789.00
18.3.3	3.00	0.00	0.00	3100.00	32768.00	32788.00
18.3.4	4.00	0.00	0.00	3114.00	32848.00	32868.00
18.3.5	5.00	0.00	0.00	3128.00	32768.00	32789.00
18.3.6	6.00	0.00	0.00	3128.00	32768.00	32788.00
18.3.7	7.00	0.00	0.00	3128.00	32768.00	32789.00
18.3.8	8.00	0.00	0.00	3102.00	32768.00	32788.00
18.3.9	9.00	0.00	0.00	3102.00	32768.00	32788.00
18.3.10	10.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.11	11.00	0.00	0.00	3102.00	32848.00	32869.00
18.3.12	12.00	0.00	0.00	3102.00	32848.00	32869.00
18.3.13	13.00	0.00	1.00	3102.00	32848.00	32869.00
18.3.14	14.00	0.00	0.00	3102.00	32848.00	32869.00
18.3.15	15.00	0.00	0.00	3102.00	32848.00	32869.00
18.3.16	16.00	0.00	0.00	3102.00	32768.00	32788.00
18.3.17	17.00	0.00	0.00	3102.00	32768.00	32788.00
18.3.18	18.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.19	19.00	0.00	0.00	3102.00	32848.00	32869.00
18.3.20	20.00	0.00	0.00	3102.00	32848.00	32869.00

18.3.21	21.00	0.00	1.00	3102.00	32848.00	32869.00
18.3.22	22.00	0.00	0.00	3102.00	32848.00	32869.00
18.3.23	23.00	0.00	0.00	3102.00	32848.00	32869.00
18.3.24	24.00	0.00	0.00	3121.00	32768.00	32789.00
18.3.25	25.00	0.00	0.00	3121.00	32768.00	32788.00
18.3.26	26.00	0.00	0.00	3121.00	32768.00	32789.00
18.3.27	27.00	0.00	0.00	3100.00	32768.00	32788.00
18.3.28	28.00	0.00	0.00	3114.00	32848.00	32868.00
18.3.29	29.00	0.00	0.00	3121.50	32768.00	32789.00
18.3.30	30.00	0.00	0.00	3120.00	32768.00	32788.00
18.3.31	31.00	0.00	0.00	3120.50	32768.00	32789.00
18.3.32	32.00	0.00	0.00	3128.00	32768.00	32789.00
18.3.33	33.00	0.00	0.00	3128.00	32768.00	32788.00
18.3.34	34.00	0.00	0.00	3128.00	32768.00	32789.00
18.3.35	35.00	0.00	0.00	3100.00	32768.00	32788.00
18.3.36	36.00	0.00	0.00	3114.00	32848.00	32868.00
18.3.37	37.00	0.00	0.00	3128.00	32768.00	32789.00
18.3.38	38.00	0.00	0.00	3128.00	32768.00	32788.00
18.3.39	39.00	0.00	0.00	3128.00	32768.00	32789.00
18.3.40	40.00	0.00	0.00	3102.00	32768.00	32788.00
18.3.41	41.00	0.00	0.00	3102.00	32768.00	32788.00
18.3.42	42.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.43	43.00	0.00	0.00	3102.00	32848.00	32869.00
18.3.44	44.00	0.00	0.00	3102.00	32848.00	32869.00

18.3.45	45.00	0.00	0.75	3102.00	32848.00	32869.00
18.3.46	46.00	0.00	0.00	3102.00	32848.00	32869.00
18.3.47	47.00	0.00	0.00	3102.00	32848.00	32869.00
18.3.48	48.00	0.00	0.00	3098.00	32768.00	32788.00
18.3.49	49.00	0.00	0.00	3098.00	32768.00	32788.00
18.3.50	50.00	0.00	0.00	3098.00	32768.00	32789.00
18.3.51	51.00	0.00	0.00	3098.00	32848.00	32869.00
18.3.52	52.00	0.00	0.00	3098.00	32848.00	32869.00
18.3.53	53.00	0.00	0.75	3099.00	32848.00	32869.00
18.3.54	54.00	0.00	0.00	3098.00	32848.00	32869.00
18.3.55	55.00	0.00	0.00	3098.00	32848.00	32869.00
18.3.56	56.00	0.00	0.00	3120.00	32768.00	32789.00
18.3.57	57.00	0.00	0.00	3120.00	32768.00	32788.00
18.3.58	58.00	0.00	0.00	3120.00	32768.00	32789.00
18.3.59	59.00	0.00	0.00	3096.00	32768.00	32788.00
18.3.60	60.00	0.00	0.00	3108.00	32848.00	32868.00
18.3.61	61.00	0.00	0.00	3121.00	32768.00	32789.00
18.3.62	62.00	0.00	0.00	3120.00	32768.00	32788.00
18.3.63	63.00	0.00	0.00	3120.00	32768.00	32789.00
18.3.64	64.00	0.00	0.00	3099.00	32768.00	32788.00
18.3.65	65.00	0.00	0.00	3098.00	32768.00	32788.00
18.3.66	66.00	0.00	0.00	3098.00	32768.00	32789.00
18.3.67	67.00	0.00	0.00	3098.00	32848.00	32869.00
18.3.68	68.00	0.00	0.00	3099.00	32848.00	32869.00

18.3.69	69.00	0.00	0.00	3099.00	32848.00	32869.00
18.3.70	70.00	0.00	0.00	3098.50	32848.00	32869.00
18.3.71	71.00	0.00	0.00	3098.00	32848.00	32869.00
18.3.72	72.00	0.00	0.00	3120.00	32768.00	32789.00
18.3.73	73.00	0.00	0.00	3120.00	32768.00	32788.00
18.3.74	74.00	0.00	0.00	3120.00	32768.00	32789.00
18.3.75	75.00	0.00	0.00	3096.00	32768.00	32788.00
18.3.76	76.00	0.00	0.00	3108.00	32848.00	32868.00
18.3.77	77.00	0.00	0.00	3121.00	32768.00	32789.00
18.3.78	78.00	0.00	0.00	3121.00	32768.00	32788.00
18.3.79	79.00	0.00	0.00	3120.00	32768.00	32789.00
18.3.80	80.00	0.00	0.00	3124.00	32768.00	32789.00
18.3.81	81.00	0.00	0.00	3124.50	32768.00	32788.00
18.3.82	82.00	0.00	0.00	3124.50	32768.00	32789.00
18.3.83	83.00	0.00	0.00	3100.00	32768.00	32788.00
18.3.84	84.00	0.00	0.00	3114.00	32848.00	32868.00
18.3.85	85.00	0.00	0.00	3124.50	32768.00	32789.00
18.3.86	86.00	0.00	0.00	3124.50	32768.00	32788.00
18.3.87	87.00	0.00	0.00	3124.50	32768.00	32789.00
18.3.88	88.00	0.00	0.00	3102.00	32768.00	32788.00
18.3.89	89.00	0.00	0.00	3102.00	32768.00	32788.00
18.3.90	90.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.91	91.00	0.00	0.00	3102.00	32848.00	32869.00
18.3.92	92.00	0.00	0.00	3102.00	32848.00	32869.00

18.3.93	93.00	0.00	0.00	3102.00	32848.00	32869.00
18.3.94	94.00	0.00	0.00	3102.00	32848.00	32869.00
18.3.95	95.00	0.00	0.00	3102.00	32848.00	32869.00
18.3.96	96.00	0.00	4.00	3102.00	32768.00	32788.00
18.3.97	97.00	0.00	4.00	3102.00	32768.00	32788.00
18.3.98	98.00	0.00	4.00	3102.00	32768.00	32789.00
18.3.99	99.00	0.00	5.00	3102.00	32848.00	32869.00
18.3.100	100.00	0.00	4.00	3102.00	32848.00	32869.00
18.3.101	101.00	0.00	5.00	3102.00	32848.00	32869.00
18.3.102	102.00	0.00	6.00	3102.00	32848.00	32869.00
18.3.103	103.00	0.00	4.00	3102.00	32848.00	32869.00
18.3.104	104.00	0.00	4.00	3128.00	32768.00	32789.00
18.3.105	105.00	0.00	5.00	3128.00	32768.00	32788.00
18.3.106	106.00	0.00	4.00	3128.00	32768.00	32789.00
18.3.107	107.00	0.00	5.00	3100.00	32768.00	32788.00
18.3.108	108.00	0.00	4.00	3114.00	32848.00	32868.00
18.3.109	109.00	0.00	4.00	3128.00	32768.00	32789.00
18.3.110	110.00	0.00	4.00	3128.00	32768.00	32788.00
18.3.111	111.00	0.00	4.75	3128.00	32768.00	32789.00
18.3.112	112.00	0.00	0.00	3120.00	32768.00	32789.00
18.3.113	113.00	0.00	0.00	3120.00	32768.00	32788.00
18.3.114	114.00	0.00	0.00	3120.00	32768.00	32789.00
18.3.115	115.00	0.00	0.00	3096.00	32768.00	32788.00
18.3.116	116.00	0.00	0.00	3108.00	32848.00	32868.00

18.3.117	117.00	0.00	1.00	3121.00	32768.00	32789.00
18.3.118	118.00	0.00	0.00	3120.00	32768.00	32788.00
18.3.119	119.00	0.00	0.00	3120.00	32768.00	32789.00
18.3.120	120.00	0.00	0.00	3098.00	32768.00	32788.00
18.3.121	121.00	0.00	0.00	3098.00	32768.00	32788.00
18.3.122	122.00	0.00	0.00	3098.00	32768.00	32789.00
18.3.123	123.00	0.00	0.00	3098.00	32848.00	32869.00
18.3.124	124.00	0.00	0.00	3098.00	32848.00	32869.00
18.3.125	125.00	0.00	1.00	3099.00	32848.00	32869.00
18.3.126	126.00	0.00	0.00	3098.00	32848.00	32869.00
18.3.127	127.00	0.00	0.00	3098.00	32848.00	32869.00

18.4 L2 Requests (per normUnit)

Metric_ID	Channel	L2 Read	L2 Write	L2 Atomic
18.4.0	0.00	829.85	10876.44	0.00
18.4.1	1.00	829.84	10876.44	0.00
18.4.2	2.00	829.85	10876.44	0.00
18.4.3	3.00	827.73	10876.44	0.00
18.4.4	4.00	855.38	10876.44	0.00
18.4.5	5.00	829.96	10876.44	0.00
18.4.6	6.00	829.84	10876.44	0.00
18.4.7	7.00	829.86	10876.44	0.00
18.4.8	8.00	832.01	10876.44	0.00
18.4.9	9.00	835.60	10876.44	0.00
18.4.10	10.00	837.67	10876.45	0.00

18.4.11	11.00	863.21	10876.46	0.00
18.4.12	12.00	863.50	10876.44	0.00
18.4.13	13.00	863.64	10876.44	0.00
18.4.14	14.00	863.27	10876.44	0.00
18.4.15	15.00	856.32	10876.44	0.00
18.4.16	16.00	832.80	10876.45	0.00
18.4.17	17.00	836.62	10876.45	0.00
18.4.18	18.00	838.73	10876.44	0.00
18.4.19	19.00	864.28	10876.45	0.00
18.4.20	20.00	864.57	10876.44	0.00
18.4.21	21.00	864.71	10876.44	0.00
18.4.22	22.00	864.34	10876.44	0.00
18.4.23	23.00	856.92	10876.44	0.00
18.4.24	24.00	830.45	10876.44	0.00
18.4.25	25.00	830.44	10876.44	0.00
18.4.26	26.00	830.46	10876.44	0.00
18.4.27	27.00	828.29	10876.44	0.00
18.4.28	28.00	855.98	10876.44	0.00
18.4.29	29.00	830.56	10876.44	0.00
18.4.30	30.00	830.44	10876.44	0.00
18.4.31	31.00	830.47	10876.44	0.00
18.4.32	32.00	830.10	10876.44	0.00
18.4.33	33.00	830.09	10876.44	0.00
18.4.34	34.00	830.10	10876.44	0.00

18.4.35	35.00	827.98	10876.44	0.00
18.4.36	36.00	855.62	10876.44	0.00
18.4.37	37.00	830.20	10876.44	0.00
18.4.38	38.00	830.09	10876.44	0.00
18.4.39	39.00	830.12	10876.44	0.00
18.4.40	40.00	832.39	10876.44	0.00
18.4.41	41.00	836.11	10876.44	0.00
18.4.42	42.00	838.18	10876.44	0.00
18.4.43	43.00	863.72	10876.45	0.00
18.4.44	44.00	864.01	10876.44	0.00
18.4.45	45.00	864.16	10876.44	0.00
18.4.46	46.00	863.78	10876.44	0.00
18.4.47	47.00	856.57	10876.44	0.00
18.4.48	48.00	833.08	10876.44	0.00
18.4.49	49.00	836.96	10876.44	0.00
18.4.50	50.00	839.13	10876.44	0.00
18.4.51	51.00	864.67	10876.45	0.00
18.4.52	52.00	864.97	10876.44	0.00
18.4.53	53.00	865.11	10876.44	0.00
18.4.54	54.00	864.73	10876.44	0.00
18.4.55	55.00	857.20	10876.44	0.00
18.4.56	56.00	830.72	10876.44	0.00
18.4.57	57.00	830.72	10876.44	0.00
18.4.58	58.00	830.73	10876.44	0.00

18.4.59	59.00	828.51	10876.44	0.00
18.4.60	60.00	856.25	10876.44	0.00
18.4.61	61.00	830.83	10876.44	0.00
18.4.62	62.00	830.72	10876.44	0.00
18.4.63	63.00	830.75	10876.44	0.00
18.4.64	64.00	832.62	10876.44	0.00
18.4.65	65.00	836.35	10876.44	0.00
18.4.66	66.00	838.52	10876.44	0.00
18.4.67	67.00	864.06	10876.44	0.00
18.4.68	68.00	864.36	10876.44	0.00
18.4.69	69.00	864.50	10876.44	0.00
18.4.70	70.00	864.13	10876.44	0.00
18.4.71	71.00	856.91	10876.44	0.00
18.4.72	72.00	830.44	10876.44	0.00
18.4.73	73.00	830.43	10876.44	0.00
18.4.74	74.00	830.44	10876.44	0.00
18.4.75	75.00	828.21	10876.44	0.00
18.4.76	76.00	855.97	10876.44	0.00
18.4.77	77.00	830.55	10876.44	0.00
18.4.78	78.00	830.43	10876.44	0.00
18.4.79	79.00	830.45	10876.44	0.00
18.4.80	80.00	829.77	10876.44	0.00
18.4.81	81.00	829.76	10876.44	0.00
18.4.82	82.00	829.78	10876.44	0.00

18.4.83	83.00	827.72	10876.44	0.00
18.4.84	84.00	855.30	10876.44	0.00
18.4.85	85.00	829.88	10876.44	0.00
18.4.86	86.00	829.76	10876.44	0.00
18.4.87	87.00	829.79	10876.44	0.00
18.4.88	88.00	832.06	10876.44	0.00
18.4.89	89.00	835.72	10876.44	0.00
18.4.90	90.00	837.71	10876.44	0.00
18.4.91	91.00	863.26	10876.44	0.00
18.4.92	92.00	863.56	10876.44	0.00
18.4.93	93.00	863.70	10876.44	0.00
18.4.94	94.00	863.32	10876.44	0.00
18.4.95	95.00	856.25	10876.44	0.00
18.4.96	96.00	833.78	10876.45	0.00
18.4.97	97.00	837.60	10876.46	0.00
18.4.98	98.00	838.90	10876.45	0.00
18.4.99	99.00	864.48	10876.46	0.00
18.4.100	100.00	865.32	10876.44	0.00
18.4.101	101.00	865.68	10876.45	0.00
18.4.102	102.00	865.25	10876.62	0.00
18.4.103	103.00	857.92	10876.47	0.00
18.4.104	104.00	830.67	10876.46	0.00
18.4.105	105.00	830.75	10876.54	0.00
18.4.106	106.00	830.66	10876.45	0.00

18.4.107	107.00	828.47	10876.47	0.00
18.4.108	108.00	856.15	10876.45	0.00
18.4.109	109.00	830.72	10876.48	0.00
18.4.110	110.00	830.61	10876.46	0.00
18.4.111	111.00	830.65	10876.57	0.00
18.4.112	112.00	830.43	10876.44	0.00
18.4.113	113.00	830.42	10876.45	0.00
18.4.114	114.00	830.43	10876.44	0.00
18.4.115	115.00	828.22	10876.45	0.00
18.4.116	116.00	855.96	10876.44	0.00
18.4.117	117.00	830.53	10876.45	0.00
18.4.118	118.00	830.42	10876.44	0.00
18.4.119	119.00	830.44	10876.45	0.00
18.4.120	120.00	832.67	10876.44	0.00
18.4.121	121.00	836.41	10876.44	0.00
18.4.122	122.00	838.62	10876.44	0.00
18.4.123	123.00	864.17	10876.45	0.00
18.4.124	124.00	864.41	10876.45	0.00
18.4.125	125.00	864.54	10876.46	0.00
18.4.126	126.00	864.18	10876.44	0.00
18.4.127	127.00	856.89	10876.45	0.00

18.5 L2-Fabric Requests (per normUnit)

Metric_ID	Channel	L2-Fabric Read	L2-Fabric Write and Atomic
L2-Fabric Atomic			

18.5.0 0.00	0.00	461.09	10570.67
18.5.1 0.00	1.00	461.08	10570.67
18.5.2 0.00	2.00	461.09	10570.67
18.5.3 0.00	3.00	460.80	10570.67
18.5.4 0.00	4.00	462.34	10570.67
18.5.5 0.00	5.00	461.20	10570.67
18.5.6 0.00	6.00	461.08	10570.67
18.5.7 0.00	7.00	461.10	10570.67
18.5.8 0.00	8.00	462.15	10570.67
18.5.9 0.00	9.00	462.73	10570.67
18.5.10 0.00	10.00	463.11	10570.67
18.5.11 0.00	11.00	464.38	10570.67

18.5.12 0.00	12.00	464.40	10570.68
18.5.13 0.00	13.00	464.59	10570.68
18.5.14 0.00	14.00	464.39	10570.67
18.5.15 0.00	15.00	463.15	10570.67
18.5.16 0.00	16.00	462.15	10570.67
18.5.17 0.00	17.00	462.73	10570.67
18.5.18 0.00	18.00	463.11	10570.67
18.5.19 0.00	19.00	464.38	10570.67
18.5.20 0.00	20.00	464.40	10570.67
18.5.21 0.00	21.00	464.59	10570.67
18.5.22 0.00	22.00	464.39	10570.68
18.5.23 0.00	23.00	463.15	10570.68

18.5.24 0.00	24.00	461.09	10570.67
18.5.25 0.00	25.00	461.08	10570.67
18.5.26 0.00	26.00	461.09	10570.67
18.5.27 0.00	27.00	460.80	10570.67
18.5.28 0.00	28.00	462.34	10570.67
18.5.29 0.00	29.00	461.19	10570.67
18.5.30 0.00	30.00	461.08	10570.67
18.5.31 0.00	31.00	461.11	10570.67
18.5.32 0.00	32.00	461.08	10570.67
18.5.33 0.00	33.00	461.08	10570.67
18.5.34 0.00	34.00	461.09	10570.67
18.5.35 0.00	35.00	460.79	10570.67

18.5.36 0.00	36.00	462.33	10570.67
18.5.37 0.00	37.00	461.19	10570.67
18.5.38 0.00	38.00	461.07	10570.67
18.5.39 0.00	39.00	461.10	10570.67
18.5.40 0.00	40.00	462.14	10570.67
18.5.41 0.00	41.00	462.72	10570.67
18.5.42 0.00	42.00	463.10	10570.67
18.5.43 0.00	43.00	464.37	10570.67
18.5.44 0.00	44.00	464.39	10570.67
18.5.45 0.00	45.00	464.58	10570.67
18.5.46 0.00	46.00	464.38	10570.67
18.5.47 0.00	47.00	463.14	10570.67

18.5.48 0.00	48.00	462.14	10570.67
18.5.49 0.00	49.00	462.72	10570.67
18.5.50 0.00	50.00	463.10	10570.67
18.5.51 0.00	51.00	464.37	10570.67
18.5.52 0.00	52.00	464.39	10570.67
18.5.53 0.00	53.00	464.58	10570.67
18.5.54 0.00	54.00	464.38	10570.67
18.5.55 0.00	55.00	463.14	10570.67
18.5.56 0.00	56.00	461.08	10570.67
18.5.57 0.00	57.00	461.07	10570.67
18.5.58 0.00	58.00	461.09	10570.67
18.5.59 0.00	59.00	460.79	10570.67

18.5.60 0.00	60.00	462.33	10570.67
18.5.61 0.00	61.00	461.19	10570.67
18.5.62 0.00	62.00	461.07	10570.67
18.5.63 0.00	63.00	461.10	10570.67
18.5.64 0.00	64.00	462.14	10570.67
18.5.65 0.00	65.00	462.72	10570.67
18.5.66 0.00	66.00	463.09	10570.67
18.5.67 0.00	67.00	464.36	10570.67
18.5.68 0.00	68.00	464.39	10570.67
18.5.69 0.00	69.00	464.58	10570.67
18.5.70 0.00	70.00	464.38	10570.67
18.5.71 0.00	71.00	463.14	10570.67

18.5.72 0.00	72.00	461.08	10570.67
18.5.73 0.00	73.00	461.07	10570.67
18.5.74 0.00	74.00	461.08	10570.67
18.5.75 0.00	75.00	460.78	10570.67
18.5.76 0.00	76.00	462.33	10570.67
18.5.77 0.00	77.00	461.19	10570.67
18.5.78 0.00	78.00	461.06	10570.67
18.5.79 0.00	79.00	461.09	10570.67
18.5.80 0.00	80.00	461.08	10570.67
18.5.81 0.00	81.00	461.07	10570.67
18.5.82 0.00	82.00	461.08	10570.67
18.5.83 0.00	83.00	460.78	10570.67

18.5.84 0.00	84.00	462.33	10570.67
18.5.85 0.00	85.00	461.19	10570.67
18.5.86 0.00	86.00	461.07	10570.67
18.5.87 0.00	87.00	461.09	10570.67
18.5.88 0.00	88.00	462.14	10570.67
18.5.89 0.00	89.00	462.72	10570.67
18.5.90 0.00	90.00	463.09	10570.67
18.5.91 0.00	91.00	464.36	10570.67
18.5.92 0.00	92.00	464.39	10570.67
18.5.93 0.00	93.00	464.58	10570.67
18.5.94 0.00	94.00	464.38	10570.67
18.5.95 0.00	95.00	463.14	10570.67

18.5.96 0.00	96.00	463.22	10570.68
18.5.97 0.00	97.00	463.85	10570.69
18.5.98 0.00	98.00	463.40	10570.67
18.5.99 0.00	99.00	464.70	10570.68
18.5.100 0.00	100.00	465.27	10570.68
18.5.101 0.00	101.00	465.68	10570.69
18.5.102 0.00	102.00	465.43	10570.68
18.5.103 0.00	103.00	464.19	10570.77
18.5.104 0.00	104.00	461.35	10570.69
18.5.105 0.00	105.00	461.43	10570.70
18.5.106 0.00	106.00	461.34	10570.68
18.5.107 0.00	107.00	461.04	10570.69

18.5.108 0.00	108.00	462.56	10570.68
18.5.109 0.00	109.00	461.41	10570.69
18.5.110 0.00	110.00	461.29	10570.68
18.5.111 0.00	111.00	461.33	10570.86
18.5.112 0.00	112.00	461.10	10570.67
18.5.113 0.00	113.00	461.09	10570.67
18.5.114 0.00	114.00	461.10	10570.67
18.5.115 0.00	115.00	460.80	10570.67
18.5.116 0.00	116.00	462.35	10570.67
18.5.117 0.00	117.00	461.20	10570.67
18.5.118 0.00	118.00	461.08	10570.67
18.5.119 0.00	119.00	461.11	10570.67

18.5.120 0.00	120.00	462.16	10570.67
18.5.121 0.00	121.00	462.73	10570.67
18.5.122 0.00	122.00	463.17	10570.68
18.5.123 0.00	123.00	464.44	10570.68
18.5.124 0.00	124.00	464.42	10570.67
18.5.125 0.00	125.00	464.59	10570.67
18.5.126 0.00	126.00	464.40	10570.67
18.5.127 0.00	127.00	463.15	10570.67

18.6 L2-Fabric Read Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.6.0	0.00	0.00	507.48	517.13	528.93	1152.00
18.6.1	1.00	281.00	523.85	533.72	544.77	1567.00
18.6.2	2.00	0.00	533.75	543.79	556.40	1824.00
18.6.3	3.00	0.00	532.94	543.16	566.12	1853.00
18.6.4	4.00	0.00	0.00	509.40	539.15	698.23

18.6.5	5.00	0.00	526.59	535.65	564.62	1452.00
18.6.6	6.00	0.00	557.17	568.22	578.58	732.48
18.6.7	7.00	0.00	537.69	552.67	561.16	959.00
18.6.8	8.00	0.00	495.73	514.99	524.04	885.00
18.6.9	9.00	0.00	519.78	536.78	548.34	705.52
18.6.10	10.00	0.00	523.11	533.85	545.28	1184.00
18.6.11	11.00	0.00	0.00	514.83	540.45	885.75
18.6.12	12.00	0.00	0.00	509.36	542.98	697.66
18.6.13	13.00	0.00	0.00	507.35	540.29	697.79
18.6.14	14.00	0.00	0.00	532.64	564.03	718.32
18.6.15	15.00	0.00	0.00	518.90	552.41	705.65
18.6.16	16.00	0.00	507.07	518.30	525.28	884.00
18.6.17	17.00	0.00	525.98	534.38	541.90	710.13
18.6.18	18.00	0.00	528.74	534.02	544.61	1168.00
18.6.19	19.00	0.00	0.00	519.62	540.96	911.75
18.6.20	20.00	0.00	0.00	516.54	535.26	713.79
18.6.21	21.00	0.00	0.00	511.05	532.11	703.51
18.6.22	22.00	0.00	0.00	542.04	558.77	733.74
18.6.23	23.00	0.00	0.00	523.82	547.22	737.74
18.6.24	24.00	0.00	509.34	514.83	527.99	1355.00
18.6.25	25.00	200.00	524.68	530.74	543.83	1579.00
18.6.26	26.00	0.00	536.60	542.59	552.17	1822.00
18.6.27	27.00	0.00	535.46	542.19	564.38	1756.00
18.6.28	28.00	0.00	0.00	518.02	537.00	701.46

18.6.29	29.00	318.75	529.02	534.77	563.23	2103.00
18.6.30	30.00	0.00	563.38	570.33	577.55	747.01
18.6.31	31.00	0.00	544.66	551.96	559.10	1023.00
18.6.32	32.00	0.00	509.18	519.18	529.62	878.00
18.6.33	33.00	83.88	525.72	537.92	552.18	1012.00
18.6.34	34.00	0.00	522.78	535.54	546.08	1326.00
18.6.35	35.00	0.00	533.13	542.68	558.90	1340.00
18.6.36	36.00	0.00	0.00	513.63	541.79	708.31
18.6.37	37.00	204.75	528.68	539.27	558.98	1453.00
18.6.38	38.00	0.00	544.04	561.29	573.11	753.74
18.6.39	39.00	0.00	526.37	546.28	555.66	812.00
18.6.40	40.00	0.00	489.83	509.40	518.61	1276.00
18.6.41	41.00	0.00	510.84	526.41	535.95	690.94
18.6.42	42.00	0.00	525.56	537.70	548.84	728.00
18.6.43	43.00	0.00	0.00	513.09	538.89	1325.00
18.6.44	44.00	0.00	0.00	511.69	541.25	713.06
18.6.45	45.00	0.00	0.00	509.01	539.37	704.67
18.6.46	46.00	0.00	0.00	543.13	573.34	752.24
18.6.47	47.00	0.00	0.00	529.75	559.74	726.96
18.6.48	48.00	0.00	498.83	519.53	525.17	1285.00
18.6.49	49.00	0.00	520.28	533.58	539.29	682.58
18.6.50	50.00	0.00	534.01	544.22	551.03	718.00
18.6.51	51.00	0.00	0.00	515.64	545.16	1182.50
18.6.52	52.00	0.00	0.00	517.50	545.01	691.50

18.6.53	53.00	0.00	0.00	513.19	543.84	694.05
18.6.54	54.00	0.00	0.00	551.21	577.86	728.75
18.6.55	55.00	0.00	0.00	530.62	566.97	724.57
18.6.56	56.00	0.00	510.21	518.55	527.54	809.00
18.6.57	57.00	89.14	530.33	540.58	547.57	1042.00
18.6.58	58.00	0.00	530.36	537.88	545.22	1330.00
18.6.59	59.00	0.00	538.64	546.47	558.28	1496.00
18.6.60	60.00	0.00	0.00	517.81	543.84	700.65
18.6.61	61.00	170.00	536.55	543.90	554.34	1716.00
18.6.62	62.00	0.00	552.82	569.63	575.72	717.70
18.6.63	63.00	0.00	535.14	551.16	557.65	831.00
18.6.64	64.00	0.00	502.20	514.29	523.83	687.99
18.6.65	65.00	0.00	519.25	530.87	539.78	707.44
18.6.66	66.00	451.00	535.32	542.64	550.84	1522.00
18.6.67	67.00	0.00	0.00	517.71	540.27	732.22
18.6.68	68.00	0.00	0.00	514.78	537.70	708.41
18.6.69	69.00	0.00	0.00	511.51	537.17	718.06
18.6.70	70.00	0.00	0.00	549.04	571.52	759.94
18.6.71	71.00	0.00	0.00	531.93	558.71	744.88
18.6.72	72.00	277.75	510.39	518.65	531.87	1452.00
18.6.73	73.00	284.00	534.99	542.54	555.32	1548.00
18.6.74	74.00	0.00	529.37	537.28	546.59	1686.00
18.6.75	75.00	0.00	537.60	545.41	562.37	1823.00
18.6.76	76.00	0.00	0.00	516.41	539.14	709.68

18.6.77	77.00	323.00	530.98	538.56	558.64	2108.00
18.6.78	78.00	0.00	551.69	560.09	568.24	748.22
18.6.79	79.00	0.00	535.10	546.99	556.24	724.50
18.6.80	80.00	261.75	513.69	518.95	532.86	1470.00
18.6.81	81.00	276.75	535.10	541.00	555.35	1569.00
18.6.82	82.00	0.00	529.97	535.32	547.97	1950.00
18.6.83	83.00	0.00	536.42	541.44	567.19	1963.00
18.6.84	84.00	0.00	0.00	523.86	535.05	684.06
18.6.85	85.00	389.33	527.89	533.76	563.25	2379.00
18.6.86	86.00	0.00	551.25	556.67	564.97	716.37
18.6.87	87.00	0.00	537.26	543.49	551.93	708.63
18.6.88	88.00	0.00	502.11	508.84	517.05	653.99
18.6.89	89.00	0.00	521.25	526.57	534.44	683.58
18.6.90	90.00	454.25	531.27	535.90	552.76	1432.00
18.6.91	91.00	0.00	0.00	523.94	535.06	697.99
18.6.92	92.00	0.00	0.00	517.28	534.21	677.78
18.6.93	93.00	0.00	0.00	517.01	532.67	685.70
18.6.94	94.00	0.00	0.00	549.97	568.01	734.55
18.6.95	95.00	0.00	0.00	538.82	554.43	715.66
18.6.96	96.00	0.00	0.00	508.44	525.24	1028.00
18.6.97	97.00	0.00	0.00	525.70	545.47	783.00
18.6.98	98.00	0.00	526.03	539.66	548.71	2074.00
18.6.99	99.00	0.00	0.00	502.16	545.02	707.89
18.6.100	100.00	0.00	0.00	0.00	539.55	2354.00

18.6.101	101.00	0.00	0.00	35.00	535.38	694.92
18.6.102	102.00	0.00	0.00	0.00	562.63	1544.00
18.6.103	103.00	0.00	0.00	0.00	546.33	934.00
18.6.104	104.00	0.00	507.03	517.50	529.54	2104.00
18.6.105	105.00	0.00	527.91	538.97	570.12	2052.00
18.6.106	106.00	0.00	538.98	549.04	558.88	2518.00
18.6.107	107.00	0.00	538.74	549.07	569.58	2327.00
18.6.108	108.00	0.00	0.00	509.81	543.20	975.00
18.6.109	109.00	0.00	532.08	541.22	562.65	2724.00
18.6.110	110.00	0.00	551.31	573.60	582.83	738.06
18.6.111	111.00	0.00	529.71	556.23	563.77	796.00
18.6.112	112.00	0.00	514.23	522.98	530.33	1940.00
18.6.113	113.00	0.00	529.85	535.48	547.13	2038.00
18.6.114	114.00	0.00	539.52	546.18	555.03	2253.00
18.6.115	115.00	0.00	538.89	546.80	566.20	2267.00
18.6.116	116.00	0.00	0.00	514.10	552.61	748.00
18.6.117	117.00	0.00	541.09	548.25	561.91	2428.00
18.6.118	118.00	0.00	565.56	580.05	586.91	732.53
18.6.119	119.00	0.00	544.79	566.05	573.05	776.00
18.6.120	120.00	0.00	504.98	521.70	528.56	1162.00
18.6.121	121.00	0.00	531.60	546.45	551.26	695.82
18.6.122	122.00	0.00	529.28	538.88	546.09	2027.00
18.6.123	123.00	0.00	0.00	514.09	547.01	705.15
18.6.124	124.00	0.00	0.00	512.00	540.61	692.10

18.6.125	125.00	0.00	0.00	512.58	540.40	695.51
18.6.126	126.00	0.00	0.00	541.87	564.77	720.59
18.6.127	127.00	0.00	0.00	523.62	551.41	710.71

18.7 L2-Fabric Write and Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.7.0	0.00	304.80	310.97	340.34	349.63	488.43
18.7.1	1.00	311.34	316.90	332.27	343.18	472.95
18.7.2	2.00	316.74	322.78	350.30	360.10	504.64
18.7.3	3.00	317.04	323.06	348.82	357.67	504.74
18.7.4	4.00	312.12	319.44	366.51	379.32	547.76
18.7.5	5.00	313.07	321.11	357.57	370.36	520.76
18.7.6	6.00	327.18	333.69	380.82	393.35	549.45
18.7.7	7.00	322.84	329.06	368.16	378.32	529.67
18.7.8	8.00	305.09	310.55	371.92	458.71	655.47
18.7.9	9.00	313.26	319.84	398.05	468.85	658.29
18.7.10	10.00	310.60	316.69	380.58	450.74	641.77
18.7.11	11.00	316.75	322.98	398.17	447.96	637.72
18.7.12	12.00	0.00	318.00	382.94	416.12	578.05
18.7.13	13.00	0.00	319.83	386.65	409.28	565.31
18.7.14	14.00	321.21	327.44	388.25	434.62	650.19
18.7.15	15.00	315.63	321.48	398.29	453.61	662.25
18.7.16	16.00	304.86	310.01	368.61	451.49	648.55
18.7.17	17.00	313.62	319.47	393.21	464.62	653.37
18.7.18	18.00	309.41	315.05	375.70	442.80	633.59

18.7.19	19.00	317.19	323.06	393.67	442.26	634.38
18.7.20	20.00	311.73	317.63	378.89	411.01	573.58
18.7.21	21.00	0.00	319.34	381.15	405.03	562.01
18.7.22	22.00	0.00	326.78	381.86	431.19	642.22
18.7.23	23.00	0.00	320.32	392.20	449.75	659.10
18.7.24	24.00	304.36	310.13	337.09	349.29	483.34
18.7.25	25.00	310.94	315.61	329.02	340.31	506.21
18.7.26	26.00	316.83	323.42	347.16	357.50	515.25
18.7.27	27.00	316.50	322.49	344.22	355.17	515.44
18.7.28	28.00	314.25	319.89	364.25	376.26	529.62
18.7.29	29.00	315.48	321.10	355.53	367.79	507.33
18.7.30	30.00	327.85	334.04	379.73	389.32	538.10
18.7.31	31.00	323.00	329.03	364.74	374.79	542.01
18.7.32	32.00	307.30	311.12	370.37	460.76	654.57
18.7.33	33.00	316.79	320.49	395.95	468.66	660.22
18.7.34	34.00	311.08	316.68	376.93	460.91	629.17
18.7.35	35.00	320.11	324.42	394.93	453.86	624.19
18.7.36	36.00	313.85	318.40	384.47	424.03	577.04
18.7.37	37.00	316.14	320.21	385.37	413.45	573.76
18.7.38	38.00	322.36	327.40	398.75	430.04	593.18
18.7.39	39.00	316.95	321.38	382.41	449.26	620.29
18.7.40	40.00	306.92	310.87	341.34	352.88	490.62
18.7.41	41.00	312.71	316.38	334.12	344.28	503.25
18.7.42	42.00	319.78	323.87	349.46	360.27	511.39

18.7.43	43.00	319.97	323.37	347.20	358.32	508.02
18.7.44	44.00	315.31	319.34	368.60	383.64	525.74
18.7.45	45.00	0.00	321.87	361.43	373.28	510.60
18.7.46	46.00	328.99	334.16	385.29	398.46	542.78
18.7.47	47.00	324.20	328.53	364.20	374.96	531.46
18.7.48	48.00	305.74	309.99	339.92	349.85	477.23
18.7.49	49.00	311.72	316.02	330.86	339.64	483.48
18.7.50	50.00	316.14	321.93	346.54	356.14	505.07
18.7.51	51.00	317.99	322.51	344.75	354.22	497.96
18.7.52	52.00	312.46	317.97	367.53	380.25	521.73
18.7.53	53.00	0.00	319.68	359.36	370.53	504.11
18.7.54	54.00	326.78	331.78	383.79	394.53	532.20
18.7.55	55.00	321.82	327.01	361.63	371.72	527.24
18.7.56	56.00	305.52	310.01	364.71	453.07	651.33
18.7.57	57.00	314.84	319.56	388.80	463.49	655.18
18.7.58	58.00	311.30	315.88	371.24	454.28	622.06
18.7.59	59.00	318.18	322.86	389.52	446.94	617.60
18.7.60	60.00	312.64	317.10	377.41	420.64	582.89
18.7.61	61.00	314.75	319.21	381.91	408.77	571.08
18.7.62	62.00	321.66	326.78	392.06	424.87	604.17
18.7.63	63.00	315.52	320.53	375.82	445.46	605.77
18.7.64	64.00	304.68	310.38	335.09	344.19	508.84
18.7.65	65.00	310.49	316.58	327.77	337.44	533.62
18.7.66	66.00	315.66	322.01	342.15	351.59	528.21

18.7.67	67.00	316.69	322.95	341.24	350.52	536.79
18.7.68	68.00	311.55	317.35	361.94	372.91	559.31
18.7.69	69.00	313.24	318.83	354.92	364.04	529.63
18.7.70	70.00	326.91	331.78	377.66	388.38	555.99
18.7.71	71.00	322.39	327.35	358.93	368.96	545.86
18.7.72	72.00	304.60	309.13	369.60	453.17	652.99
18.7.73	73.00	314.00	318.70	394.78	463.27	652.91
18.7.74	74.00	311.13	315.40	375.04	443.55	650.12
18.7.75	75.00	317.46	322.44	394.35	438.90	642.96
18.7.76	76.00	311.50	315.95	381.46	408.65	604.17
18.7.77	77.00	313.86	318.10	383.94	402.44	603.34
18.7.78	78.00	321.96	326.79	393.60	420.88	609.97
18.7.79	79.00	316.20	320.97	380.22	442.88	661.52
18.7.80	80.00	304.60	309.84	371.55	458.31	654.00
18.7.81	81.00	314.71	319.80	399.19	464.88	650.74
18.7.82	82.00	310.52	315.53	378.44	447.56	646.29
18.7.83	83.00	318.71	324.35	397.41	442.61	642.01
18.7.84	84.00	311.39	317.04	383.58	413.12	595.14
18.7.85	85.00	314.37	319.28	385.37	405.76	591.14
18.7.86	86.00	322.03	327.17	398.64	424.23	603.04
18.7.87	87.00	315.16	320.90	382.83	446.02	661.09
18.7.88	88.00	306.32	311.51	336.81	348.06	493.95
18.7.89	89.00	311.42	316.94	329.91	339.52	510.33
18.7.90	90.00	318.76	324.21	343.99	354.63	514.33

18.7.91	91.00	319.23	323.85	342.22	353.19	498.30
18.7.92	92.00	314.62	319.64	363.40	375.67	562.67
18.7.93	93.00	315.86	321.04	355.92	366.77	524.99
18.7.94	94.00	328.88	334.24	380.27	390.30	560.74
18.7.95	95.00	324.32	329.02	360.06	370.34	536.31
18.7.96	96.00	0.00	309.86	369.32	459.18	658.32
18.7.97	97.00	0.00	319.49	392.58	462.21	656.18
18.7.98	98.00	310.50	316.24	377.85	447.16	661.09
18.7.99	99.00	0.00	324.41	391.03	439.62	653.09
18.7.100	100.00	0.00	317.12	380.85	412.44	604.17
18.7.101	101.00	0.00	319.17	382.21	404.11	578.90
18.7.102	102.00	0.00	326.81	385.45	427.36	614.20
18.7.103	103.00	0.00	319.67	362.03	446.72	651.91
18.7.104	104.00	0.00	311.77	333.55	342.53	494.94
18.7.105	105.00	0.00	316.70	325.84	332.81	484.86
18.7.106	106.00	0.00	324.21	344.82	352.22	500.99
18.7.107	107.00	0.00	323.44	341.24	349.48	497.23
18.7.108	108.00	0.00	319.18	362.65	373.41	526.45
18.7.109	109.00	0.00	320.68	352.06	362.64	512.53
18.7.110	110.00	0.00	333.80	379.09	390.30	542.57
18.7.111	111.00	0.00	327.75	357.85	366.45	515.58
18.7.112	112.00	306.06	310.68	334.83	344.53	492.18
18.7.113	113.00	0.00	316.70	326.67	335.35	517.91
18.7.114	114.00	318.36	323.00	344.99	353.66	528.50

18.7.115	115.00	0.00	322.71	342.62	352.34	523.11
18.7.116	116.00	312.32	317.70	364.26	375.58	527.14
18.7.117	117.00	0.00	318.88	354.20	364.06	518.10
18.7.118	118.00	327.30	331.87	381.25	392.07	548.63
18.7.119	119.00	0.00	327.86	361.88	370.84	551.65
18.7.120	120.00	305.75	310.18	369.37	460.33	660.47
18.7.121	121.00	0.00	320.01	394.07	462.81	654.29
18.7.122	122.00	0.00	316.97	377.09	450.95	656.17
18.7.123	123.00	0.00	323.99	393.00	445.17	646.81
18.7.124	124.00	312.60	317.17	381.18	415.30	612.87
18.7.125	125.00	0.00	319.55	382.98	406.53	584.10
18.7.126	126.00	322.71	327.30	383.75	428.41	618.52
18.7.127	127.00	317.07	321.70	394.66	452.57	643.92

18.8 L2-Fabric Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.8.0	0.00	0.00	0.00	0.00	0.00	0.00
18.8.1	1.00	0.00	0.00	0.00	0.00	0.00
18.8.2	2.00	0.00	0.00	0.00	0.00	0.00
18.8.3	3.00	0.00	0.00	0.00	0.00	0.00
18.8.4	4.00	0.00	0.00	0.00	0.00	0.00
18.8.5	5.00	0.00	0.00	0.00	0.00	0.00
18.8.6	6.00	0.00	0.00	0.00	0.00	0.00
18.8.7	7.00	0.00	0.00	0.00	0.00	0.00
18.8.8	8.00	0.00	0.00	0.00	0.00	0.00

18.8.9	9.00	0.00	0.00	0.00	0.00	0.00
18.8.10	10.00	0.00	0.00	0.00	0.00	0.00
18.8.11	11.00	0.00	0.00	0.00	0.00	0.00
18.8.12	12.00	0.00	0.00	0.00	0.00	0.00
18.8.13	13.00	0.00	0.00	0.00	0.00	0.00
18.8.14	14.00	0.00	0.00	0.00	0.00	0.00
18.8.15	15.00	0.00	0.00	0.00	0.00	0.00
18.8.16	16.00	0.00	0.00	0.00	0.00	0.00
18.8.17	17.00	0.00	0.00	0.00	0.00	0.00
18.8.18	18.00	0.00	0.00	0.00	0.00	0.00
18.8.19	19.00	0.00	0.00	0.00	0.00	0.00
18.8.20	20.00	0.00	0.00	0.00	0.00	0.00
18.8.21	21.00	0.00	0.00	0.00	0.00	0.00
18.8.22	22.00	0.00	0.00	0.00	0.00	0.00
18.8.23	23.00	0.00	0.00	0.00	0.00	0.00
18.8.24	24.00	0.00	0.00	0.00	0.00	0.00
18.8.25	25.00	0.00	0.00	0.00	0.00	0.00
18.8.26	26.00	0.00	0.00	0.00	0.00	0.00
18.8.27	27.00	0.00	0.00	0.00	0.00	0.00
18.8.28	28.00	0.00	0.00	0.00	0.00	0.00
18.8.29	29.00	0.00	0.00	0.00	0.00	0.00
18.8.30	30.00	0.00	0.00	0.00	0.00	0.00
18.8.31	31.00	0.00	0.00	0.00	0.00	0.00
18.8.32	32.00	0.00	0.00	0.00	0.00	0.00

18.8.33	33.00	0.00	0.00	0.00	0.00	0.00
18.8.34	34.00	0.00	0.00	0.00	0.00	0.00
18.8.35	35.00	0.00	0.00	0.00	0.00	0.00
18.8.36	36.00	0.00	0.00	0.00	0.00	0.00
18.8.37	37.00	0.00	0.00	0.00	0.00	0.00
18.8.38	38.00	0.00	0.00	0.00	0.00	0.00
18.8.39	39.00	0.00	0.00	0.00	0.00	0.00
18.8.40	40.00	0.00	0.00	0.00	0.00	0.00
18.8.41	41.00	0.00	0.00	0.00	0.00	0.00
18.8.42	42.00	0.00	0.00	0.00	0.00	0.00
18.8.43	43.00	0.00	0.00	0.00	0.00	0.00
18.8.44	44.00	0.00	0.00	0.00	0.00	0.00
18.8.45	45.00	0.00	0.00	0.00	0.00	0.00
18.8.46	46.00	0.00	0.00	0.00	0.00	0.00
18.8.47	47.00	0.00	0.00	0.00	0.00	0.00
18.8.48	48.00	0.00	0.00	0.00	0.00	0.00
18.8.49	49.00	0.00	0.00	0.00	0.00	0.00
18.8.50	50.00	0.00	0.00	0.00	0.00	0.00
18.8.51	51.00	0.00	0.00	0.00	0.00	0.00
18.8.52	52.00	0.00	0.00	0.00	0.00	0.00
18.8.53	53.00	0.00	0.00	0.00	0.00	0.00
18.8.54	54.00	0.00	0.00	0.00	0.00	0.00
18.8.55	55.00	0.00	0.00	0.00	0.00	0.00
18.8.56	56.00	0.00	0.00	0.00	0.00	0.00

18.8.57	57.00	0.00	0.00	0.00	0.00	0.00
18.8.58	58.00	0.00	0.00	0.00	0.00	0.00
18.8.59	59.00	0.00	0.00	0.00	0.00	0.00
18.8.60	60.00	0.00	0.00	0.00	0.00	0.00
18.8.61	61.00	0.00	0.00	0.00	0.00	0.00
18.8.62	62.00	0.00	0.00	0.00	0.00	0.00
18.8.63	63.00	0.00	0.00	0.00	0.00	0.00
18.8.64	64.00	0.00	0.00	0.00	0.00	0.00
18.8.65	65.00	0.00	0.00	0.00	0.00	0.00
18.8.66	66.00	0.00	0.00	0.00	0.00	0.00
18.8.67	67.00	0.00	0.00	0.00	0.00	0.00
18.8.68	68.00	0.00	0.00	0.00	0.00	0.00
18.8.69	69.00	0.00	0.00	0.00	0.00	0.00
18.8.70	70.00	0.00	0.00	0.00	0.00	0.00
18.8.71	71.00	0.00	0.00	0.00	0.00	0.00
18.8.72	72.00	0.00	0.00	0.00	0.00	0.00
18.8.73	73.00	0.00	0.00	0.00	0.00	0.00
18.8.74	74.00	0.00	0.00	0.00	0.00	0.00
18.8.75	75.00	0.00	0.00	0.00	0.00	0.00
18.8.76	76.00	0.00	0.00	0.00	0.00	0.00
18.8.77	77.00	0.00	0.00	0.00	0.00	0.00
18.8.78	78.00	0.00	0.00	0.00	0.00	0.00
18.8.79	79.00	0.00	0.00	0.00	0.00	0.00
18.8.80	80.00	0.00	0.00	0.00	0.00	0.00

18.8.81	81.00	0.00	0.00	0.00	0.00	0.00
18.8.82	82.00	0.00	0.00	0.00	0.00	0.00
18.8.83	83.00	0.00	0.00	0.00	0.00	0.00
18.8.84	84.00	0.00	0.00	0.00	0.00	0.00
18.8.85	85.00	0.00	0.00	0.00	0.00	0.00
18.8.86	86.00	0.00	0.00	0.00	0.00	0.00
18.8.87	87.00	0.00	0.00	0.00	0.00	0.00
18.8.88	88.00	0.00	0.00	0.00	0.00	0.00
18.8.89	89.00	0.00	0.00	0.00	0.00	0.00
18.8.90	90.00	0.00	0.00	0.00	0.00	0.00
18.8.91	91.00	0.00	0.00	0.00	0.00	0.00
18.8.92	92.00	0.00	0.00	0.00	0.00	0.00
18.8.93	93.00	0.00	0.00	0.00	0.00	0.00
18.8.94	94.00	0.00	0.00	0.00	0.00	0.00
18.8.95	95.00	0.00	0.00	0.00	0.00	0.00
18.8.96	96.00	0.00	0.00	0.00	0.00	0.00
18.8.97	97.00	0.00	0.00	0.00	0.00	0.00
18.8.98	98.00	0.00	0.00	0.00	0.00	0.00
18.8.99	99.00	0.00	0.00	0.00	0.00	0.00
18.8.100	100.00	0.00	0.00	0.00	0.00	0.00
18.8.101	101.00	0.00	0.00	0.00	0.00	0.00
18.8.102	102.00	0.00	0.00	0.00	0.00	0.00
18.8.103	103.00	0.00	0.00	0.00	0.00	0.00
18.8.104	104.00	0.00	0.00	0.00	0.00	0.00

18.8.105	105.00	0.00	0.00	0.00	0.00	0.00
18.8.106	106.00	0.00	0.00	0.00	0.00	0.00
18.8.107	107.00	0.00	0.00	0.00	0.00	0.00
18.8.108	108.00	0.00	0.00	0.00	0.00	0.00
18.8.109	109.00	0.00	0.00	0.00	0.00	0.00
18.8.110	110.00	0.00	0.00	0.00	0.00	0.00
18.8.111	111.00	0.00	0.00	0.00	0.00	0.00
18.8.112	112.00	0.00	0.00	0.00	0.00	0.00
18.8.113	113.00	0.00	0.00	0.00	0.00	0.00
18.8.114	114.00	0.00	0.00	0.00	0.00	0.00
18.8.115	115.00	0.00	0.00	0.00	0.00	0.00
18.8.116	116.00	0.00	0.00	0.00	0.00	0.00
18.8.117	117.00	0.00	0.00	0.00	0.00	0.00
18.8.118	118.00	0.00	0.00	0.00	0.00	0.00
18.8.119	119.00	0.00	0.00	0.00	0.00	0.00
18.8.120	120.00	0.00	0.00	0.00	0.00	0.00
18.8.121	121.00	0.00	0.00	0.00	0.00	0.00
18.8.122	122.00	0.00	0.00	0.00	0.00	0.00
18.8.123	123.00	0.00	0.00	0.00	0.00	0.00
18.8.124	124.00	0.00	0.00	0.00	0.00	0.00
18.8.125	125.00	0.00	0.00	0.00	0.00	0.00
18.8.126	126.00	0.00	0.00	0.00	0.00	0.00
18.8.127	127.00	0.00	0.00	0.00	0.00	0.00

18.12 L2-Fabric (128B read requests per normUnit)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.12.0	0.00	0.00	0.00	0.00	1152.00	1152.00
18.12.1	1.00	0.00	0.00	0.00	1152.00	1152.00
18.12.2	2.00	0.00	0.00	0.00	1152.00	1152.00
18.12.3	3.00	0.00	0.00	0.00	1152.00	1152.00
18.12.4	4.00	0.00	0.00	0.00	1152.00	1152.00
18.12.5	5.00	0.00	0.00	0.00	1152.00	1152.00
18.12.6	6.00	0.00	0.00	0.00	1152.00	1152.00
18.12.7	7.00	0.00	0.00	0.00	1152.00	1152.00
18.12.8	8.00	0.00	0.00	0.00	1154.00	1154.00
18.12.9	9.00	0.00	0.00	0.00	1154.00	1154.00
18.12.10	10.00	0.00	0.00	0.00	1154.00	1154.00
18.12.11	11.00	0.00	0.00	0.00	1154.00	1154.00
18.12.12	12.00	0.00	0.00	0.00	1154.00	1154.00
18.12.13	13.00	0.00	0.00	0.00	1154.00	1154.00
18.12.14	14.00	0.00	0.00	0.00	1154.00	1154.00
18.12.15	15.00	0.00	0.00	0.00	1154.00	1154.00
18.12.16	16.00	0.00	0.00	0.00	1154.00	1154.00
18.12.17	17.00	0.00	0.00	0.00	1154.00	1154.00
18.12.18	18.00	0.00	0.00	0.00	1154.00	1154.00
18.12.19	19.00	0.00	0.00	0.00	1154.00	1154.00
18.12.20	20.00	0.00	0.00	0.00	1154.00	1154.00
18.12.21	21.00	0.00	0.00	0.00	1154.00	1154.00
18.12.22	22.00	0.00	0.00	0.00	1154.00	1154.00

18.12.23	23.00	0.00	0.00	0.00	1154.00	1154.00
18.12.24	24.00	0.00	0.00	0.00	1152.00	1152.00
18.12.25	25.00	0.00	0.00	0.00	1152.00	1152.00
18.12.26	26.00	0.00	0.00	0.00	1152.00	1152.00
18.12.27	27.00	0.00	0.00	0.00	1152.00	1152.00
18.12.28	28.00	0.00	0.00	0.00	1152.00	1152.00
18.12.29	29.00	0.00	0.00	0.00	1152.00	1152.00
18.12.30	30.00	0.00	0.00	0.00	1152.00	1152.00
18.12.31	31.00	0.00	0.00	0.00	1152.00	1152.00
18.12.32	32.00	0.00	0.00	0.00	1152.00	1152.00
18.12.33	33.00	0.00	0.00	0.00	1152.00	1152.00
18.12.34	34.00	0.00	0.00	0.00	1152.00	1152.00
18.12.35	35.00	0.00	0.00	0.00	1152.00	1152.00
18.12.36	36.00	0.00	0.00	0.00	1152.00	1152.00
18.12.37	37.00	0.00	0.00	0.00	1152.00	1152.00
18.12.38	38.00	0.00	0.00	0.00	1152.00	1152.00
18.12.39	39.00	0.00	0.00	0.00	1152.00	1152.00
18.12.40	40.00	0.00	0.00	0.00	1154.00	1154.00
18.12.41	41.00	0.00	0.00	0.00	1154.00	1154.00
18.12.42	42.00	0.00	0.00	0.00	1154.00	1154.00
18.12.43	43.00	0.00	0.00	0.00	1154.00	1154.00
18.12.44	44.00	0.00	0.00	0.00	1154.00	1154.00
18.12.45	45.00	0.00	0.00	0.00	1154.00	1154.00
18.12.46	46.00	0.00	0.00	0.00	1154.00	1154.00

18.12.47	47.00	0.00	0.00	0.00	1154.00	1154.00
18.12.48	48.00	0.00	0.00	0.00	1154.00	1154.00
18.12.49	49.00	0.00	0.00	0.00	1154.00	1154.00
18.12.50	50.00	0.00	0.00	0.00	1154.00	1154.00
18.12.51	51.00	0.00	0.00	0.00	1154.00	1154.00
18.12.52	52.00	0.00	0.00	0.00	1154.00	1154.00
18.12.53	53.00	0.00	0.00	0.00	1154.00	1154.00
18.12.54	54.00	0.00	0.00	0.00	1154.00	1154.00
18.12.55	55.00	0.00	0.00	0.00	1154.00	1154.00
18.12.56	56.00	0.00	0.00	0.00	1152.00	1152.00
18.12.57	57.00	0.00	0.00	0.00	1152.00	1152.00
18.12.58	58.00	0.00	0.00	0.00	1152.00	1152.00
18.12.59	59.00	0.00	0.00	0.00	1152.00	1152.00
18.12.60	60.00	0.00	0.00	0.00	1152.00	1152.00
18.12.61	61.00	0.00	0.00	0.00	1152.00	1152.00
18.12.62	62.00	0.00	0.00	0.00	1152.00	1152.00
18.12.63	63.00	0.00	0.00	0.00	1152.00	1152.00
18.12.64	64.00	0.00	0.00	0.00	1154.00	1154.00
18.12.65	65.00	0.00	0.00	0.00	1154.00	1154.00
18.12.66	66.00	0.00	0.00	0.00	1154.00	1154.00
18.12.67	67.00	0.00	0.00	0.00	1154.00	1154.00
18.12.68	68.00	0.00	0.00	0.00	1154.00	1154.00
18.12.69	69.00	0.00	0.00	0.00	1154.00	1154.00
18.12.70	70.00	0.00	0.00	0.00	1154.00	1154.00

18.12.71	71.00	0.00	0.00	0.00	1154.00	1154.00
18.12.72	72.00	0.00	0.00	0.00	1152.00	1152.00
18.12.73	73.00	0.00	0.00	0.00	1152.00	1152.00
18.12.74	74.00	0.00	0.00	0.00	1152.00	1152.00
18.12.75	75.00	0.00	0.00	0.00	1152.00	1152.00
18.12.76	76.00	0.00	0.00	0.00	1152.00	1152.00
18.12.77	77.00	0.00	0.00	0.00	1152.00	1152.00
18.12.78	78.00	0.00	0.00	0.00	1152.00	1152.00
18.12.79	79.00	0.00	0.00	0.00	1152.00	1152.00
18.12.80	80.00	0.00	0.00	0.00	1152.00	1152.00
18.12.81	81.00	0.00	0.00	0.00	1152.00	1152.00
18.12.82	82.00	0.00	0.00	0.00	1152.00	1152.00
18.12.83	83.00	0.00	0.00	0.00	1152.00	1152.00
18.12.84	84.00	0.00	0.00	0.00	1152.00	1152.00
18.12.85	85.00	0.00	0.00	0.00	1152.00	1152.00
18.12.86	86.00	0.00	0.00	0.00	1152.00	1152.00
18.12.87	87.00	0.00	0.00	0.00	1152.00	1152.00
18.12.88	88.00	0.00	0.00	0.00	1154.00	1154.00
18.12.89	89.00	0.00	0.00	0.00	1154.00	1154.00
18.12.90	90.00	0.00	0.00	0.00	1154.00	1154.00
18.12.91	91.00	0.00	0.00	0.00	1154.00	1154.00
18.12.92	92.00	0.00	0.00	0.00	1154.00	1154.00
18.12.93	93.00	0.00	0.00	0.00	1154.00	1154.00
18.12.94	94.00	0.00	0.00	0.00	1154.00	1154.00

18.12.95	95.00	0.00	0.00	0.00	1154.00	1154.00
18.12.96	96.00	0.00	0.00	0.00	1154.00	1154.00
18.12.97	97.00	0.00	0.00	0.00	1154.00	1154.00
18.12.98	98.00	0.00	0.00	0.00	1154.00	1154.00
18.12.99	99.00	0.00	0.00	0.50	1154.00	1154.00
18.12.100	100.00	0.00	0.00	0.00	1154.00	1154.00
18.12.101	101.00	0.00	0.00	0.00	1154.00	1154.00
18.12.102	102.00	0.00	0.00	0.00	1154.00	1154.00
18.12.103	103.00	0.00	0.00	0.00	1154.00	1154.00
18.12.104	104.00	0.00	0.00	0.00	1152.00	1152.00
18.12.105	105.00	0.00	0.00	0.00	1152.00	1152.00
18.12.106	106.00	0.00	0.00	0.00	1152.00	1152.00
18.12.107	107.00	0.00	0.00	0.00	1152.00	1152.00
18.12.108	108.00	0.00	0.00	0.00	1152.00	1152.00
18.12.109	109.00	0.00	0.00	0.00	1152.00	1152.00
18.12.110	110.00	0.00	0.00	0.00	1152.00	1152.00
18.12.111	111.00	0.00	0.00	1.00	1152.00	1152.00
18.12.112	112.00	0.00	0.00	0.00	1152.00	1152.00
18.12.113	113.00	0.00	0.00	0.00	1152.00	1152.00
18.12.114	114.00	0.00	0.00	0.00	1152.00	1152.00
18.12.115	115.00	0.00	0.00	0.00	1152.00	1152.00
18.12.116	116.00	0.00	0.00	0.00	1152.00	1152.00
18.12.117	117.00	0.00	0.00	0.00	1152.00	1152.00
18.12.118	118.00	0.00	0.00	0.00	1152.00	1152.00

18.12.119	119.00	0.00	0.00	0.00	1152.00	1152.00
18.12.120	120.00	0.00	0.00	0.00	1154.00	1154.00
18.12.121	121.00	0.00	0.00	0.00	1154.00	1154.00
18.12.122	122.00	0.00	0.00	0.00	1154.00	1154.00
18.12.123	123.00	0.00	0.00	0.00	1154.00	1154.00
18.12.124	124.00	0.00	0.00	0.00	1154.00	1154.00
18.12.125	125.00	0.00	0.00	0.00	1154.00	1154.00
18.12.126	126.00	0.00	0.00	0.00	1154.00	1154.00
18.12.127	127.00	0.00	0.00	0.00	1154.00	1154.00

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